Azure

1. **What is Azure?**
   - Microsoft's Cloud Computing Platform
   - Offers - IaaS, PaaS and SaaS
2. **Why Azure?**
   - 2nd largest cloud provider after AWS
   - Strong integration with Microsoft
   - Outstanding hybrid cloud capabilities
3. **Why Learn Azure?**
   - High demand for Azure professional
   - Org leveraging Microsoft technologies are adapting Azure
   - Comprehensive Services - Compute, Storage, AI, Analytics, IoT
4. **Key features of Azure?**
   - Global Infra
   - Enterprise grade security and good compliance

Azure

How is this course Organized?

Beginner	Introduction and making you comfortable with azure
Intermediate	Compute, Storage, Networking, Security
Advance	Real Time projects and application (Terraform)

Regions

Azure

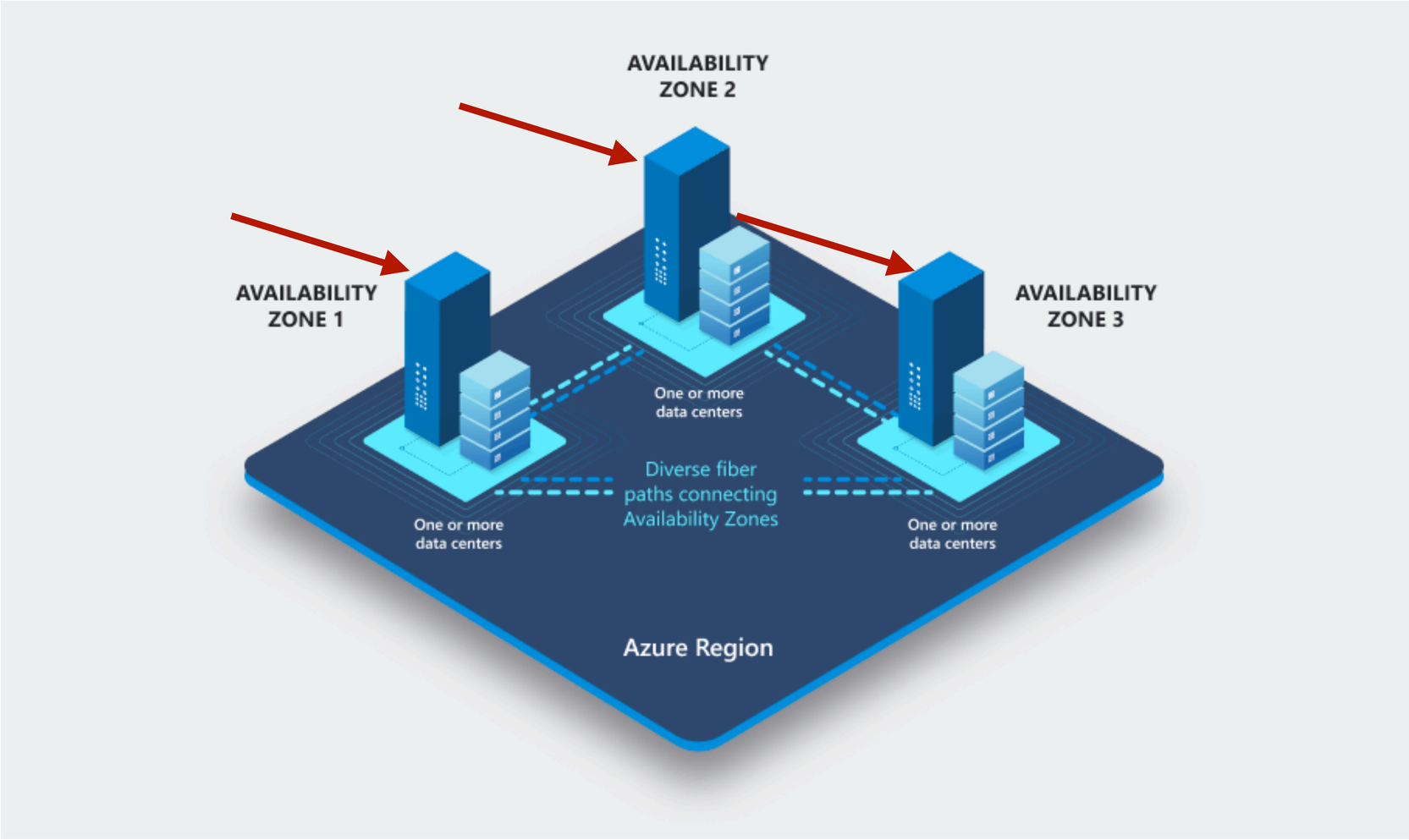
- Available region
- Announced regions
- Availability Zones
- Announced Availability Zones
- Announced region with Availability zones

Americas	Europe	Middle East	Africa	Asia Pacific
Brazil South	France Central	Qatar Central	South Africa North	Australia East
Canada Central	Italy North	UAE North		Central India
Central US	Germany West Central	Israel Central		Japan East
East US	Norway East			*Japan West
East US 2	North Europe			Southeast Asia
South Central US	UK South			East Asia
US Gov Virginia	West Europe			China North 3
West US 2	Sweden Central			Korea Central
West US 3	Switzerland North			New Zealand North
Mexico Central	Poland Central			
	Spain Central			

Source - https://learn.microsoft.com/en-us/azure/reliability/availability-zones-region-support

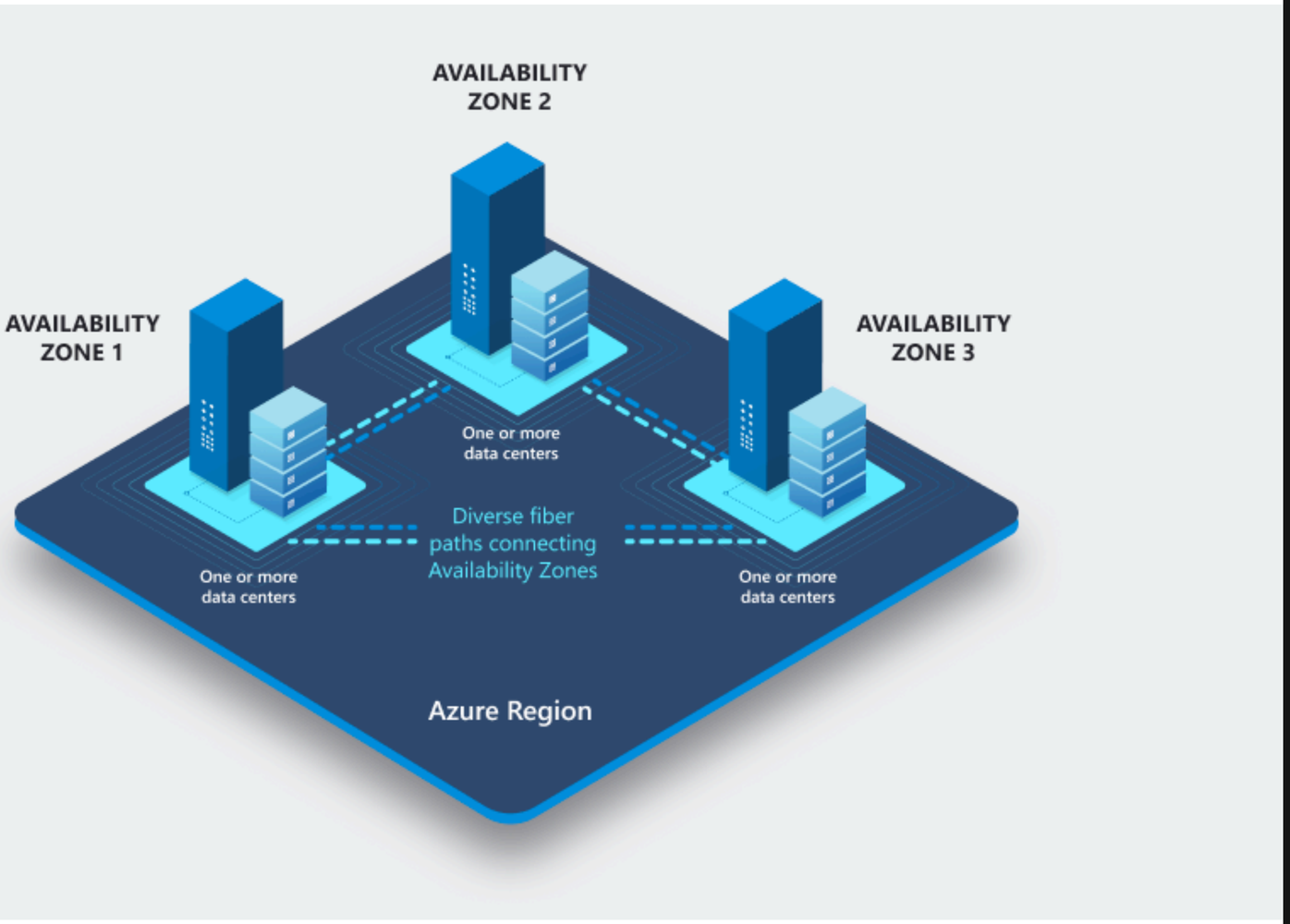
Zones

Azure


AVAILABILITY ZONE 2
AVAILABILITY ZONE 1
AVAILABILITY ZONE 3
One or more data centers
One or more data centers
One or more data centers
Diverse fiber paths connecting Availability Zones
Azure Region

Zones

Azure



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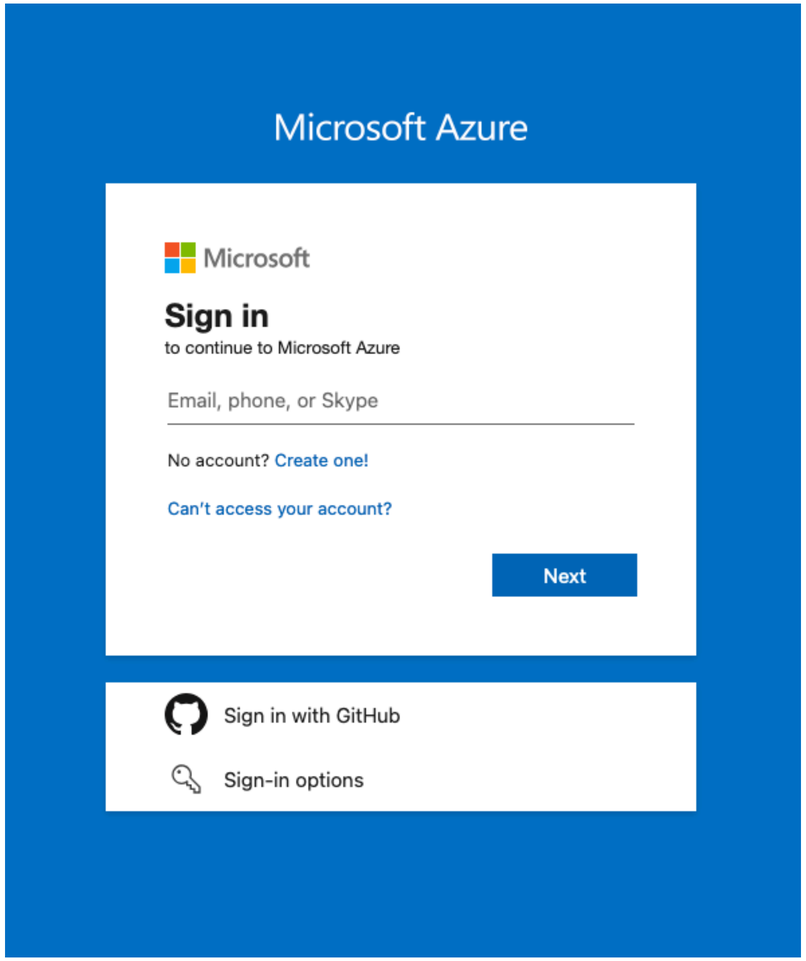
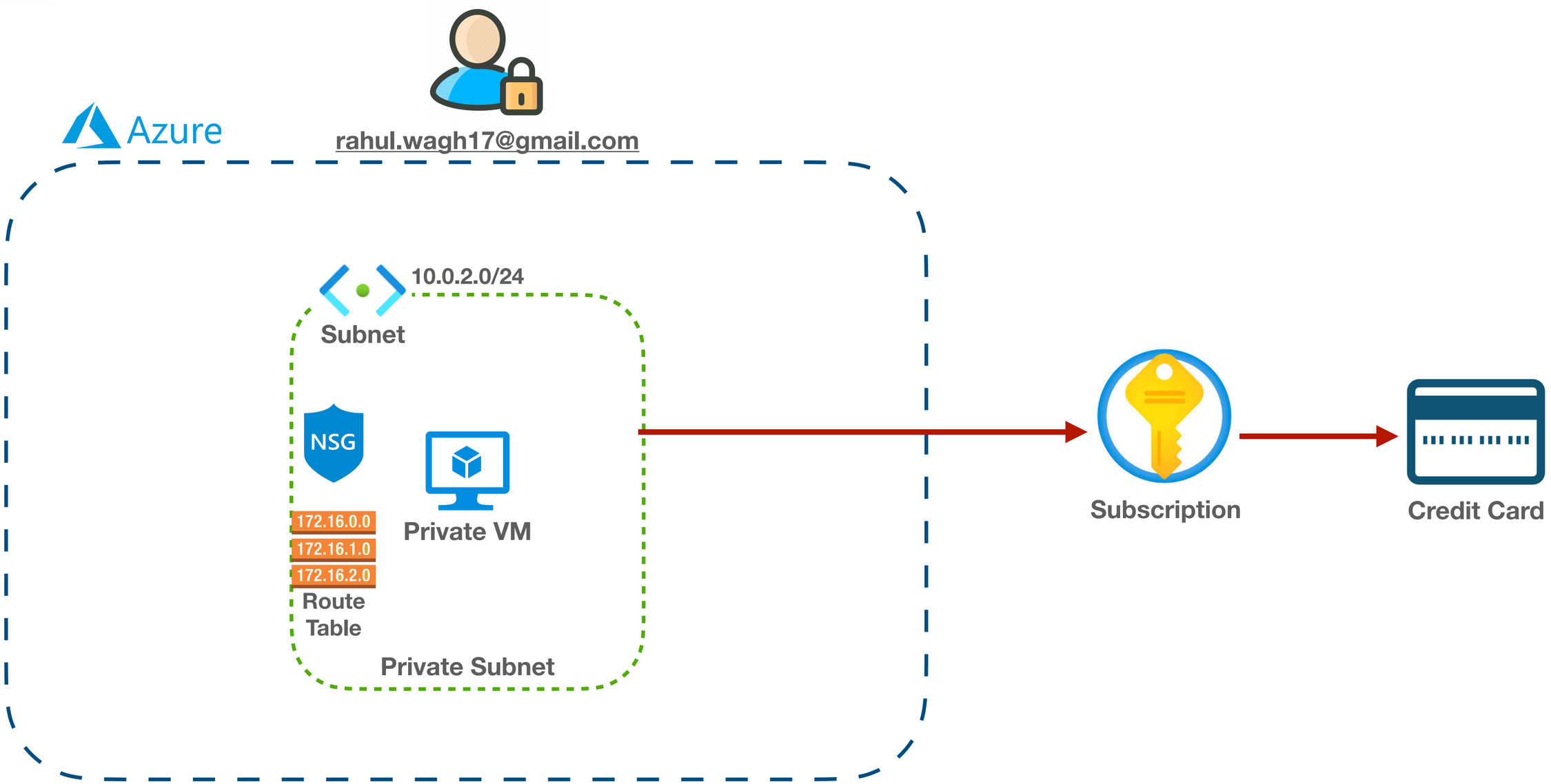
Sign up for Azure



Link - https://portal.azure.com

Account and subscription



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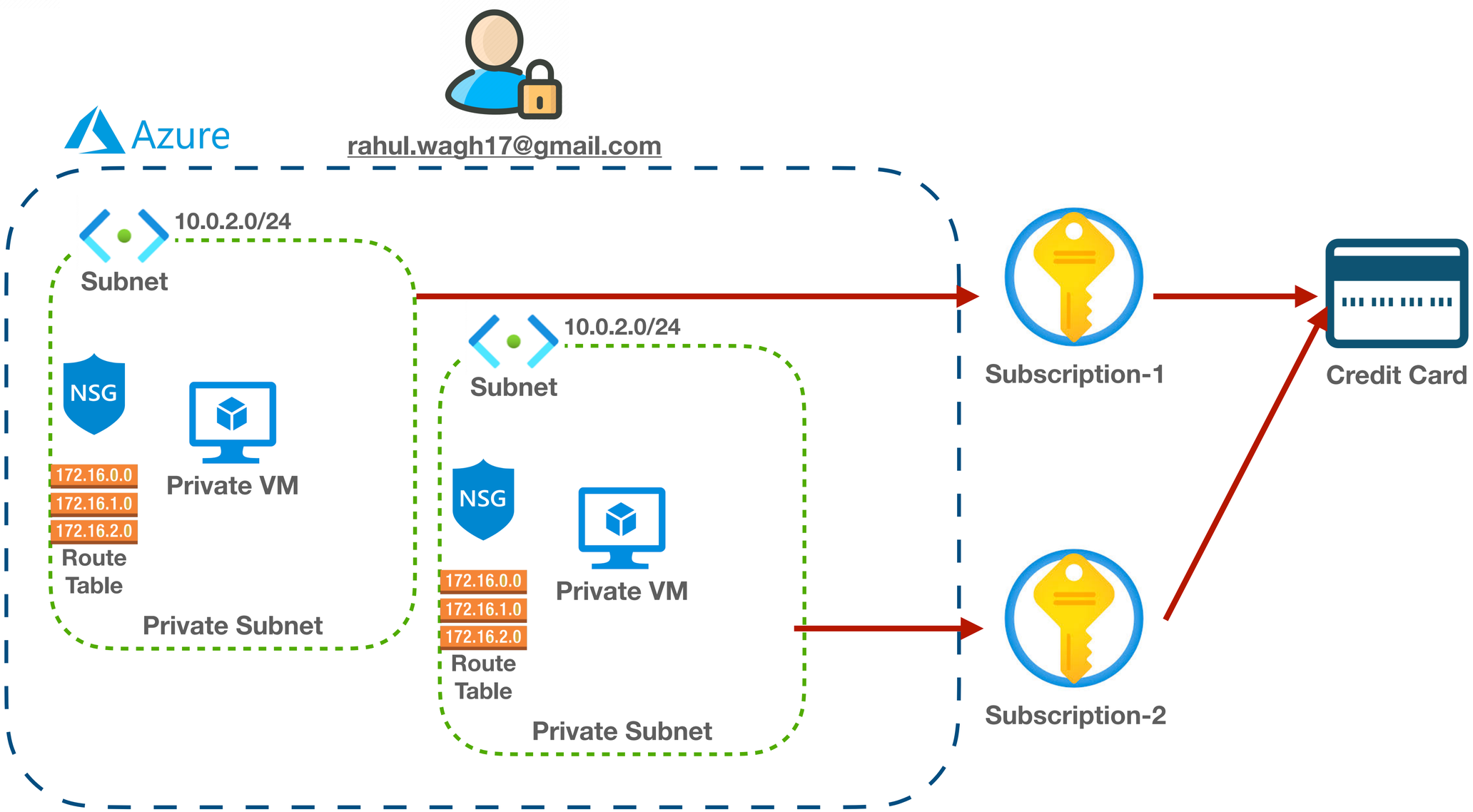
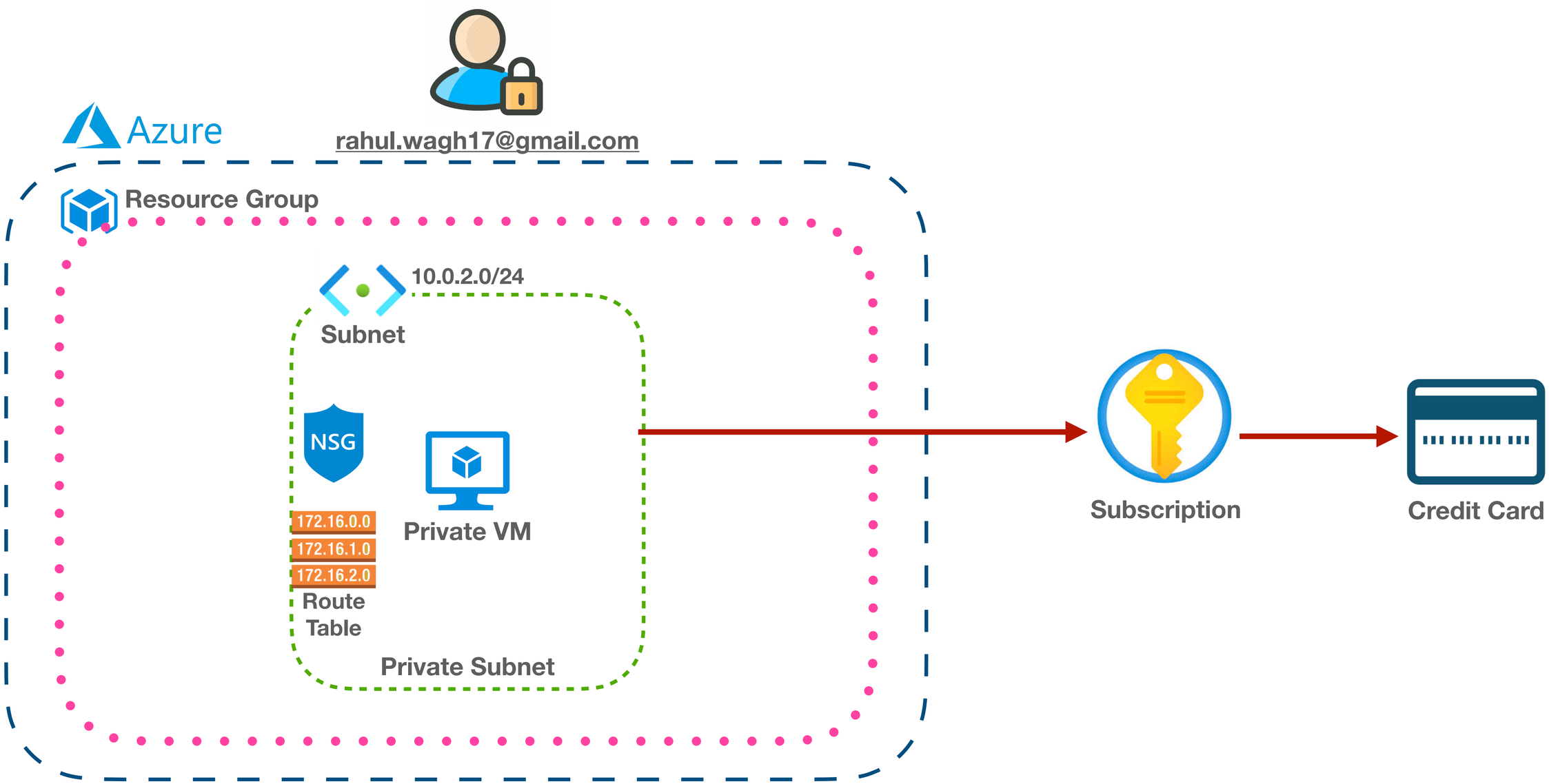
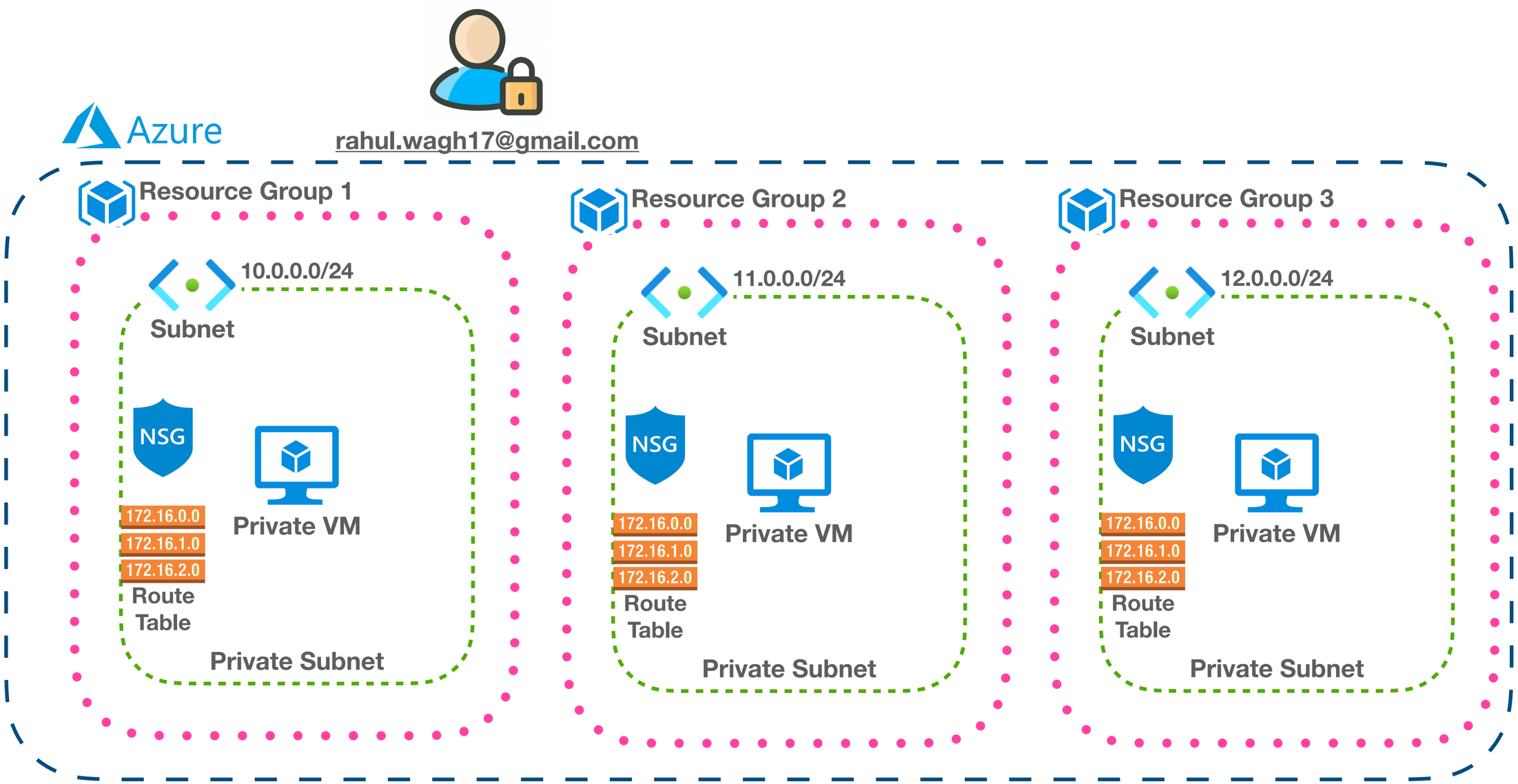


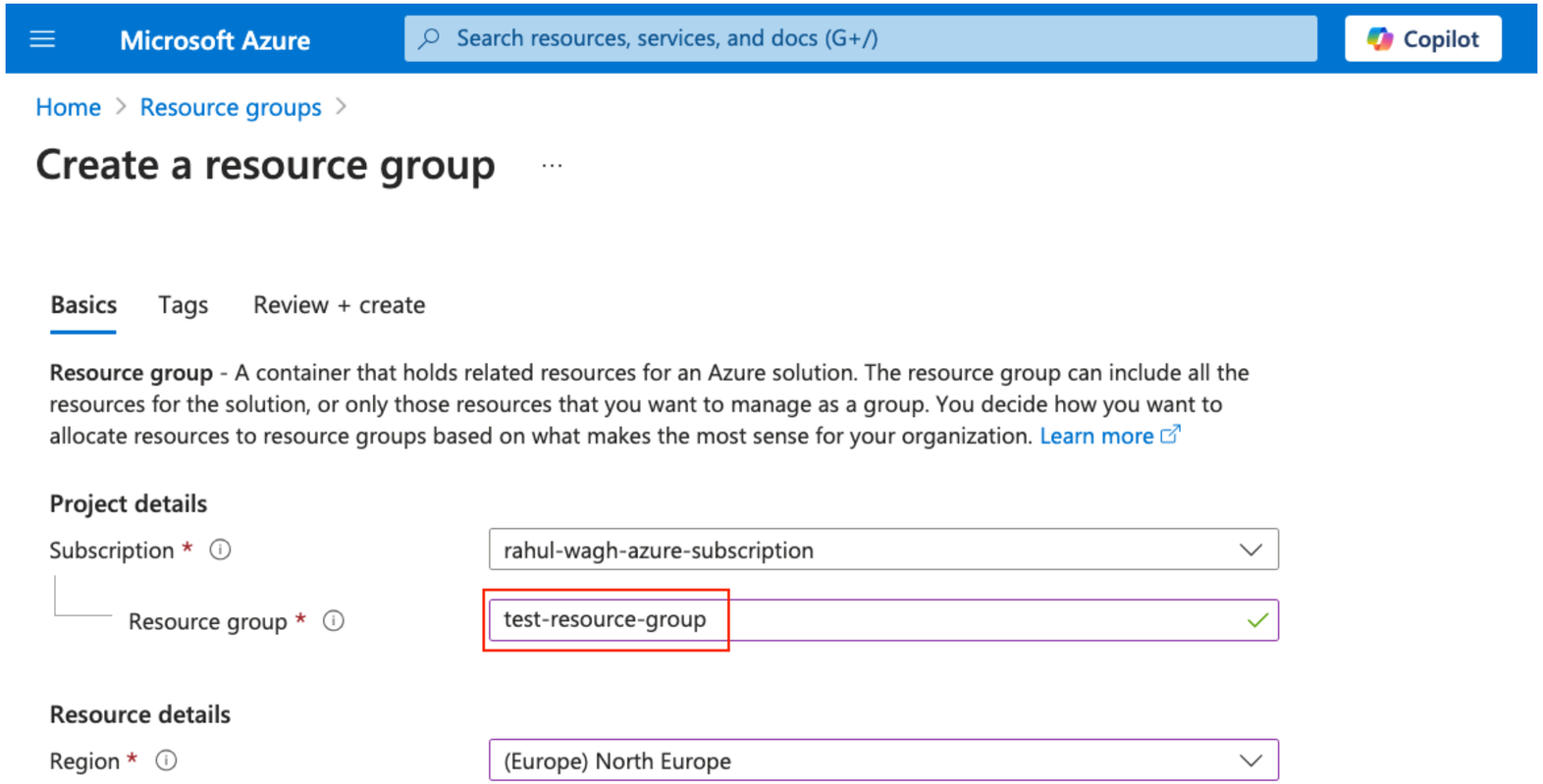
Resource Group



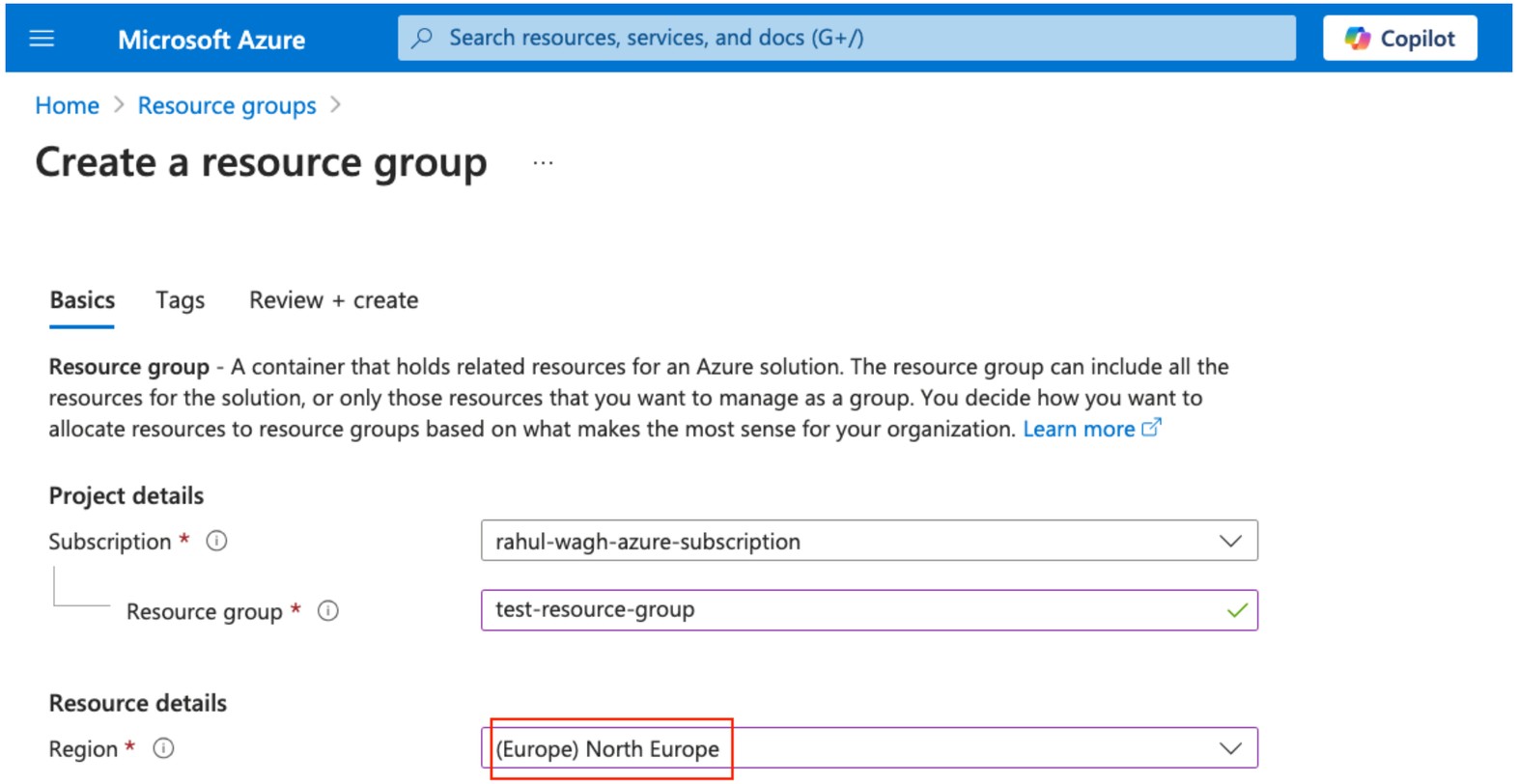
Resource Group



Azure CLI and Powershell



Azure CLI and Powershell



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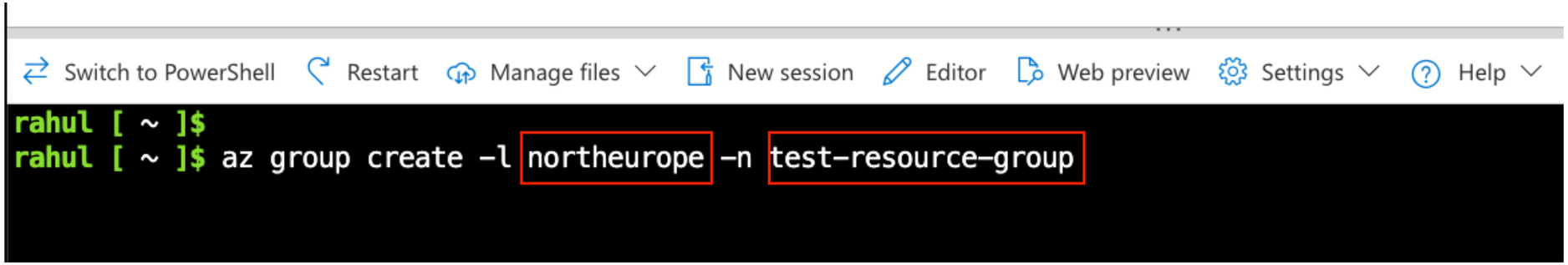
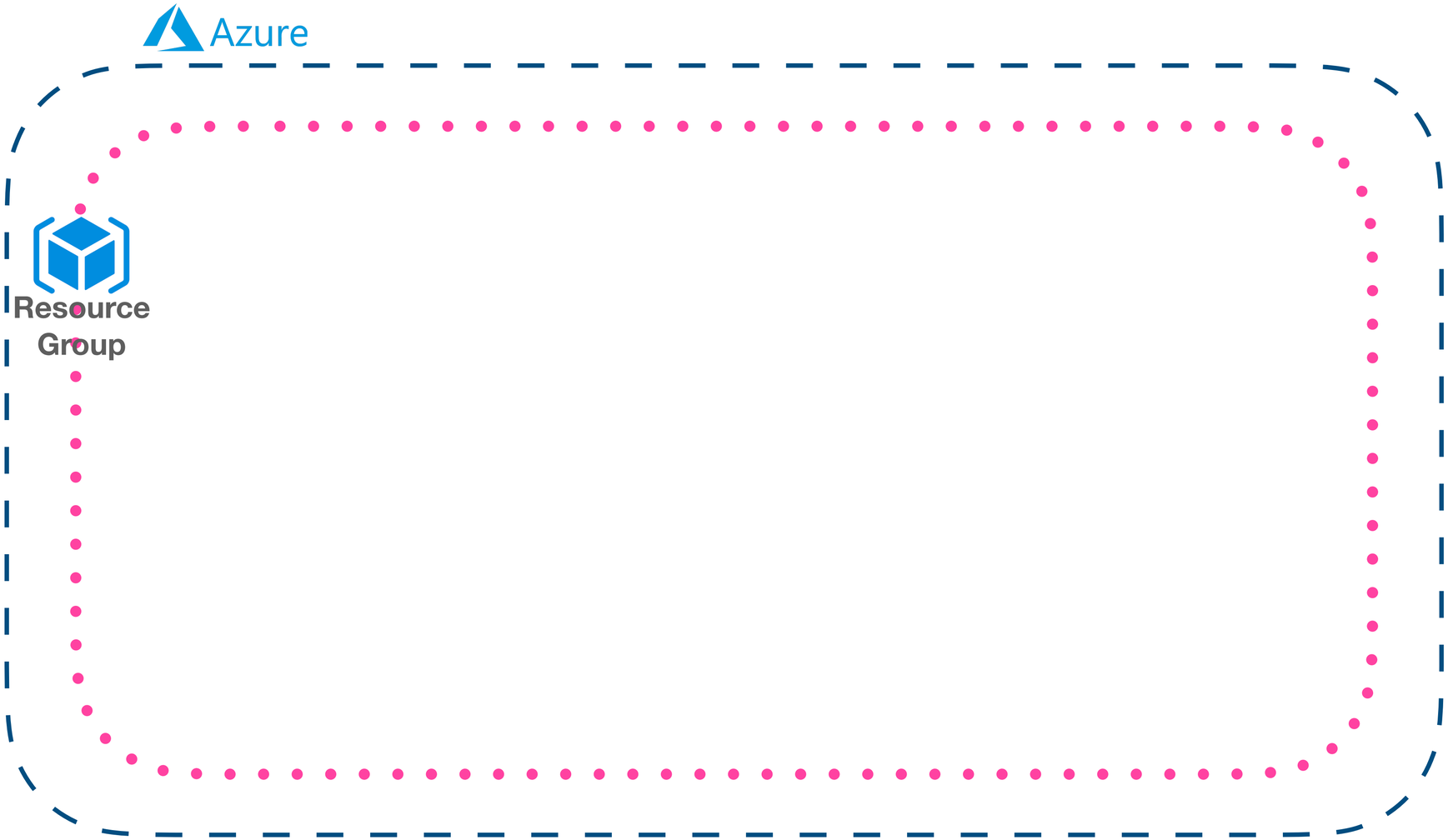


Azure Storage Account

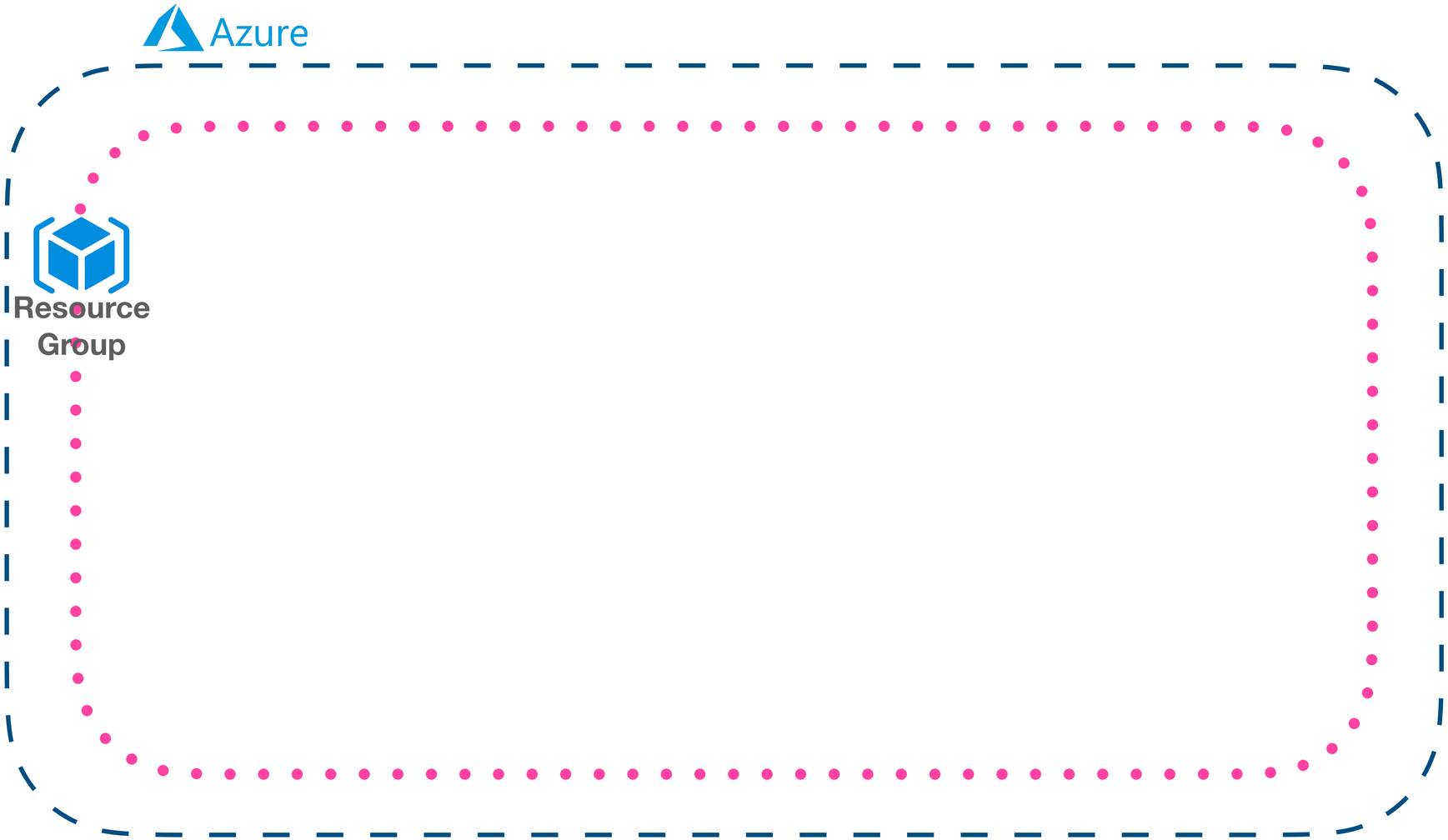
Azure

Azure

Azure Storage Account



Azure Storage Account



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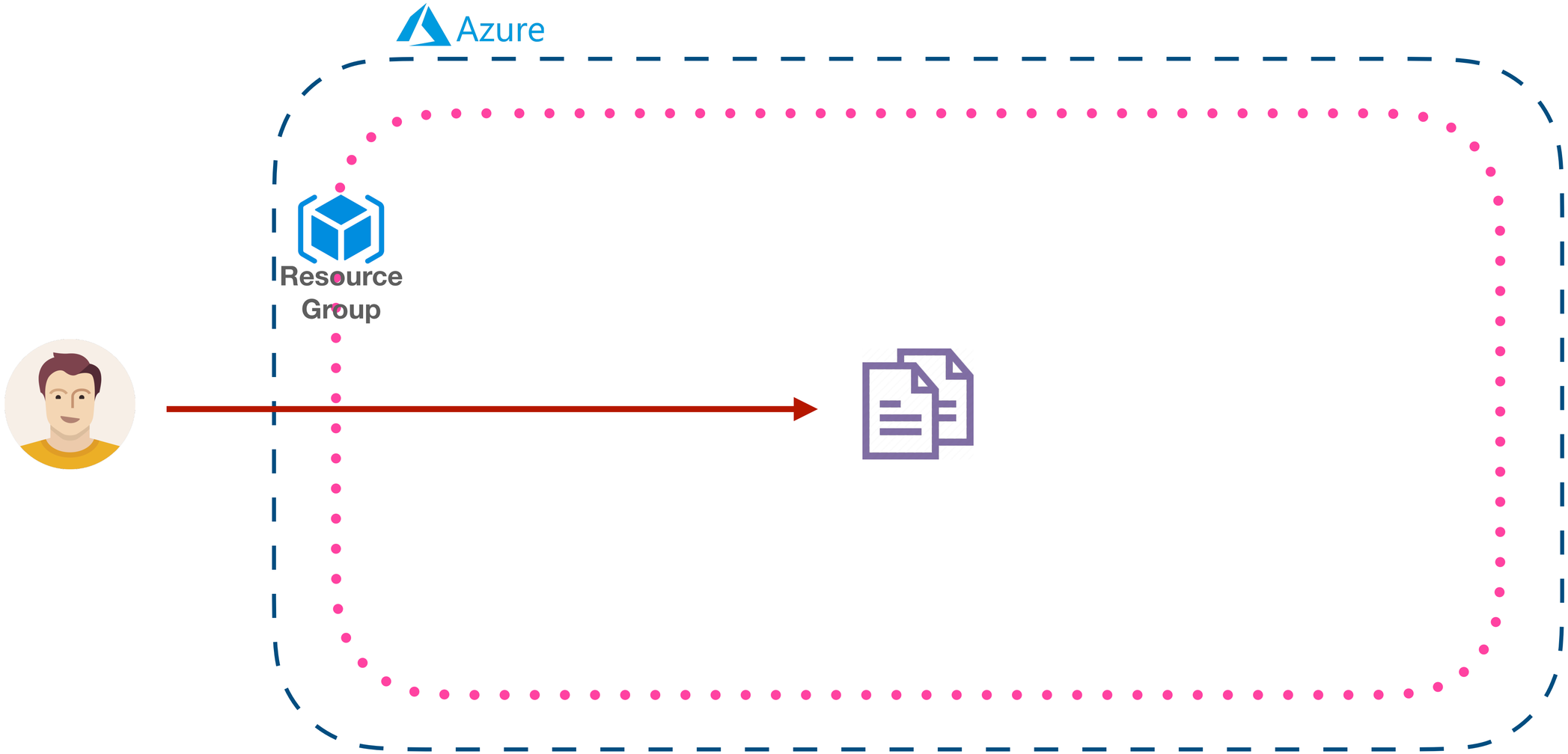
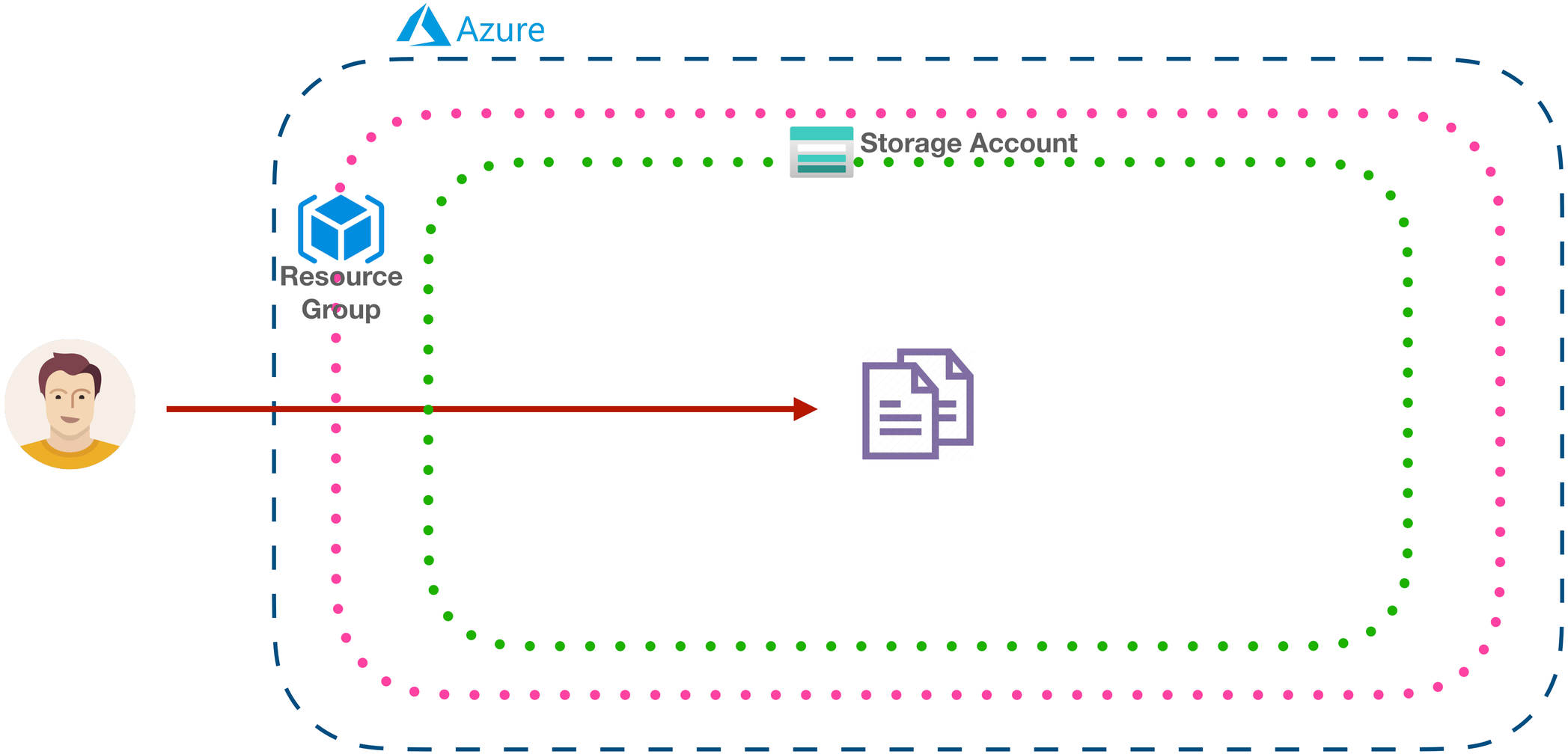


Azure Storage Account



Azure Storage Account

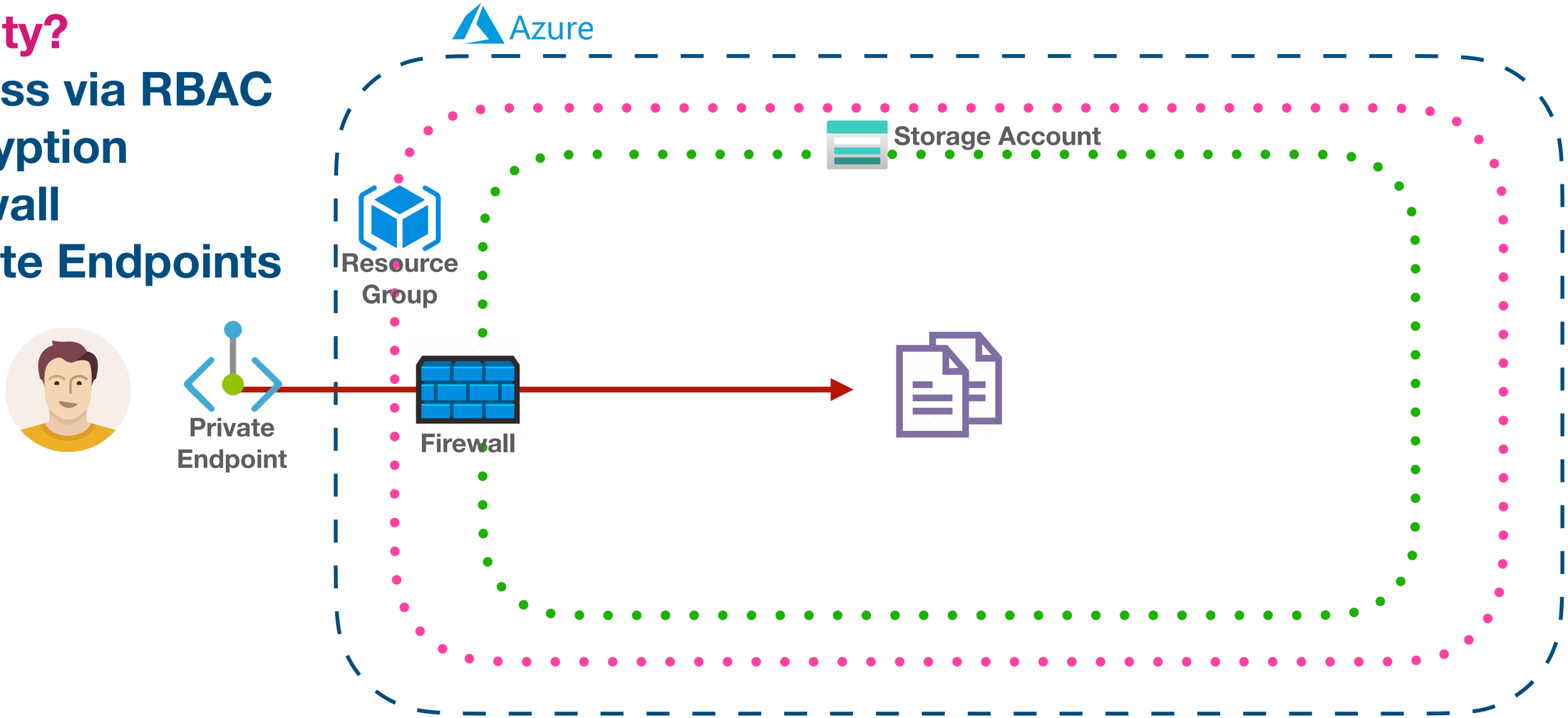
1. **What can be stored?**
   - **Blob (Object storage for unstructured data)**
   - **File (SMB-based file shares)**
   - **Table (NoSQL Key value storage)**
   - **Disk (Persistent storage for VMs)**

2. **Data availability?**
   - **LRS**(Locally Redundant Storage)
   - **ZRS**(Zero Redundant Storage)
   - **GRS**(Geo Redundant Storage)
   - **RA-GRS**(Read Access Geo Redundant Storage)

Azure Storage Account

3. Security?

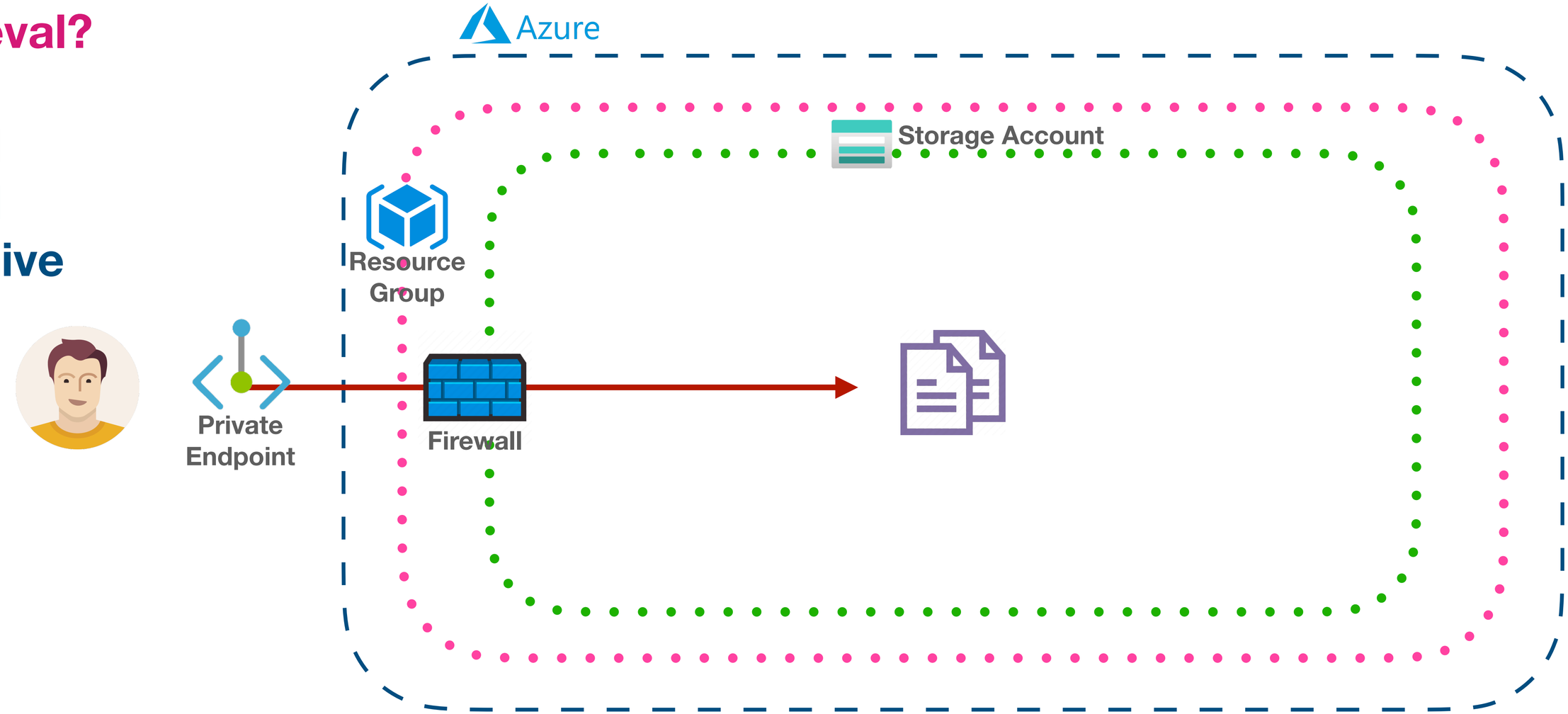
- Access via RBAC
- Encryption
- Firewall
- Private Endpoints



Azure Storage Account

4. Retrieval?

- Hot
- Cool
- Cold
- Archive

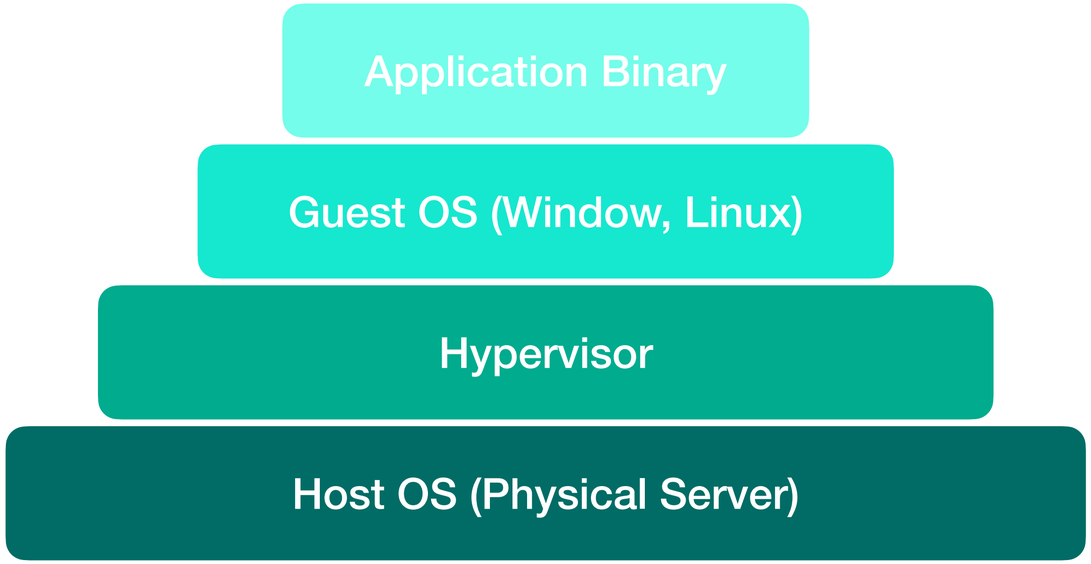


Azure Compute

1. Types of Service

- Virtual Machine
- App Services
- Azure Kubernetes Service
- Azure functions

Azure Virtual Machine



Virtual machine architecture

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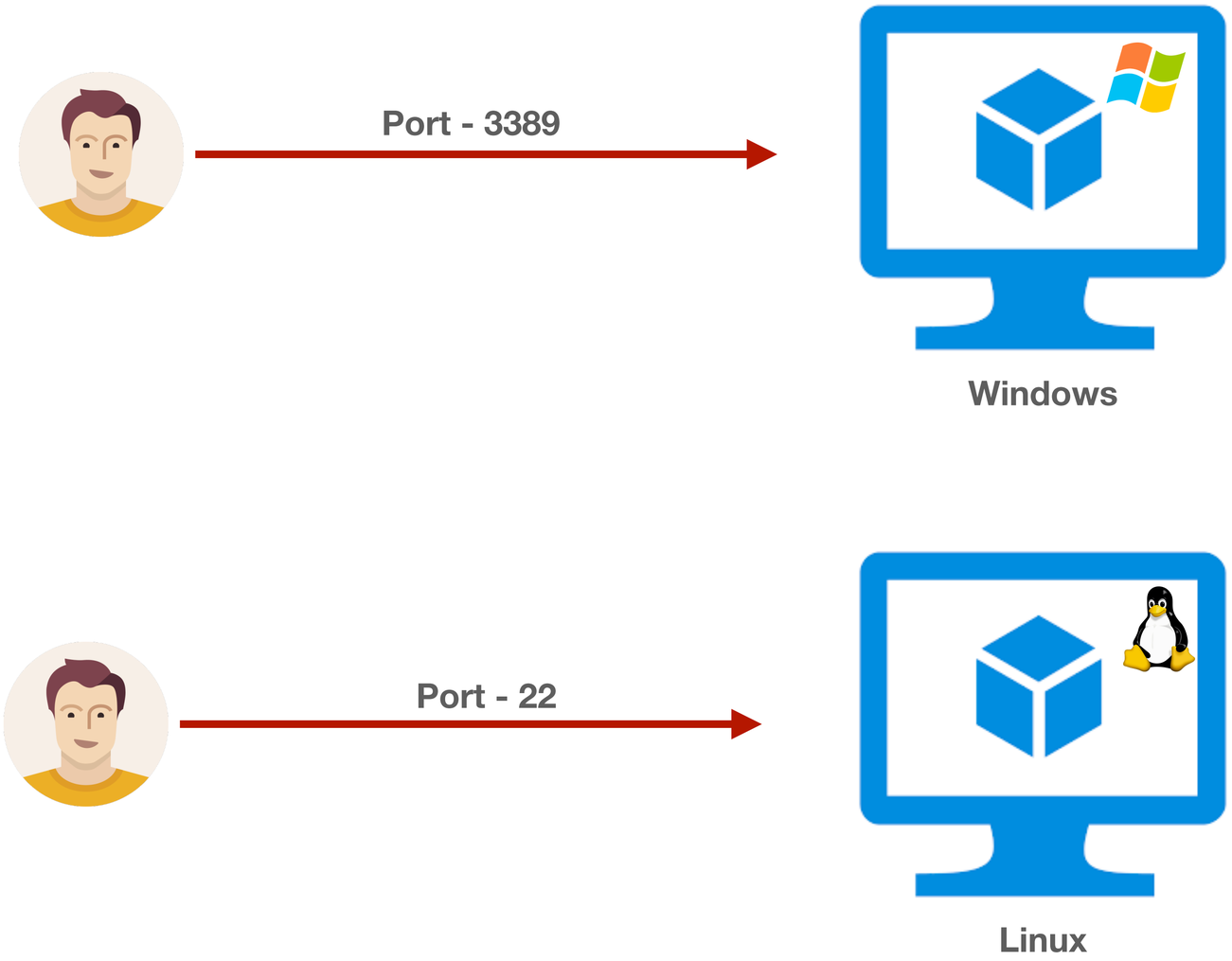
Azure Virtual Machine



Azure Virtual Machine

Azure Resource Manager (ARM) templates

Azure Virtual Machine

Azure

Azure Resource Manager (ARM) templates

JSON

```
{
    "$schema":"https://schema.management.azure.com/schemas/2015-01-01/deploymentT
    "contentVersion": "1.0.0.0",
    "parameters" :{},
    "variables" :{},
    "resources": [
        {
            "resource-1"
        },
        {
            "resource-2"
        }
    ],
    "outputs": {},
    "functions": []
}
```

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Azure Virtual Machine

```
{
    "$schema":"https://schema.management.azure.com/schemas/2015-01-01/deploymentTemplate.json#",
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```

Structure and data format.

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Azure Virtual Machine

```
{
    "$schema":"https://schema.management.azure.com/schemas/2015-01-01/deploymentTemplate.json#",
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```

Versioning system for development purposes

Azure Virtual Machine

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```

Input parameters for template

Azure Virtual Machine

```
{
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}
```

Dynamic variable for template

Azure Virtual Machine

```
{
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    "functions": []
}
```

What kind of resource you want to create Virtual Machine, Storage Account, Vent etc…

Azure Virtual Machine

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}
```

Print something on output console

Azure Virtual Machine

```
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    "variables" :{},
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        }
    ],
    "outputs": {},
    "functions": []
}
```

Custom JSON expression

Azure Virtual Machine With Terraform

Azure

Terraform + Virtual Machine

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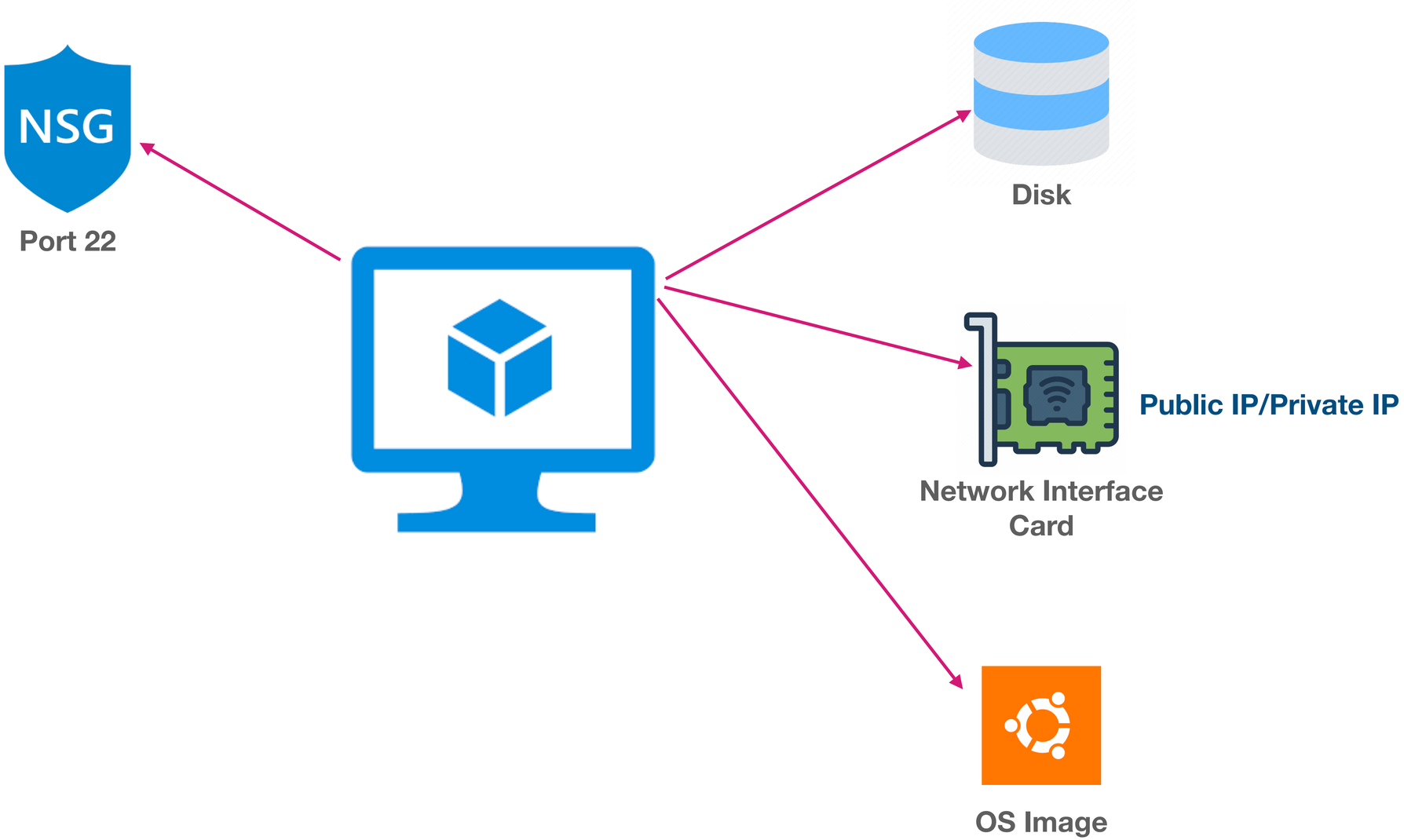
Azure Virtual Machine With Terraform

```
resource "azurerm_linux_virtual_machine" "dev_eu_north_ubuntu_vm" {
  name                            = "dev-eu-north-ubuntu2204-vm"
  resource_group_name             = azurerm_resource_group.dev_eu_north_rg.name
  location                        = azurerm_resource_group.dev_eu_north_rg.location
  size                            = "Standard_F2"
  admin_username                  = "rahulwagh"
  #admin_password                  = "G<7€YraRgk_7lnksE}yu37`Fe"
  disable_password_authentication = true
  admin_ssh_key {
    username       = "rahulwagh"
    public_key     = file("/Users/rahulwagh/.ssh/dev-vm-ssh-key-pair.pub")
  }
```

Azure Virtual Machine With Terraform



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Azure Virtual Machine With Terraform

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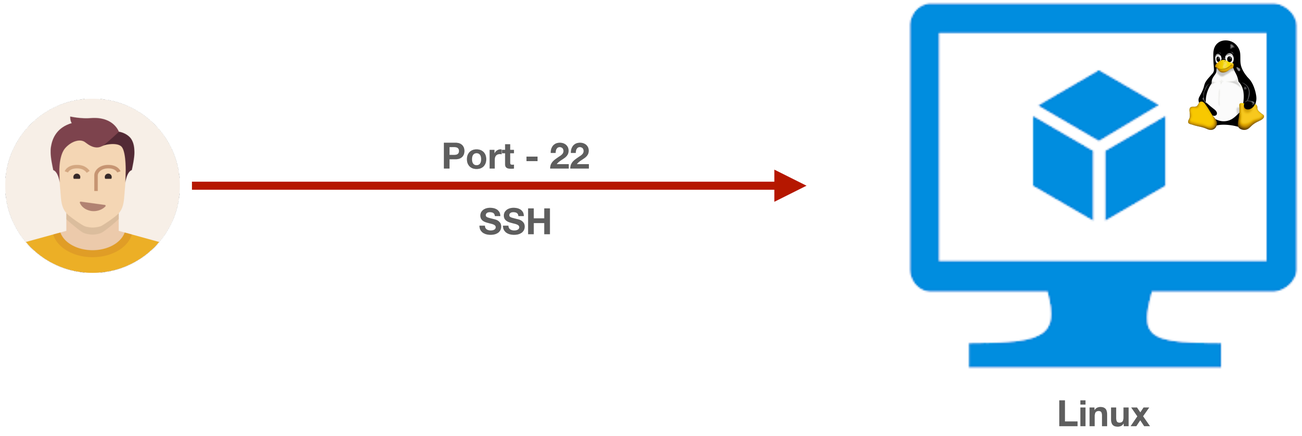
NSG

Network Security Group

Azure Virtual Machine



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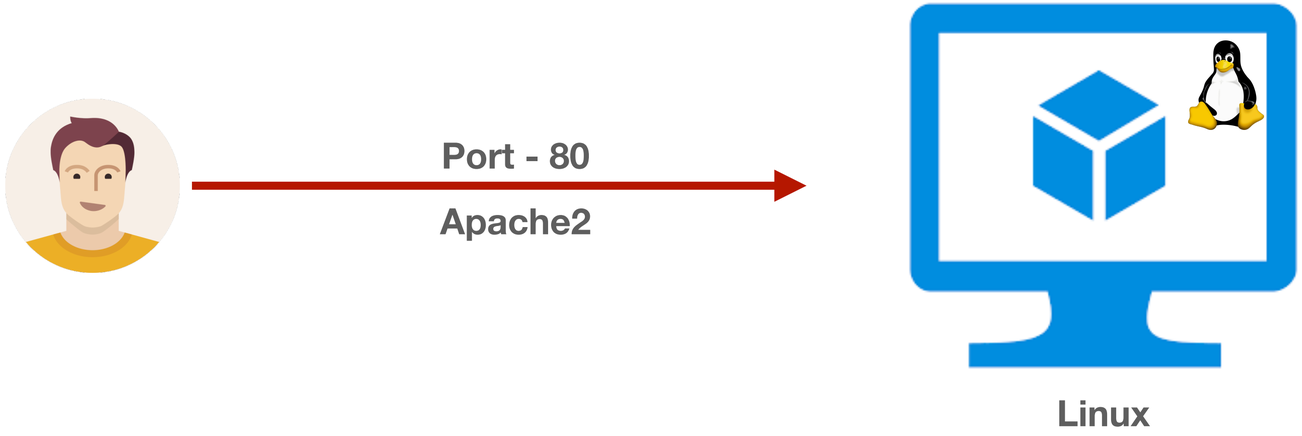
Azure Virtual Machine



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Azure Virtual Machine

Custom Data & User Data

```
#!/bin/bash
yes | sudo apt update
yes | sudo apt install apache2
```

Why we need ?

- Install packages on VM
- Pre-boot setup when machine is setup for first time
- Gives you an advantage to installing pre-define packages

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Azure Virtual Machine

Custom Data & User Data

What is "user data"

User data is a set of scripts or other metadata that's inserted to an Azure virtual machine at provision time. Any application on the virtual machine can access the user data from the Azure Instance Metadata Service (IMDS) after provision.

Azure Virtual Machine

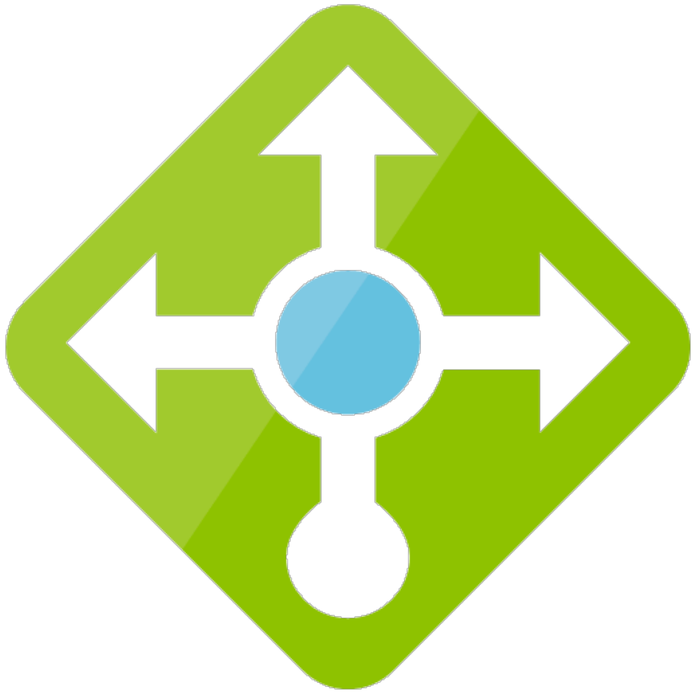
Custom Data & User Data

User data is a new version of custom data and it offers added benefits:

- User data can be retrieved from Azure Instance Metadata Service(IMDS) after provision.
- User data is persistent. It will be available during the lifetime of the VM.
- User data can be updated from outside the VM, without stopping or rebooting the VM.
- User data can be queried via GET VM/VMSS API with $expand option.

Azure Virtual Machine

Azure



Loadbalancer

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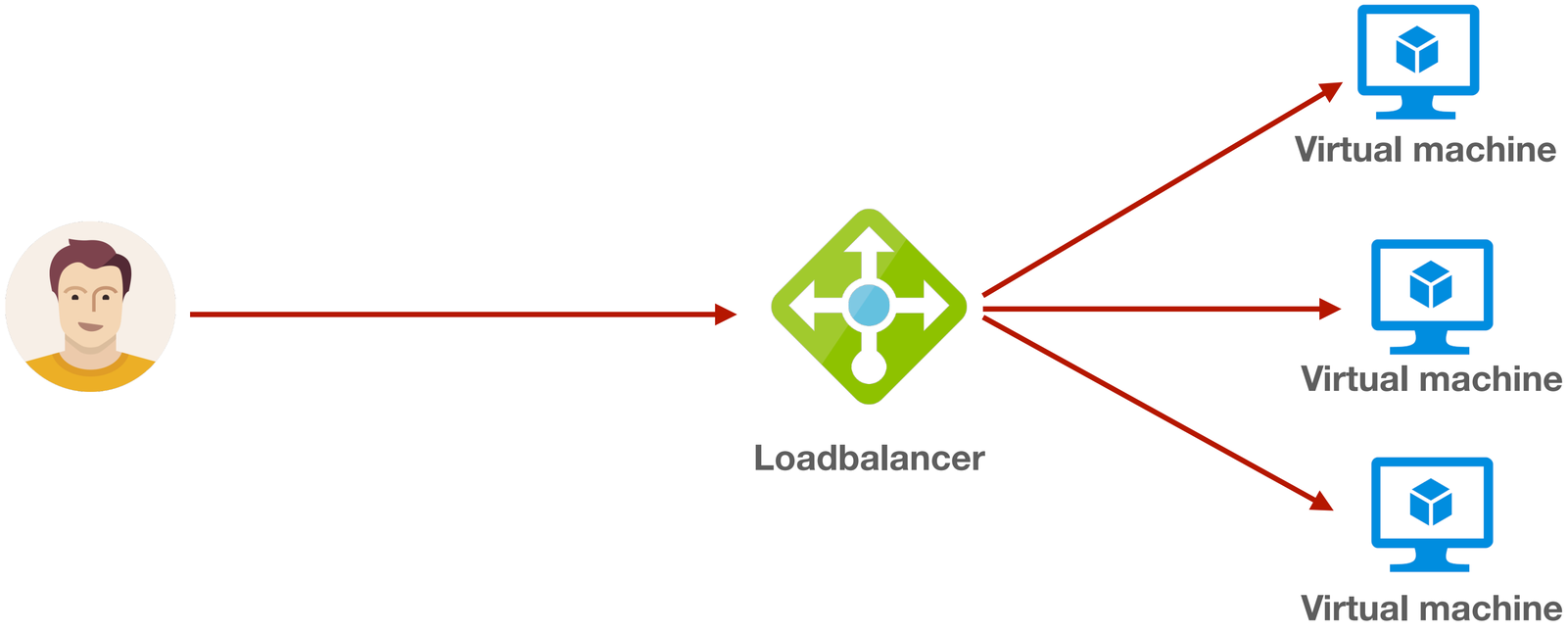
Azure Virtual Machine



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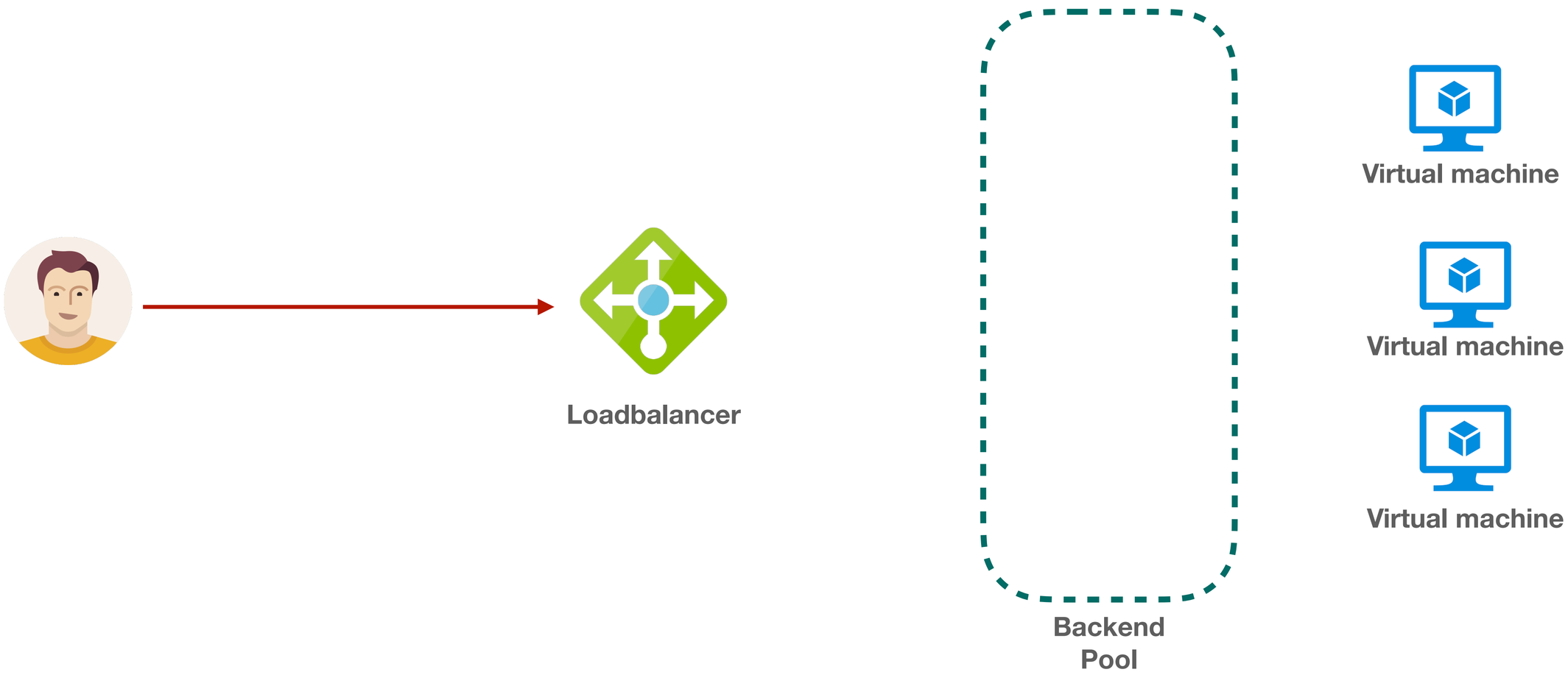
Azure Virtual Machine



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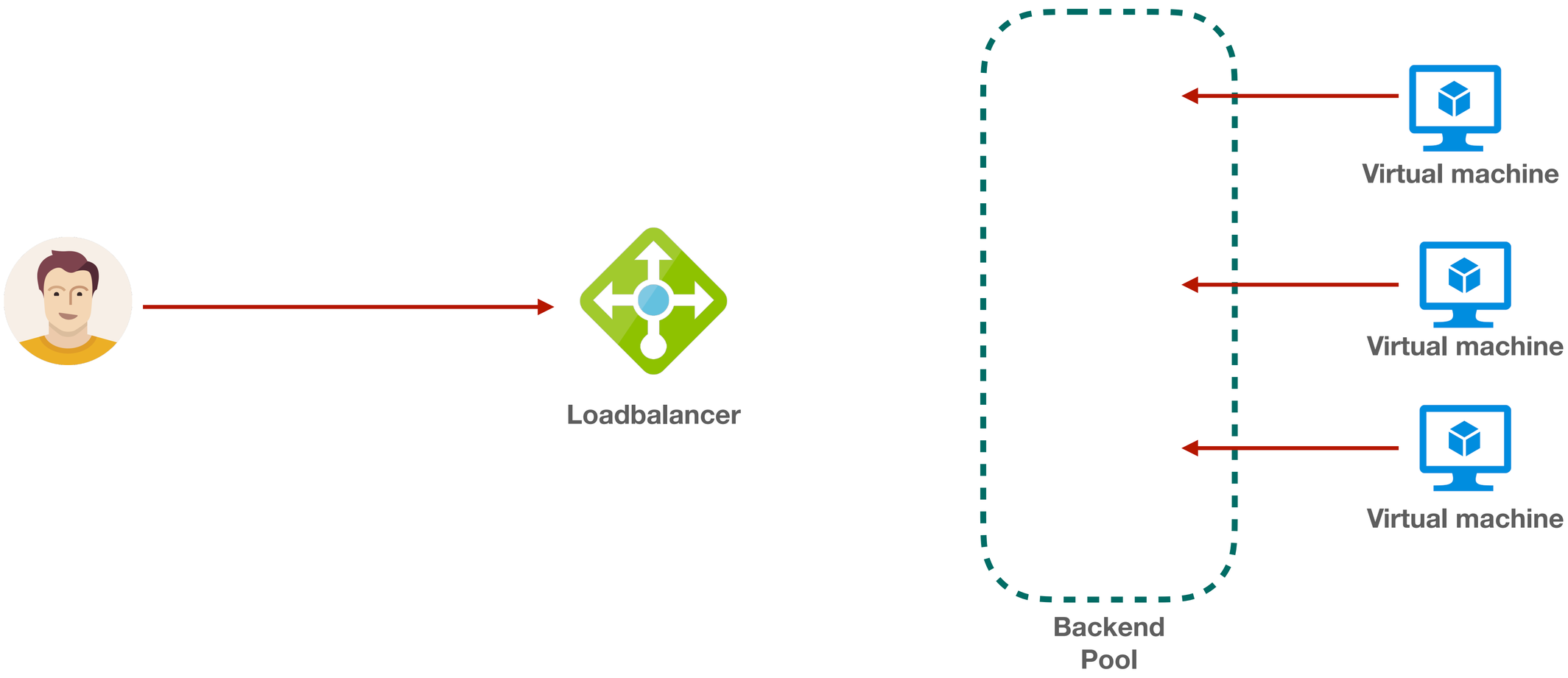
Azure Virtual Machine



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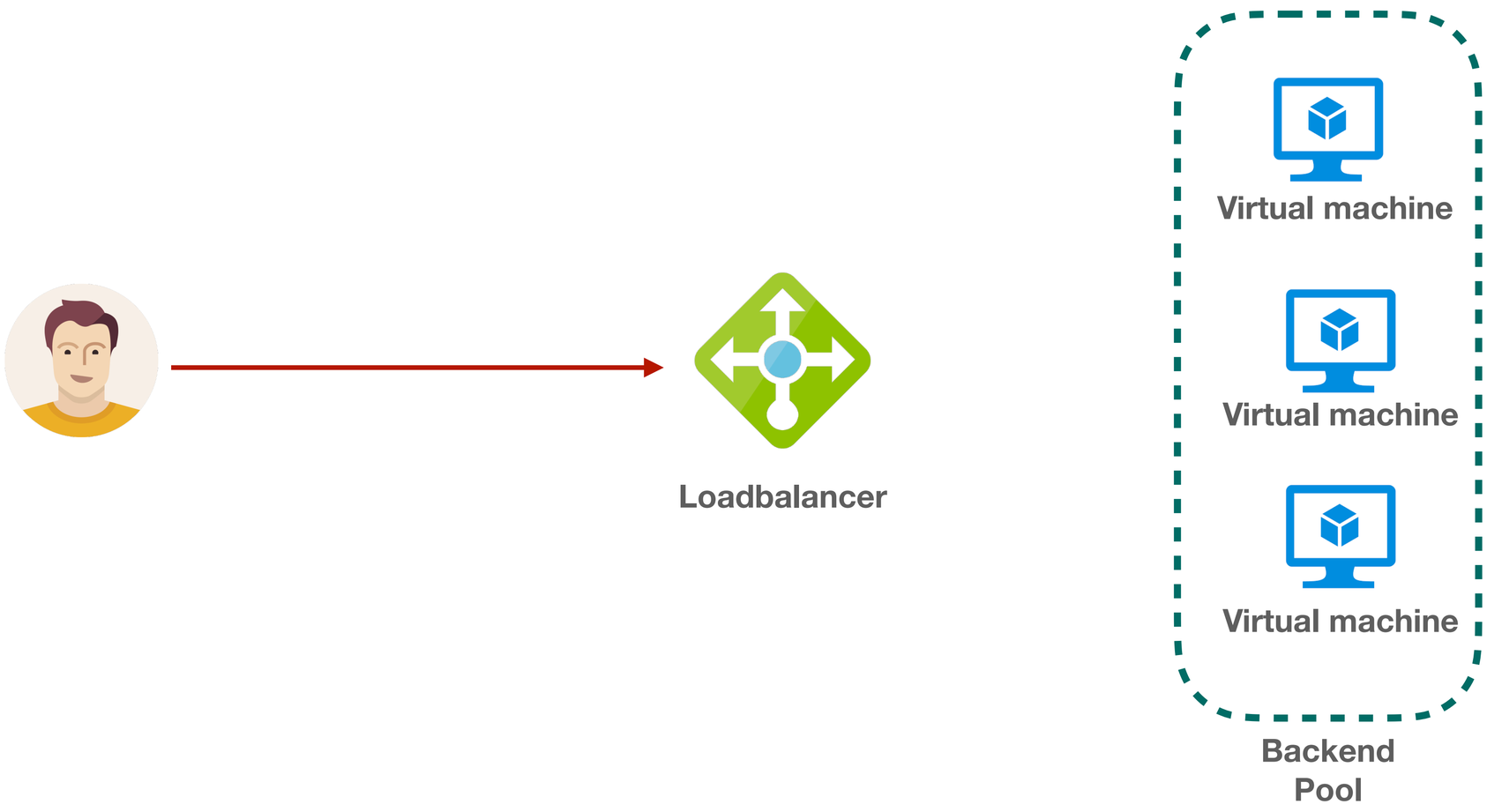

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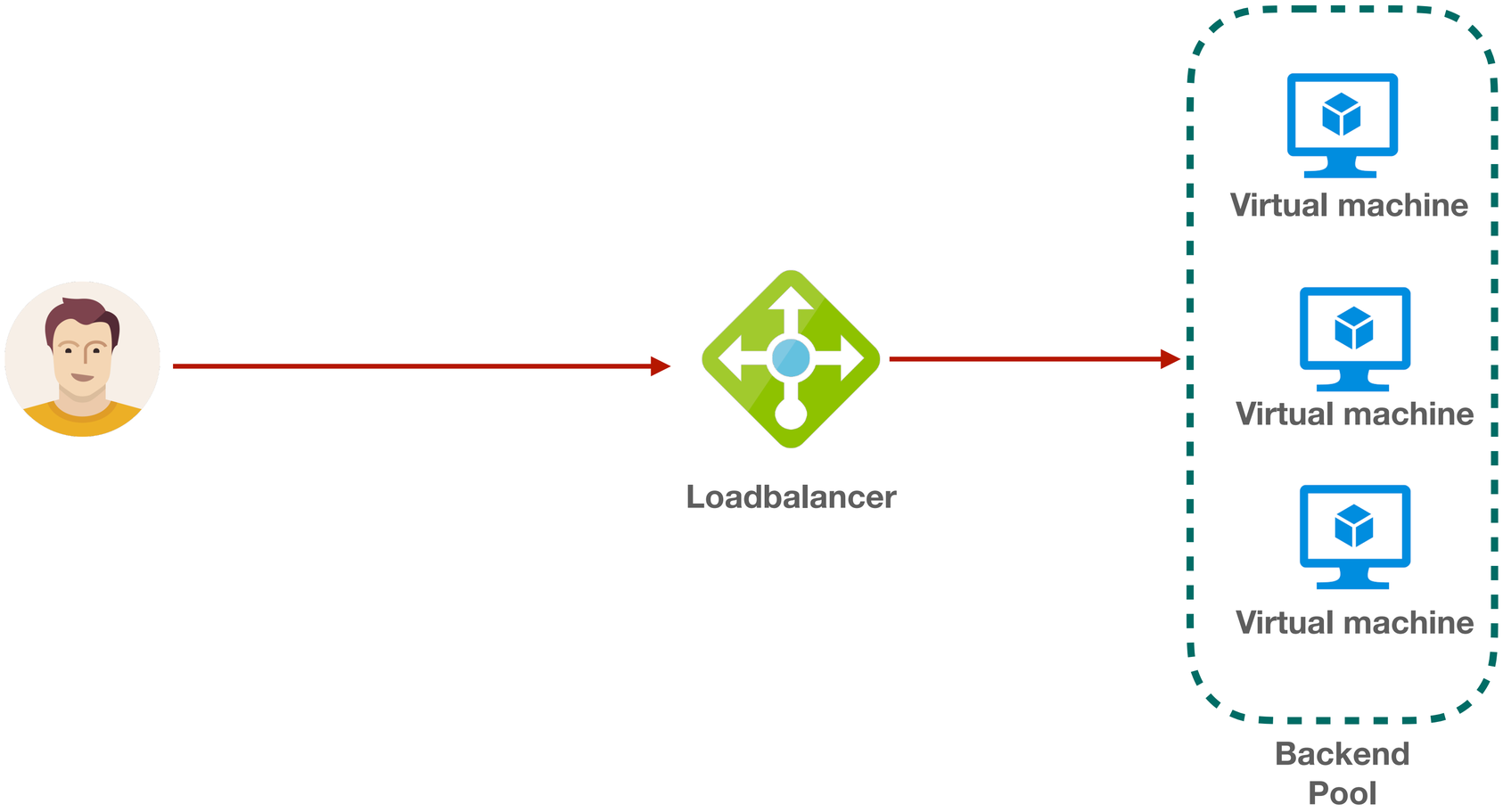

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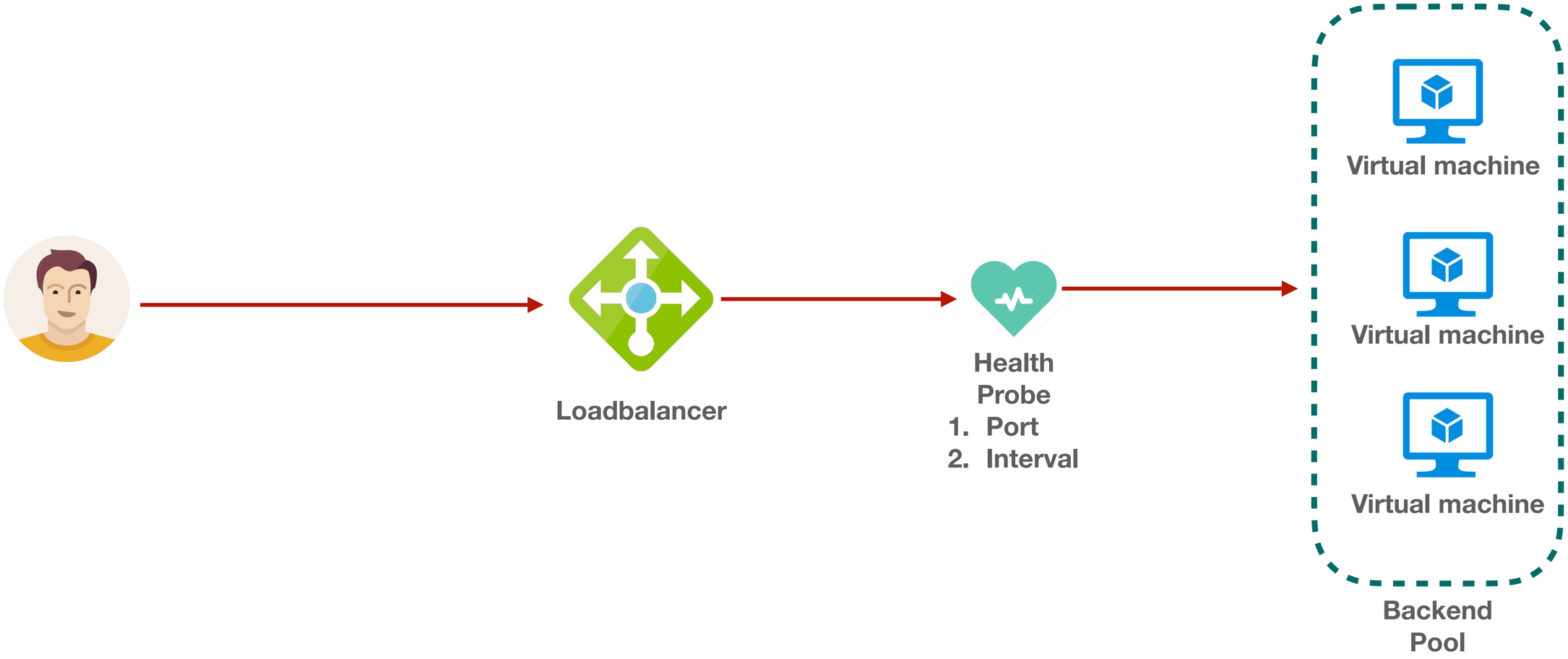

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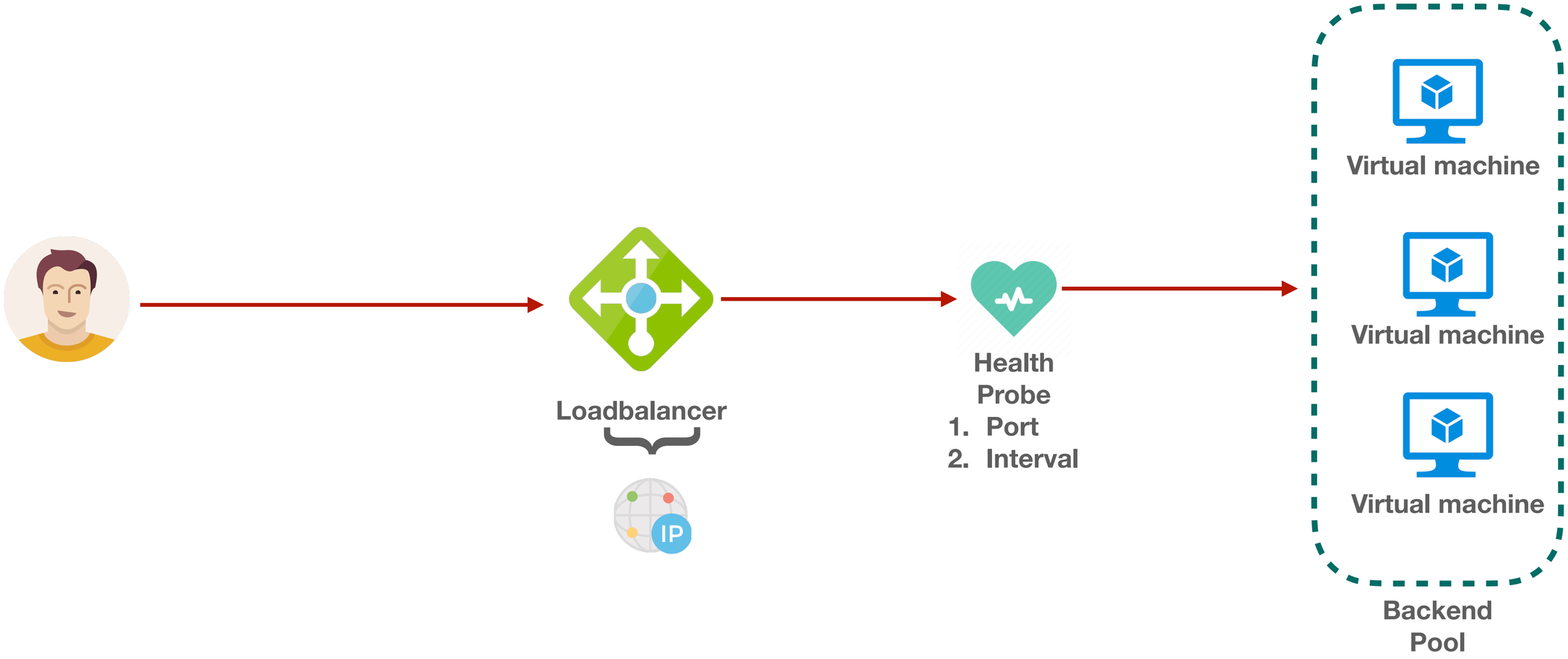

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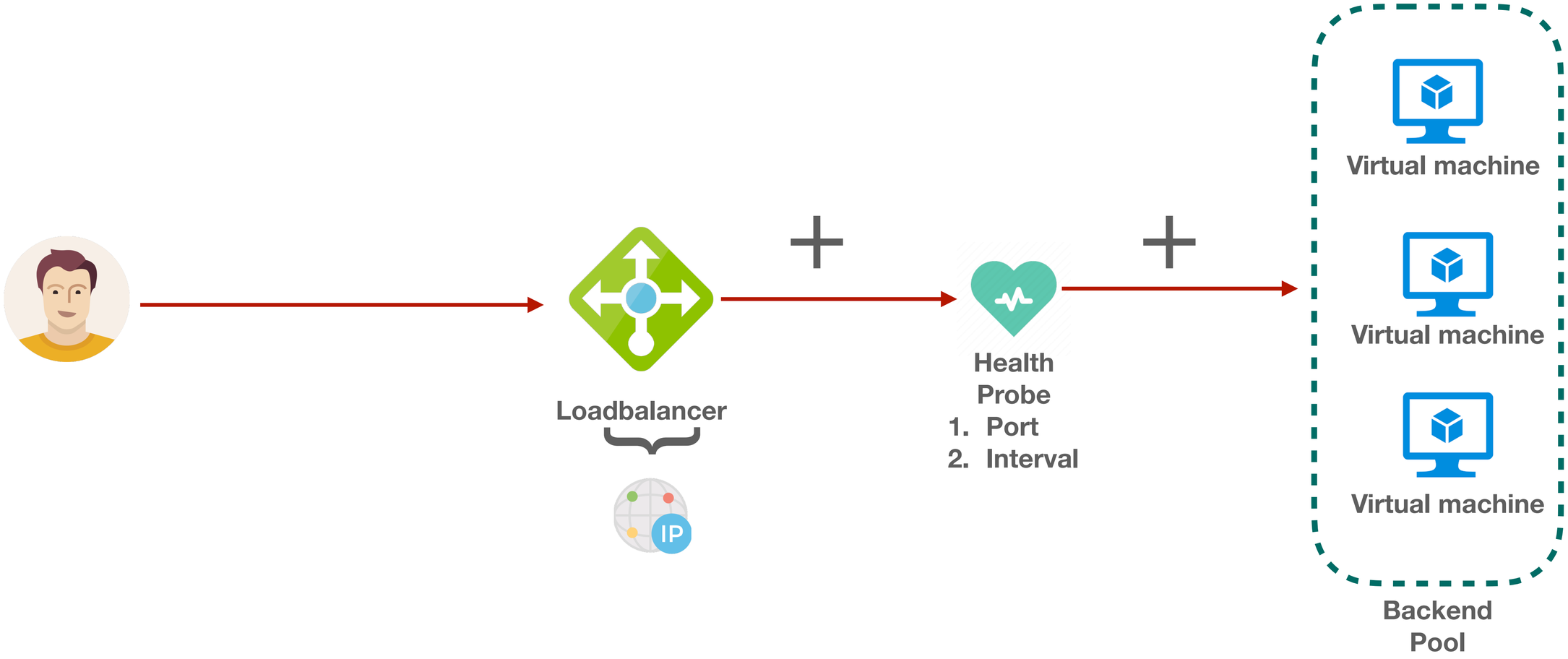

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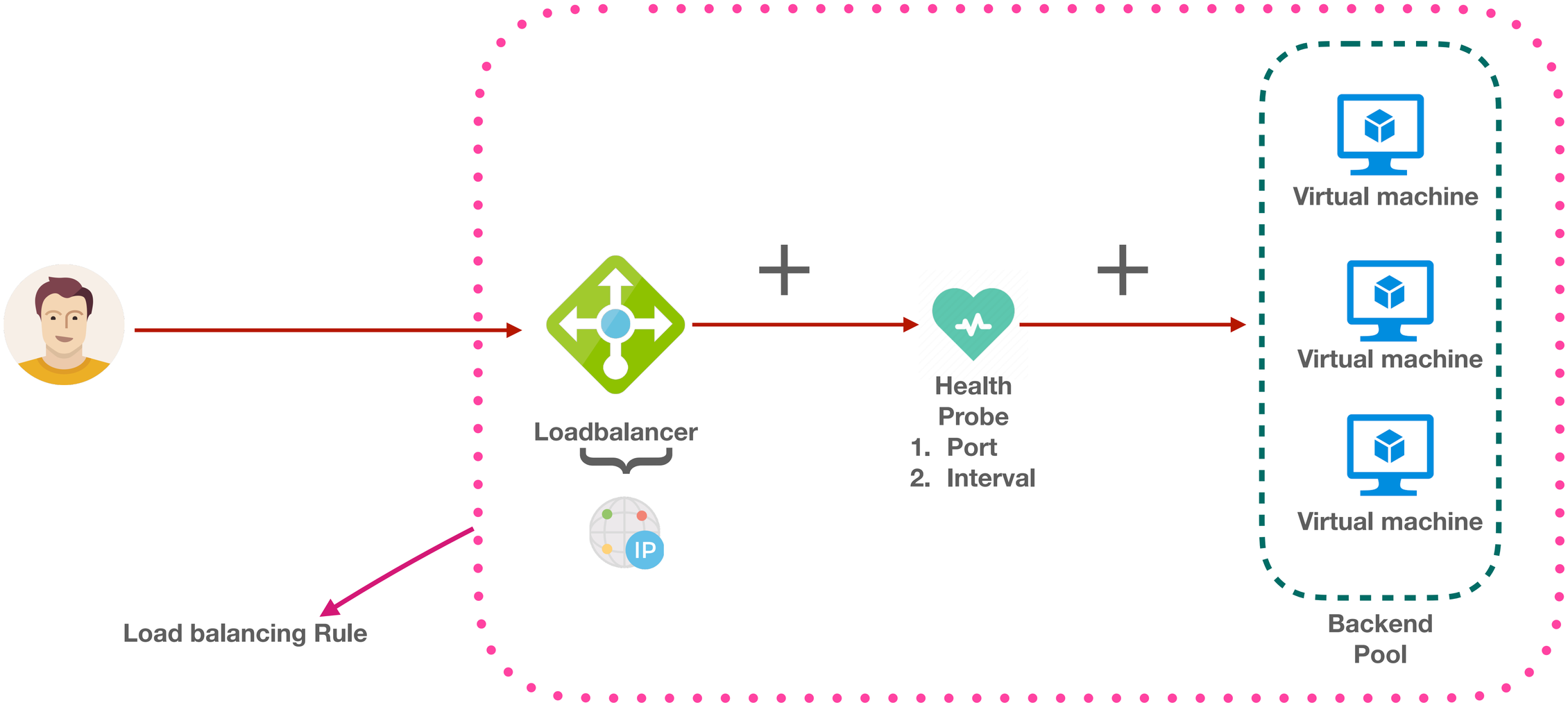

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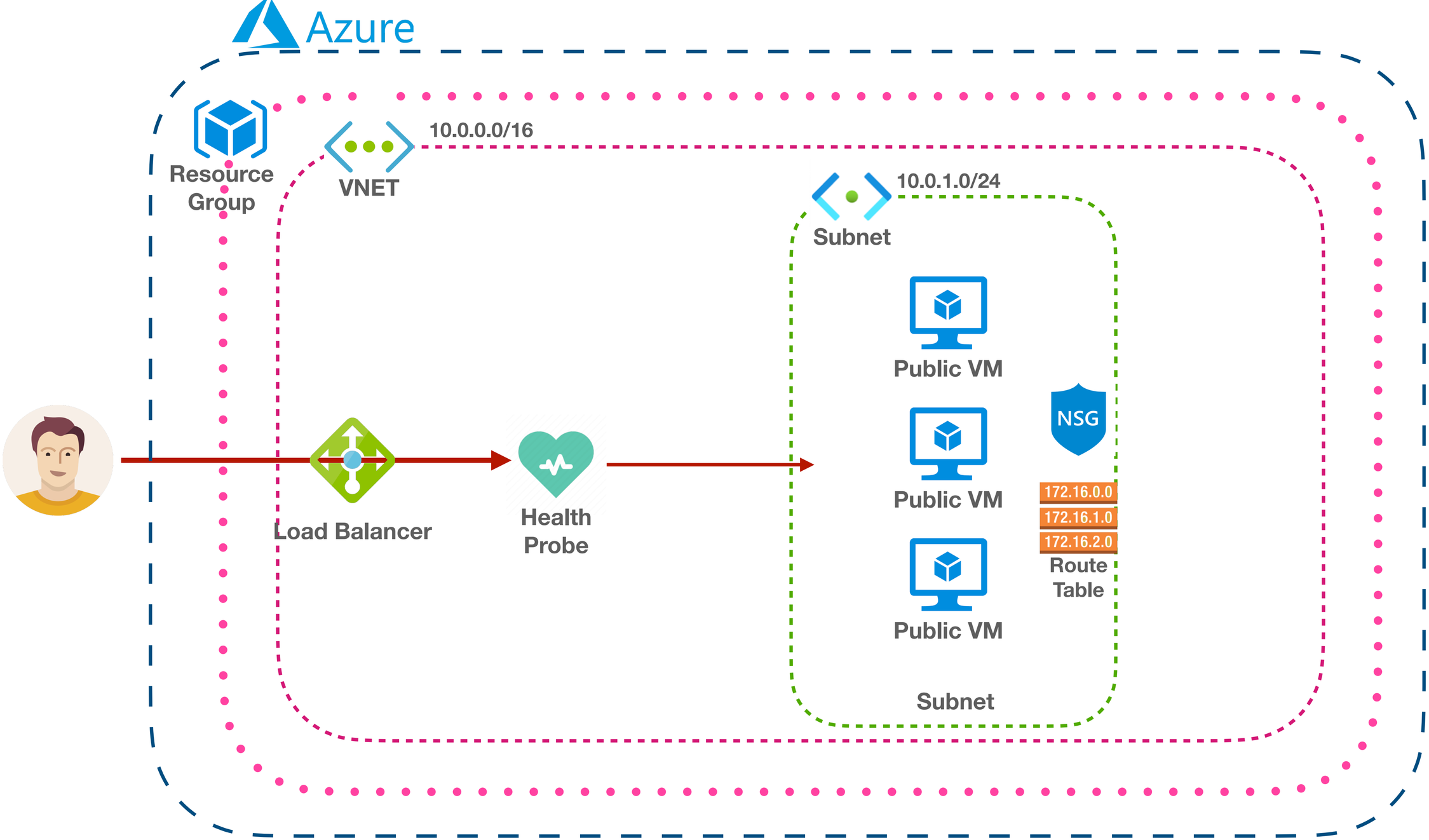

Bastion Host/Jump Host

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Virtual Machine Scale Set

Azure Virtual Machine

Azure

Virtual Machine Scale Set

Benefits

- Scale In / Scale Out
- More automatic control on VM
- Cost effective with spot instances
- Very beneficial of high work load

High Traffic —-> Increase VMs

Low Traffic —-> Decrease VMs

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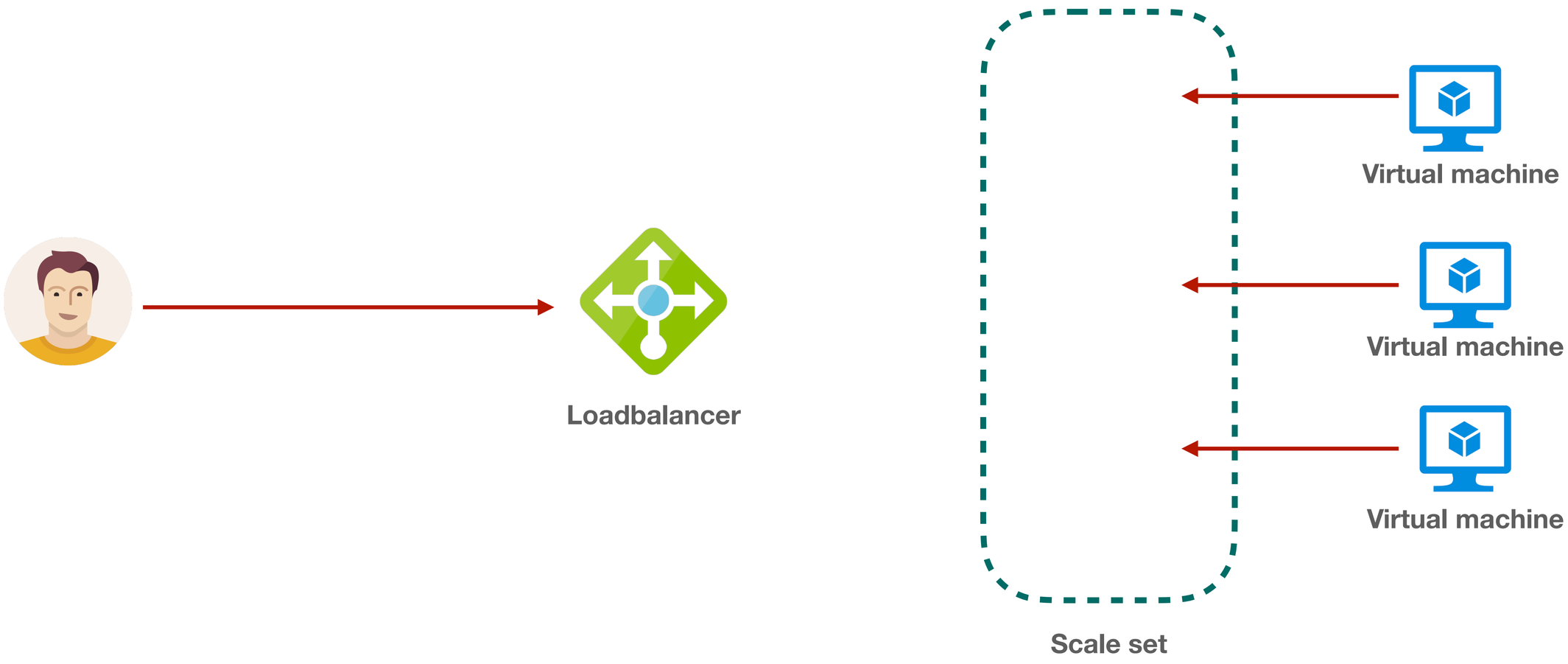
Azure Virtual Machine



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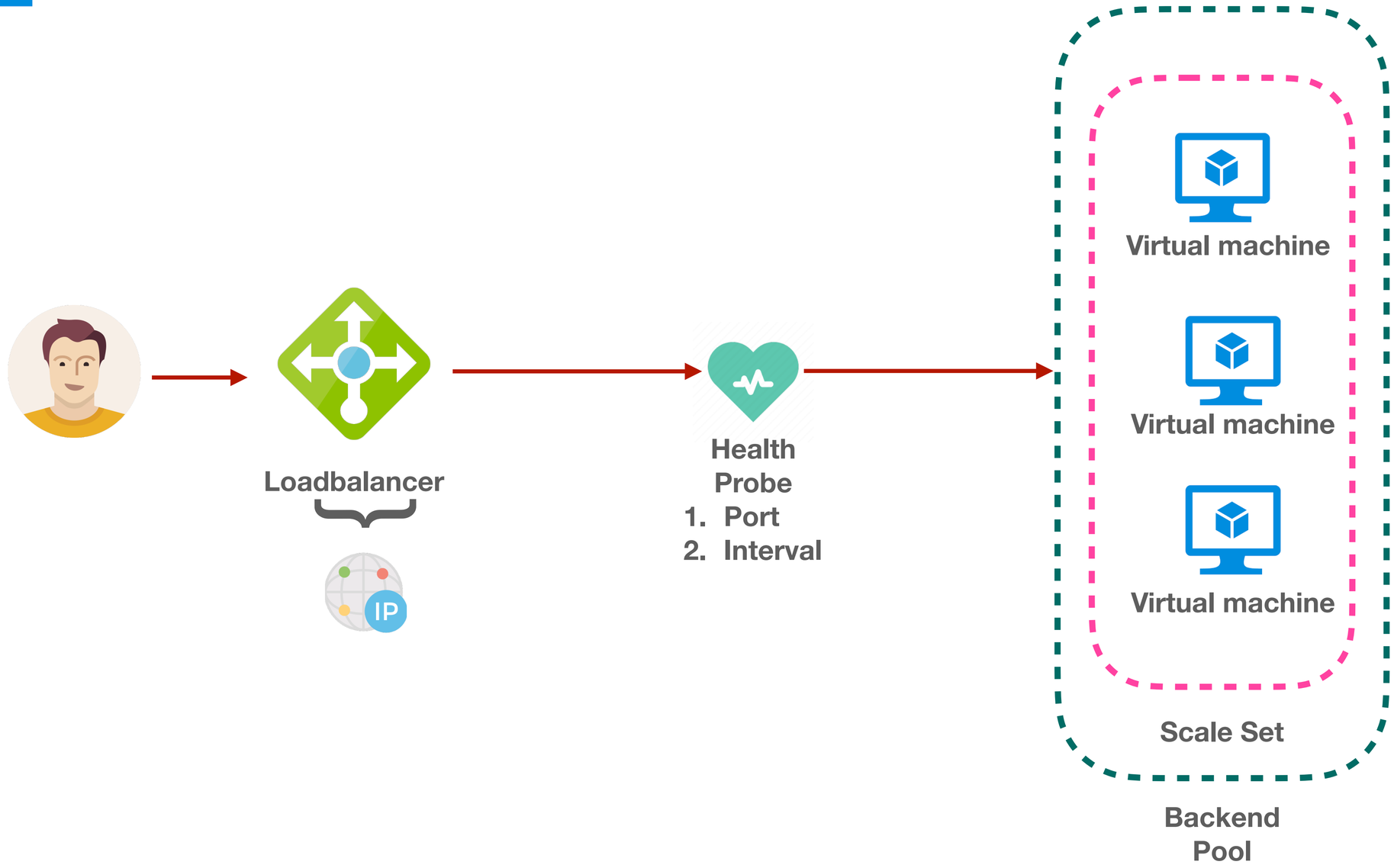

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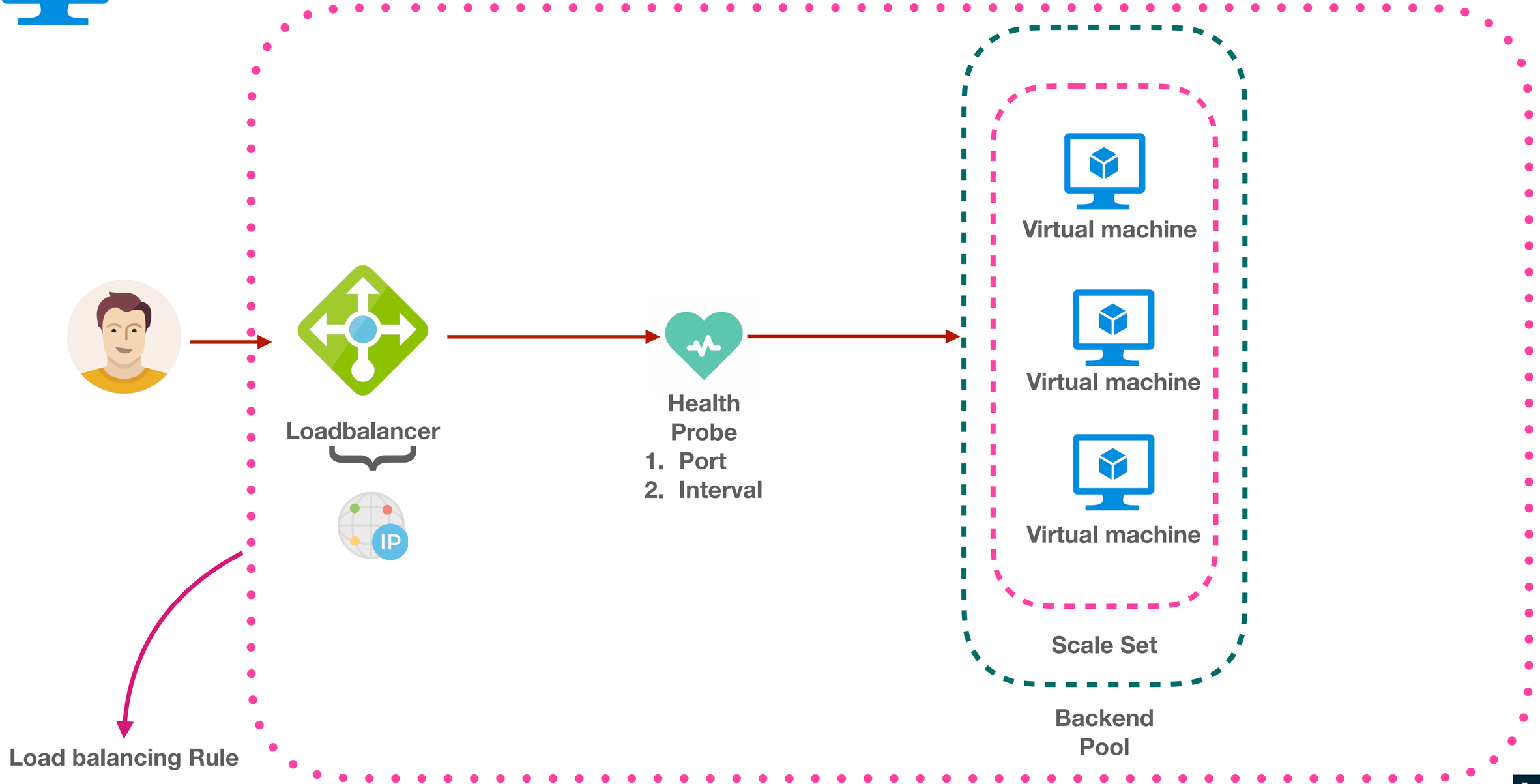
Azure Virtual Machine



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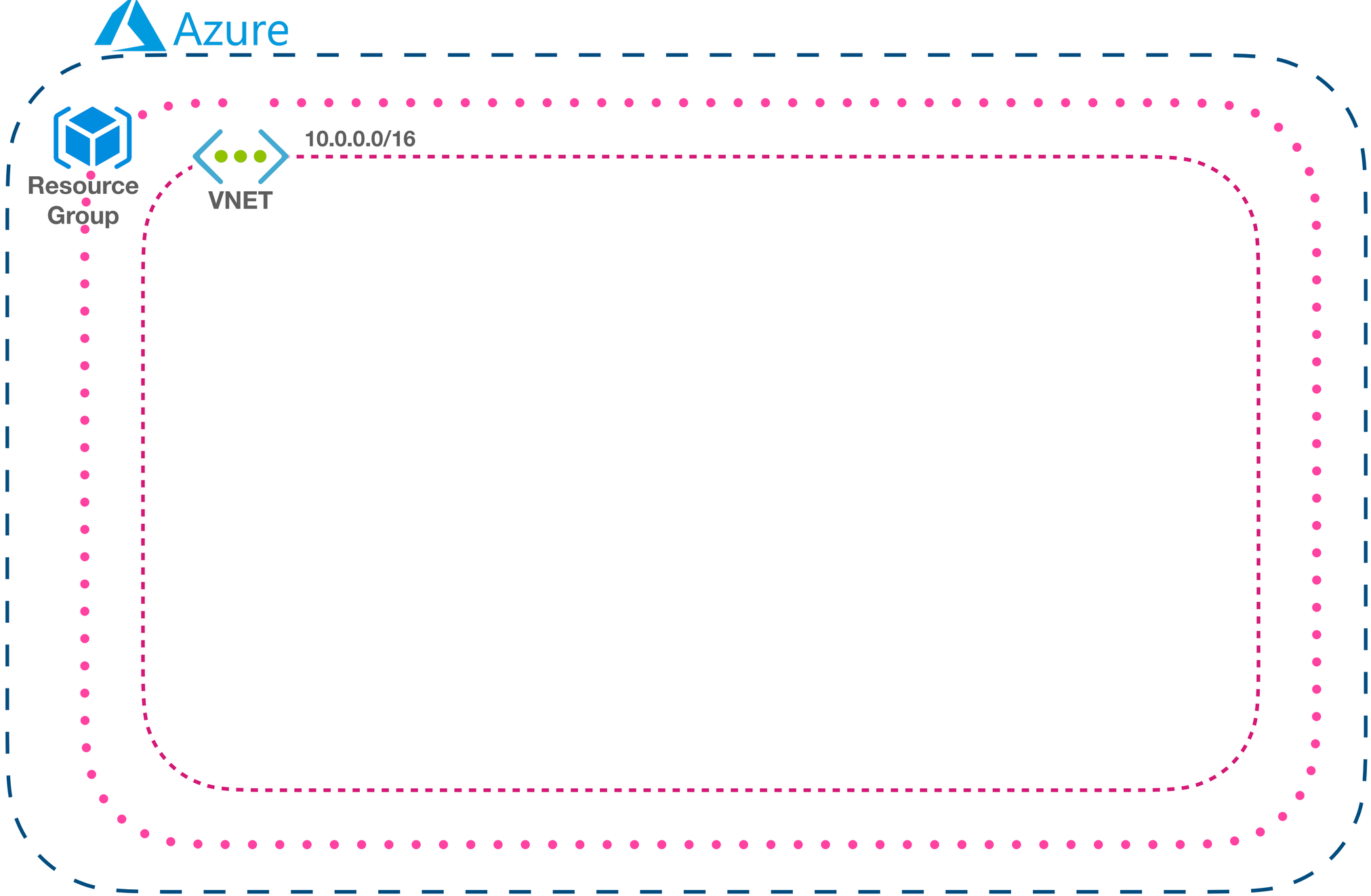
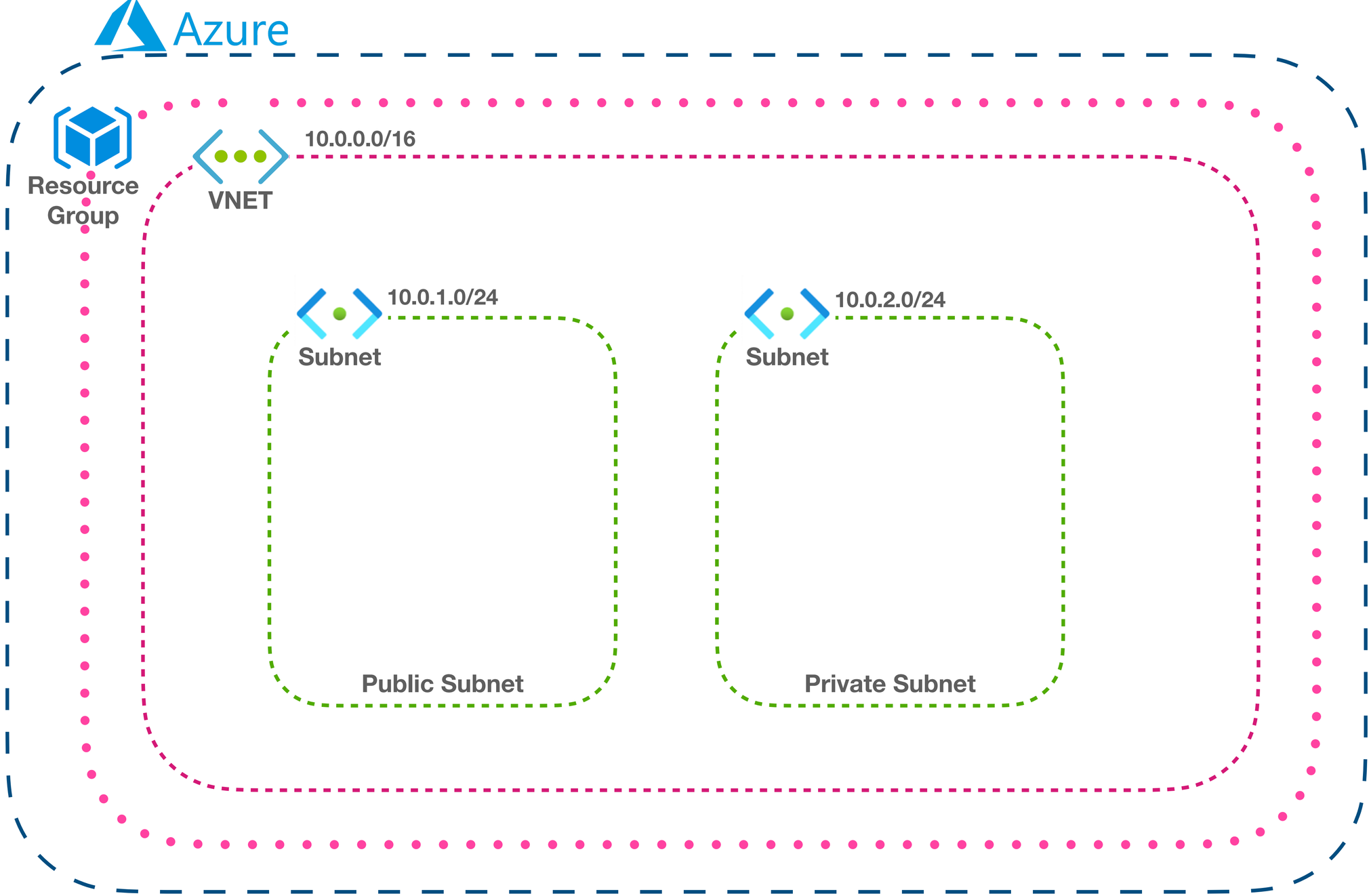
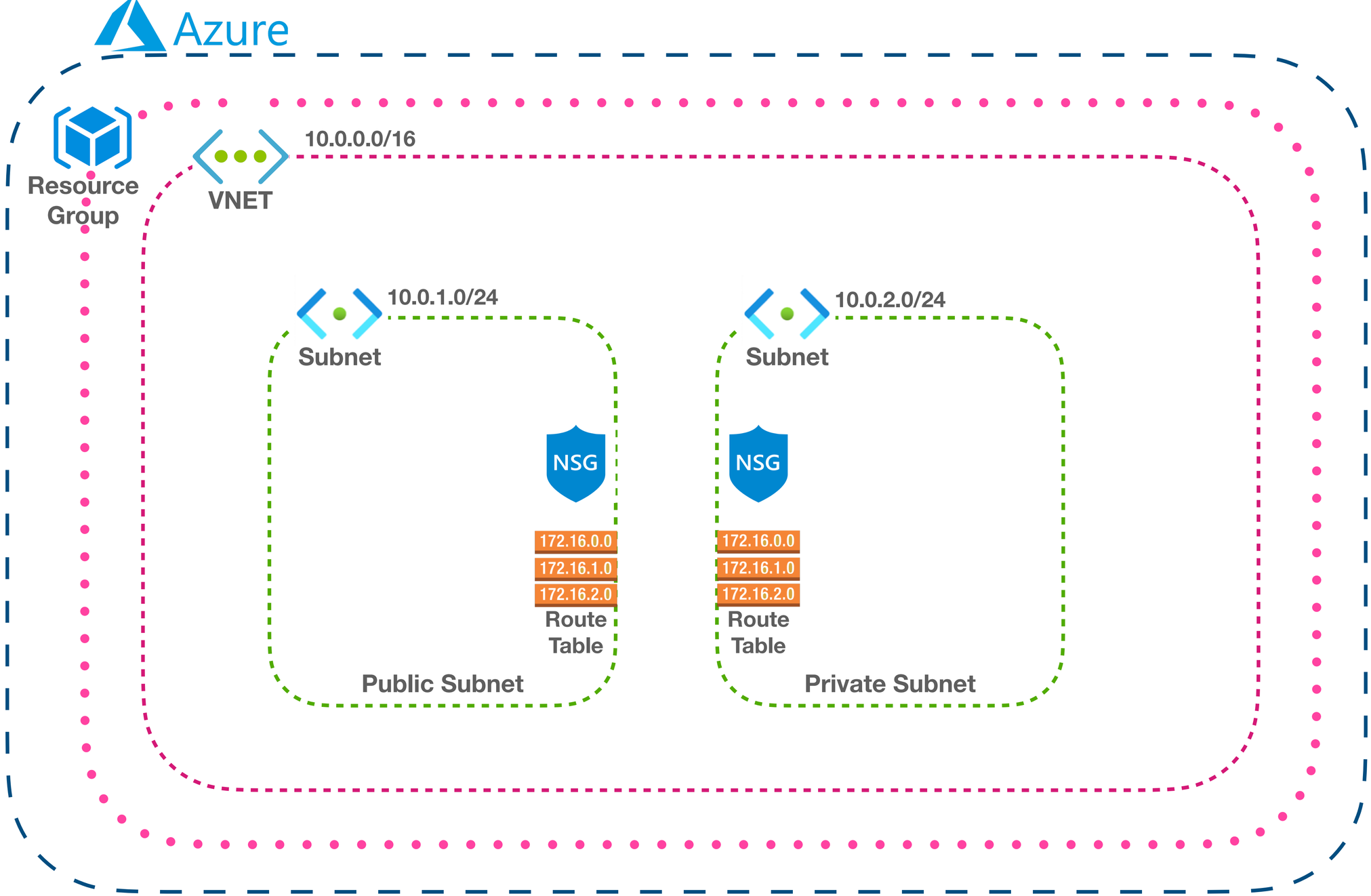
Azure
Resource Group



J->Hooq

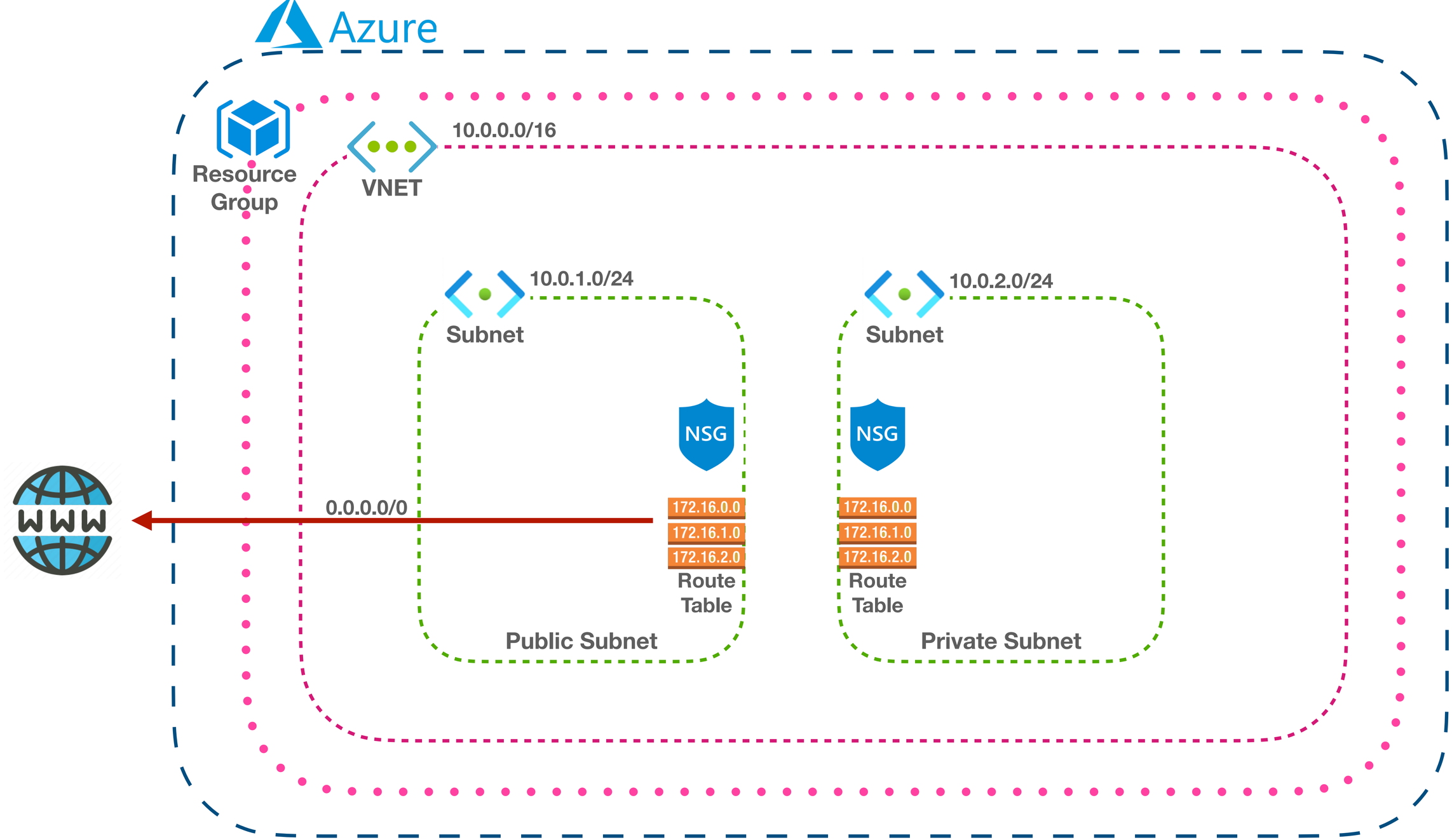

Azure
Resource Group
VNET
10.0.0.0/16
10.0.1.0/24
Subnet
Public Subnet
10.0.2.0/24
Subnet
Private Subnet

J->Hooq



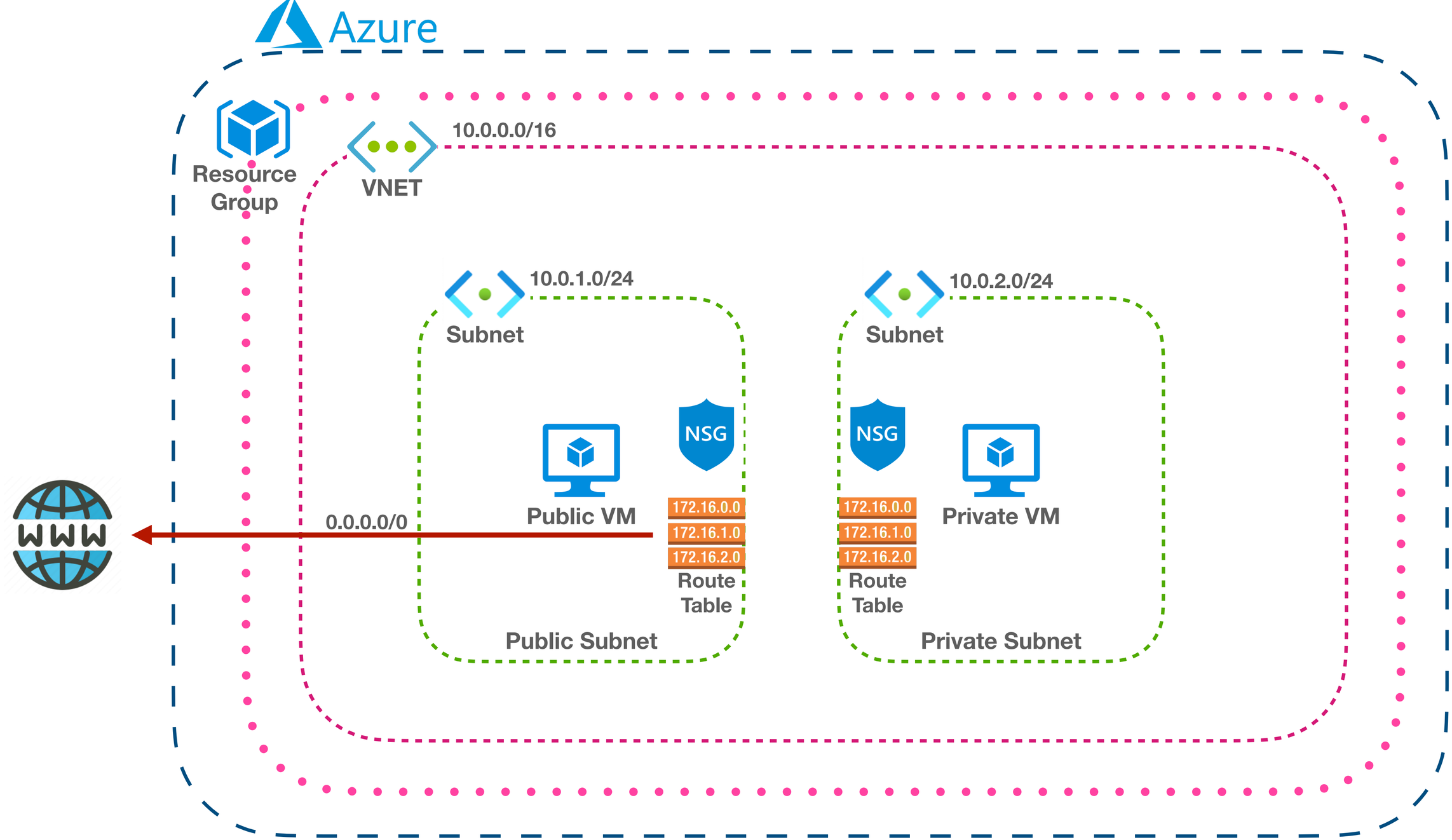
Accessing Resource in the Private Subnet

J->Hooq



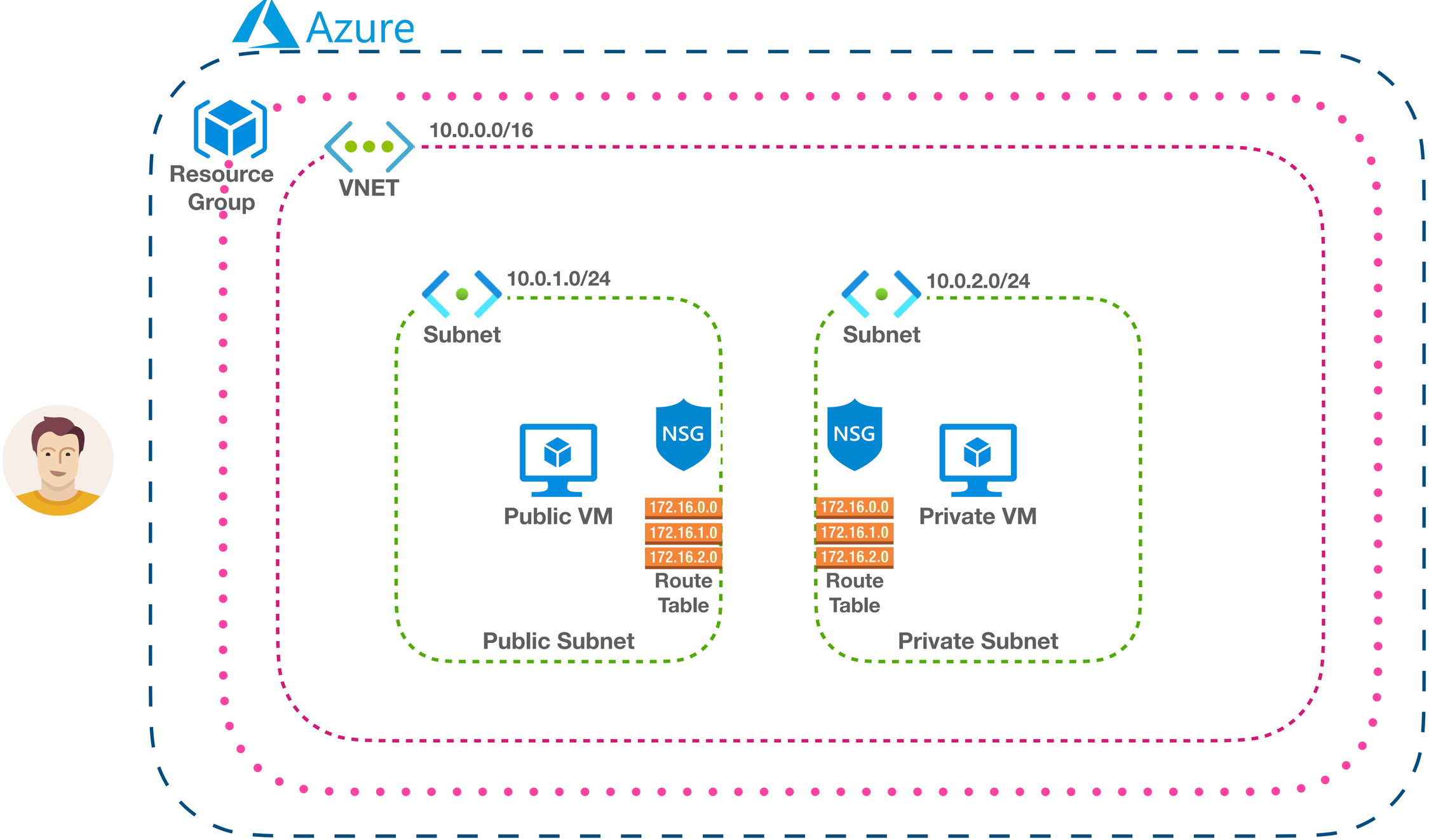
Accessing Resource in the Private Subnet

J->Hooq



Accessing Resource in the Private Subnet

J->Hooq



Accessing Resource in the Private Subnet

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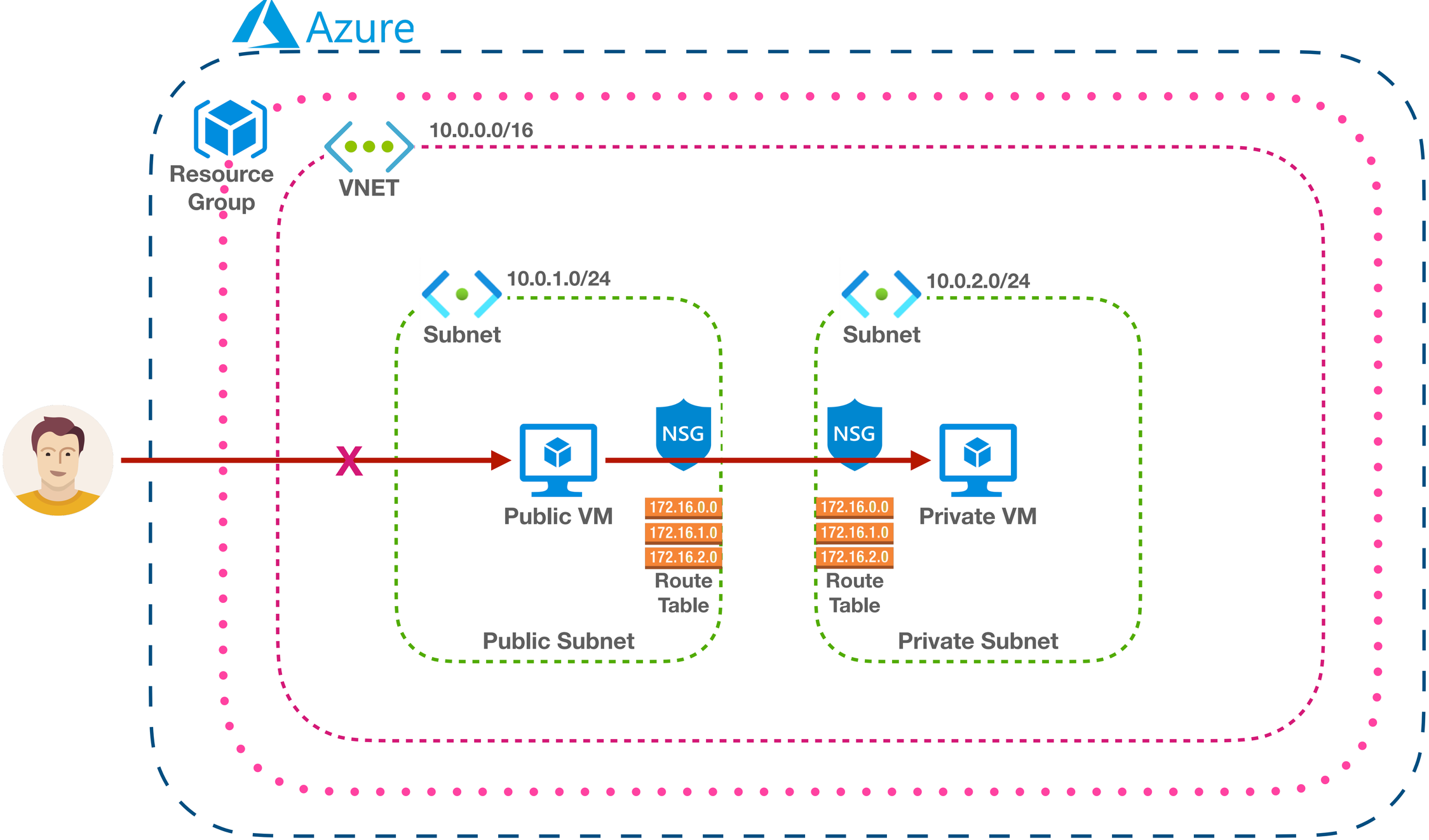

Accessing Resource in the Private Subnet

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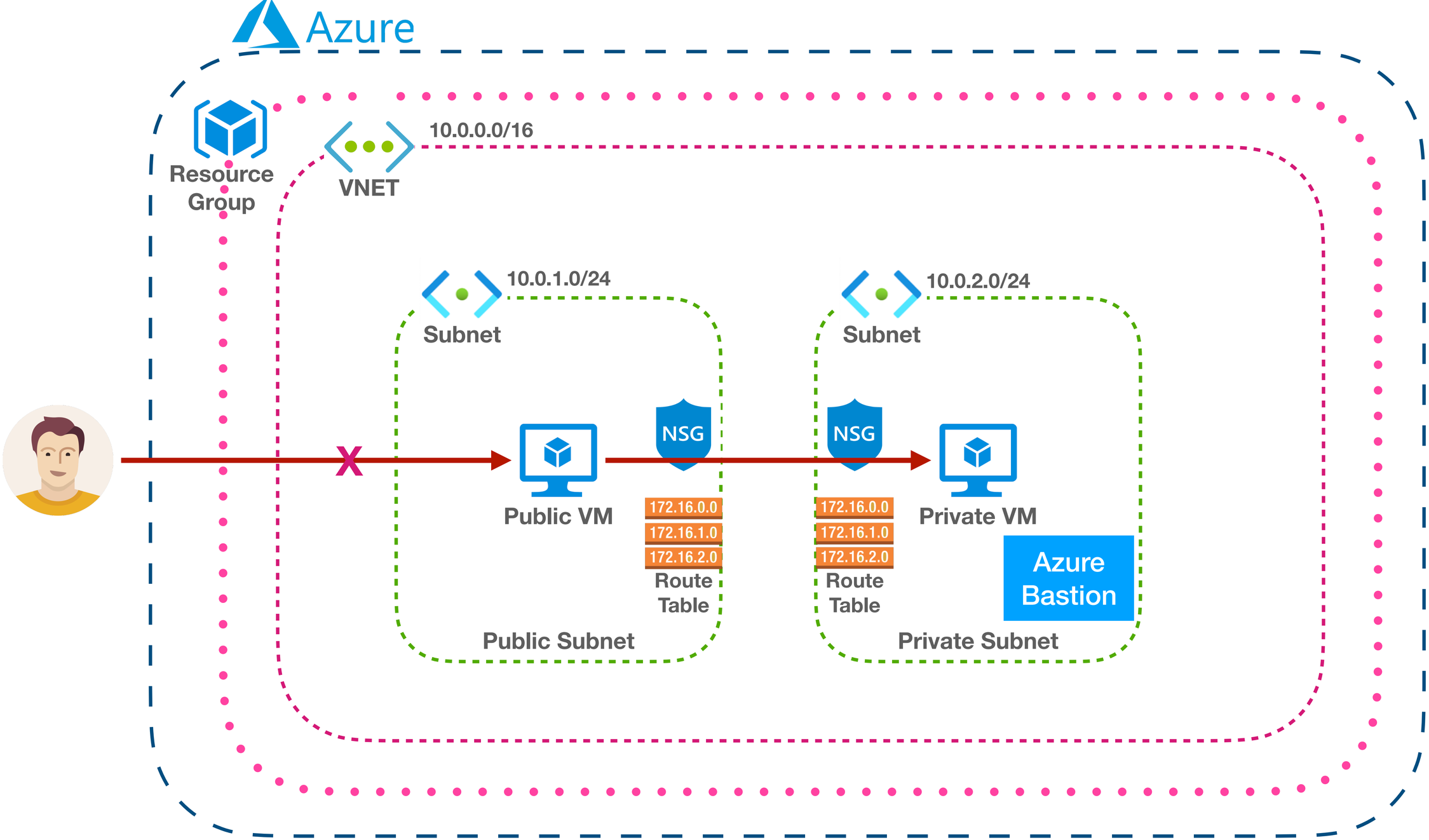

Accessing Resource in the Private Subnet

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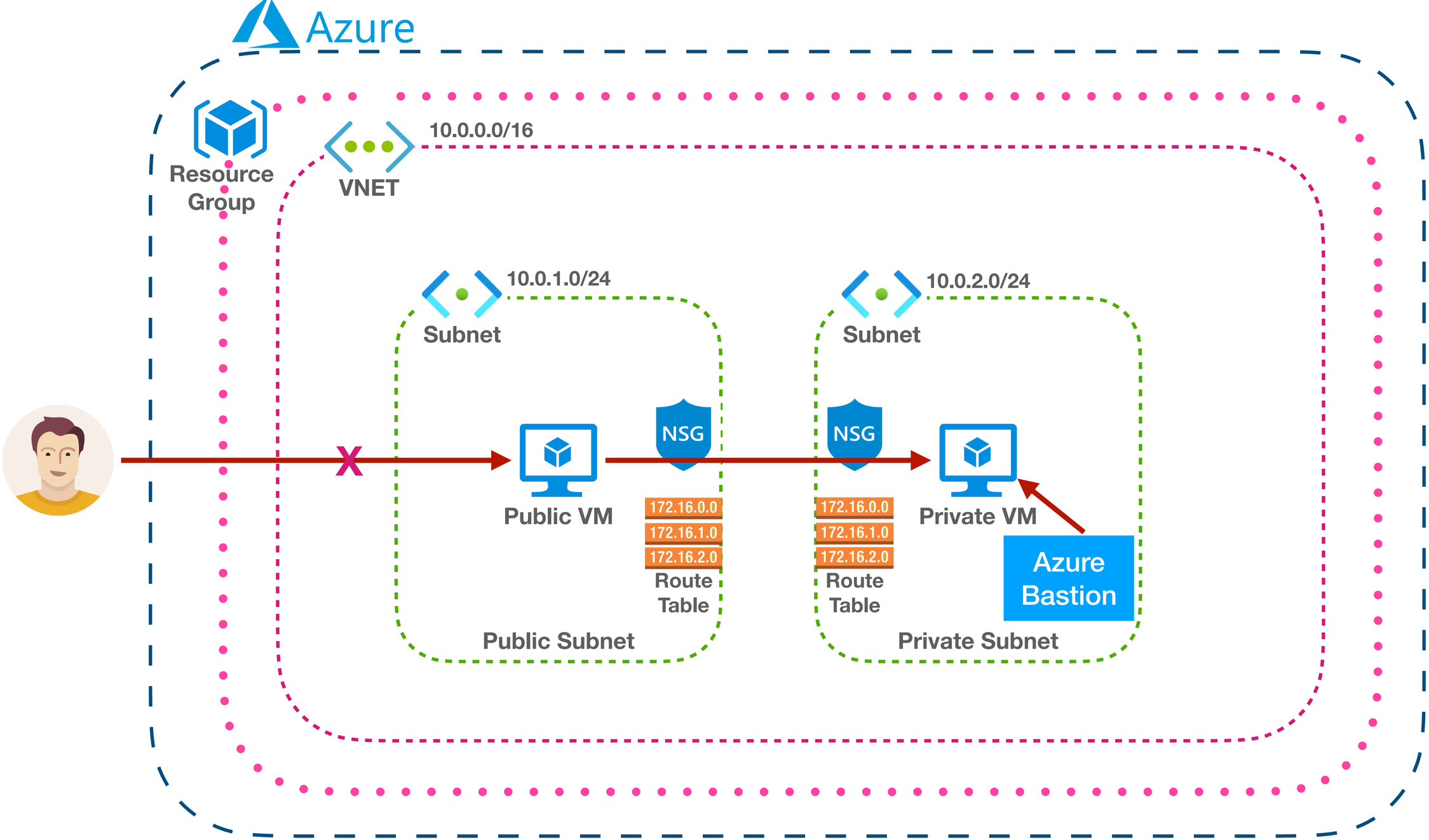
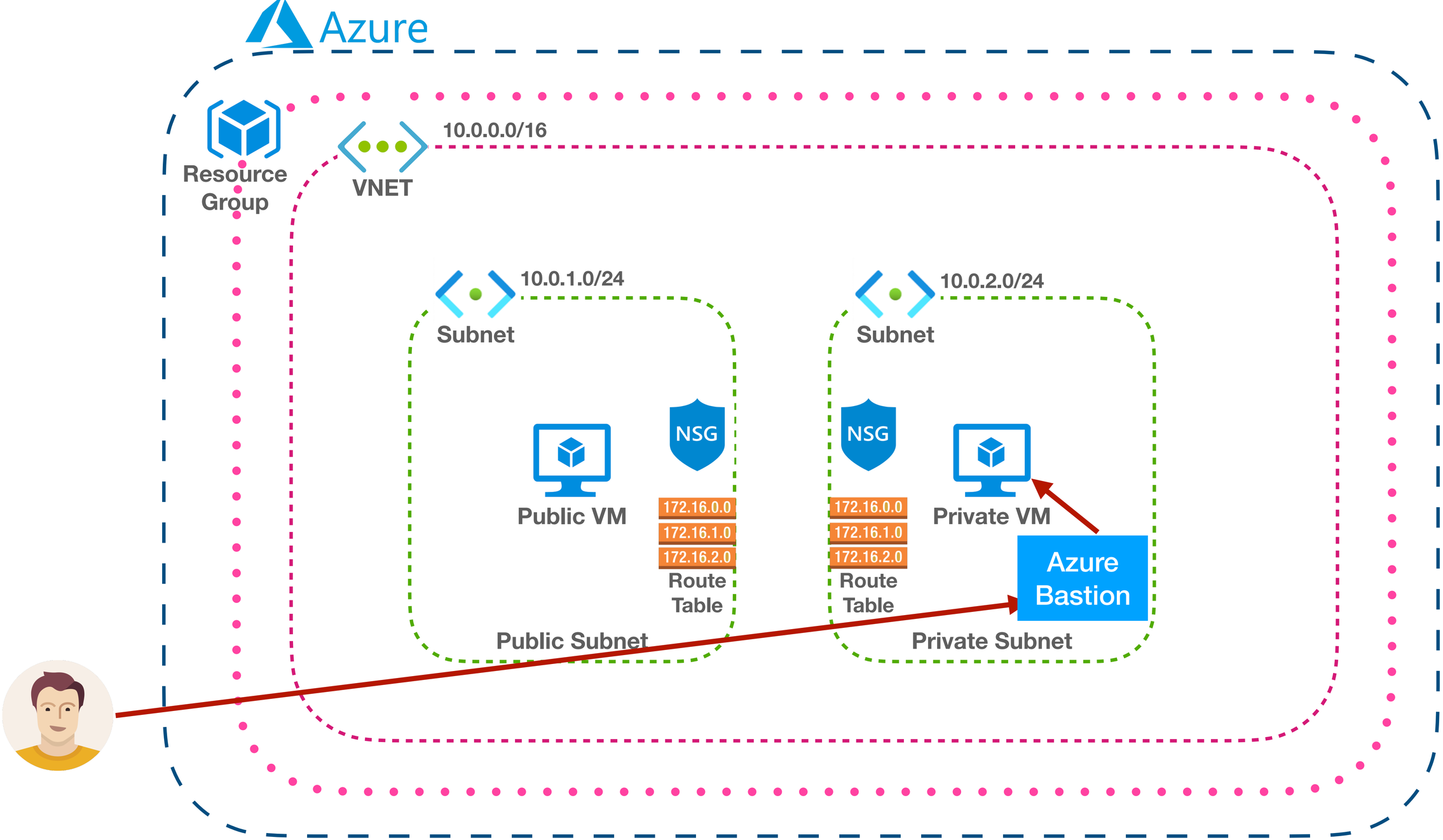


Accessing Resource in the Private Subnet

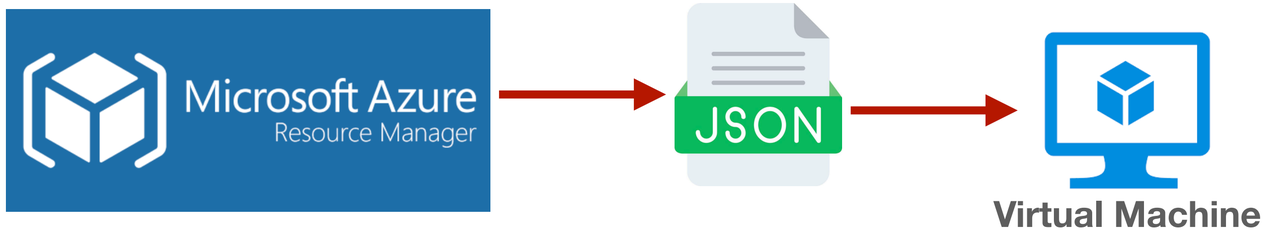
J->Hooq



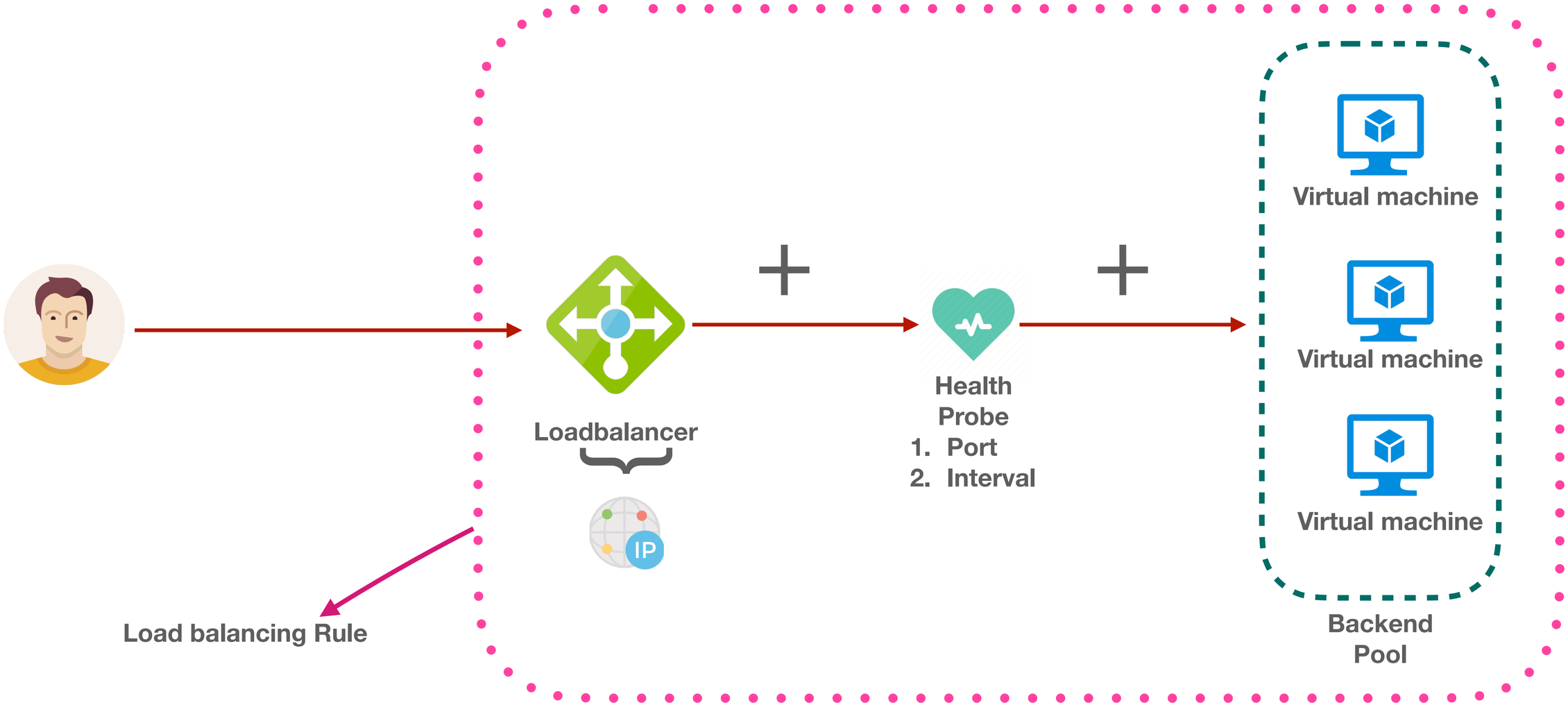
Accessing Resource in the Private Subnet

J->Hooq

Azure Virtual Machine

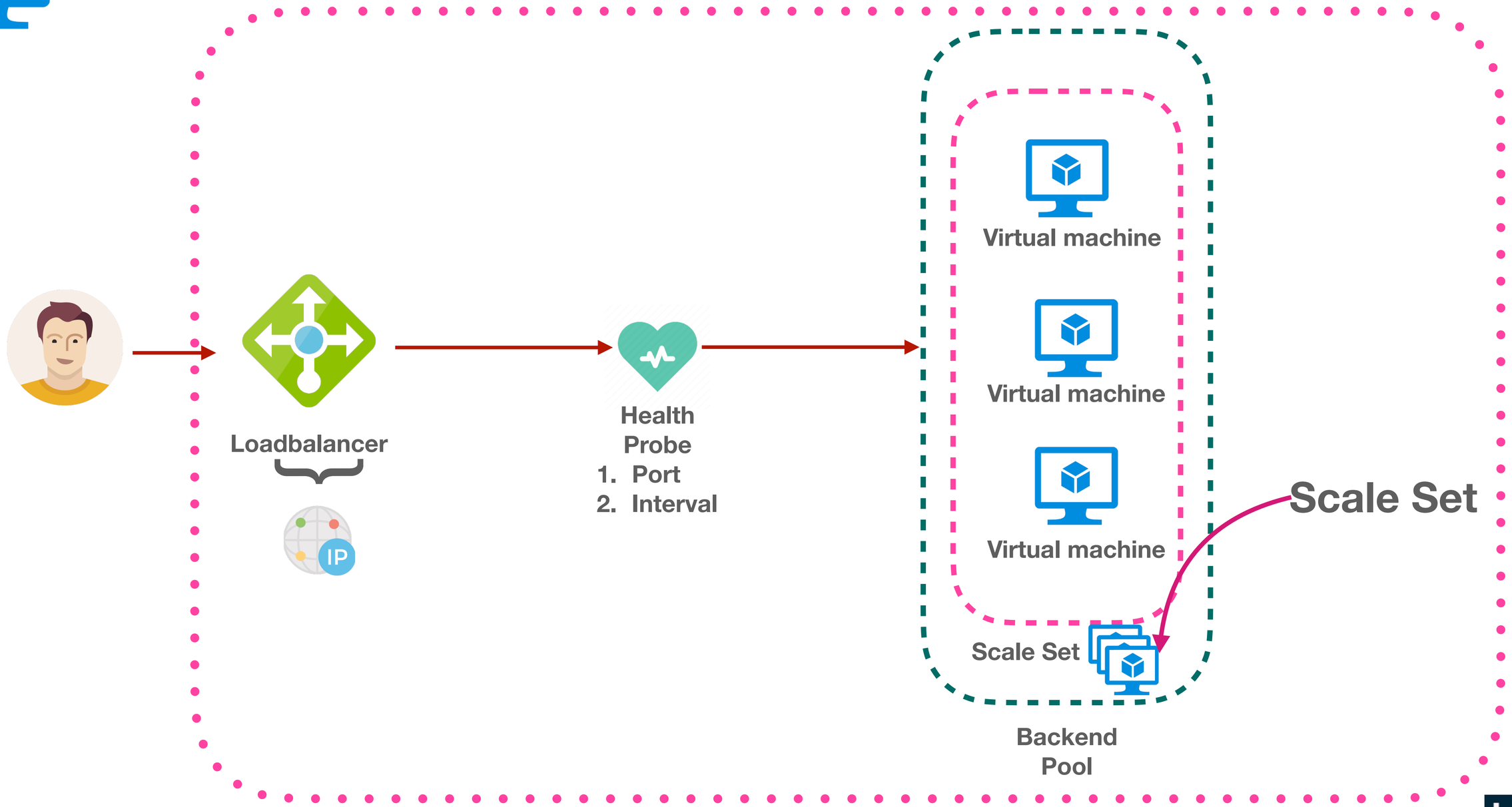


Azure Virtual Machine

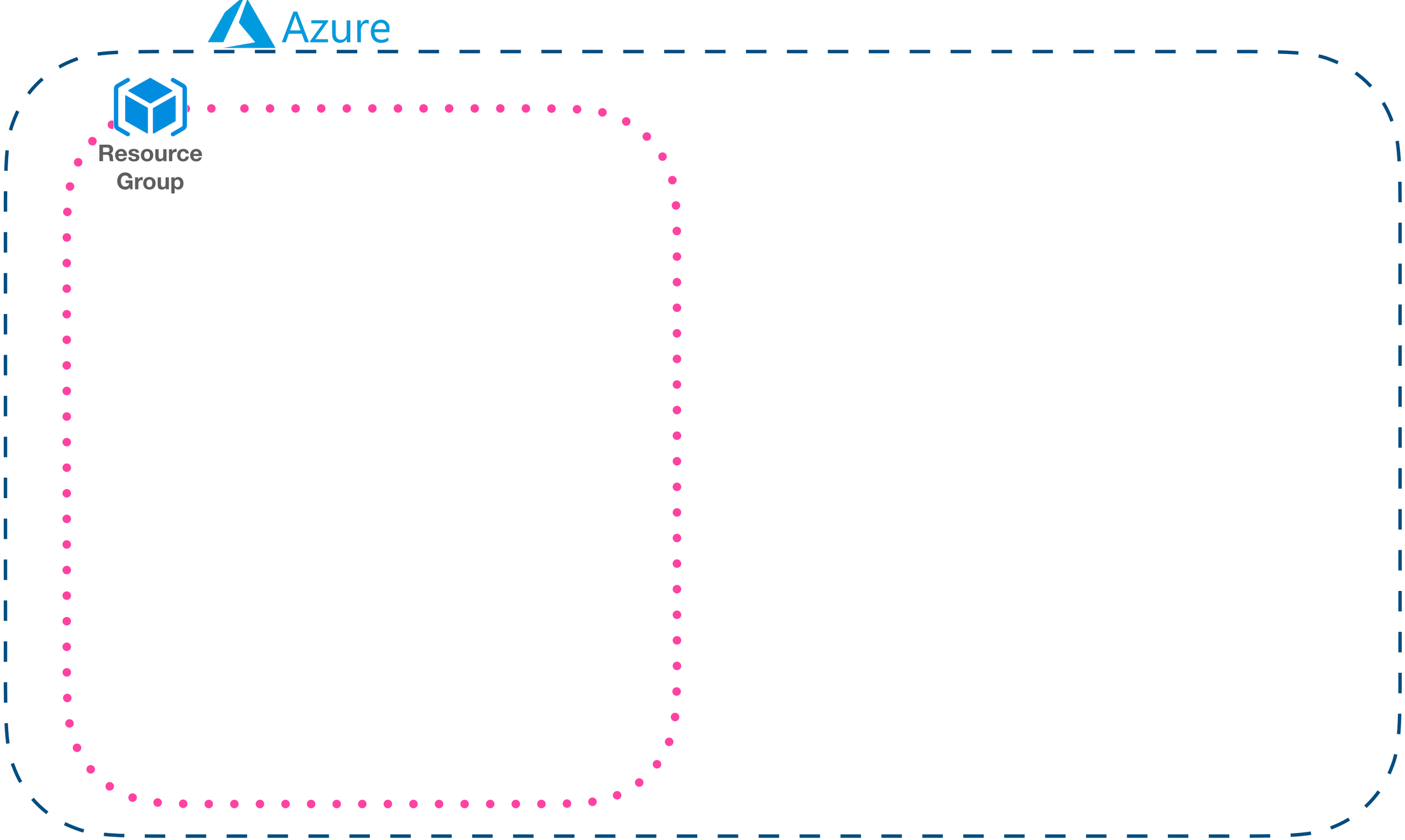


J->Hooq

Azure Virtual Machine + Scale Set

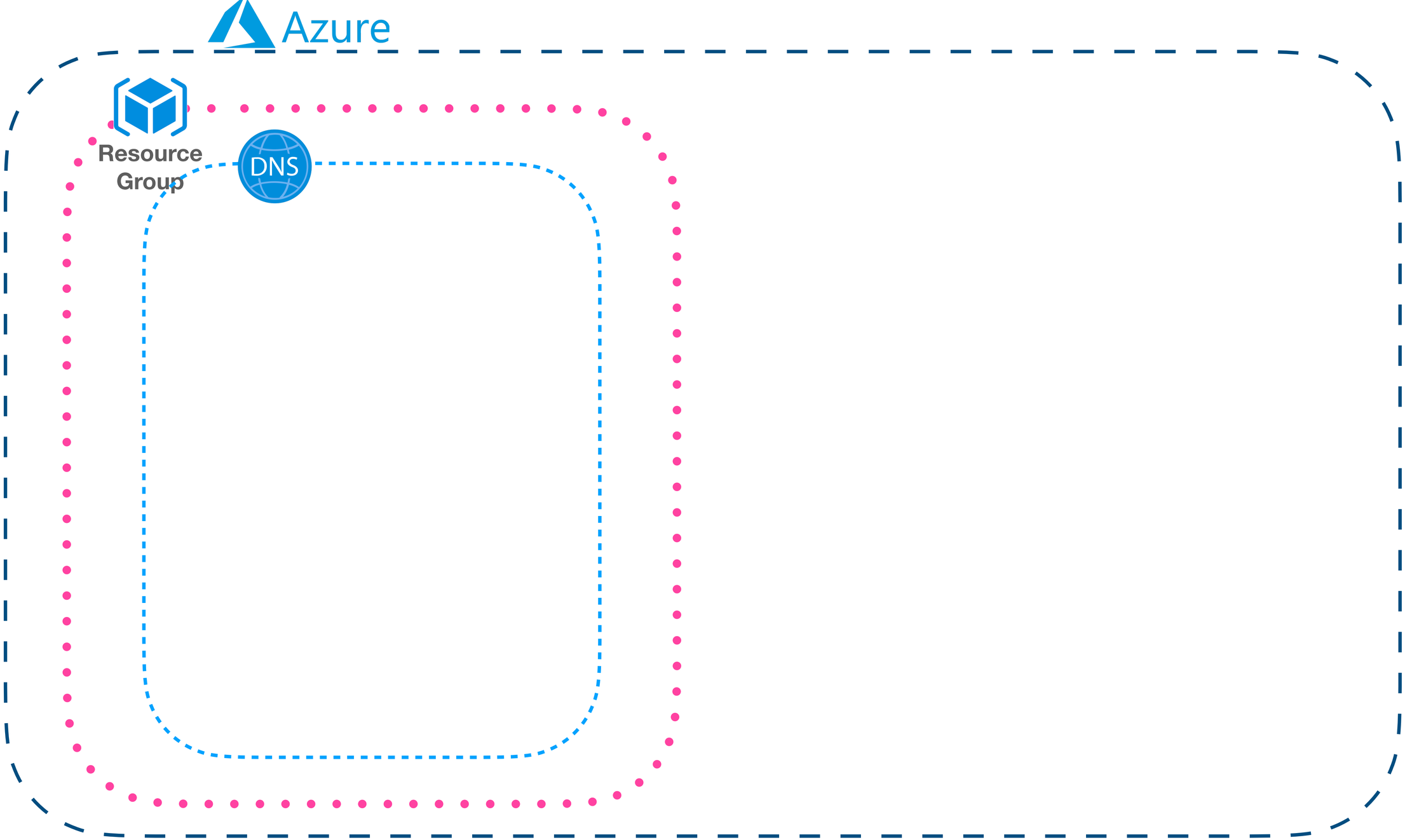


J->Hooq



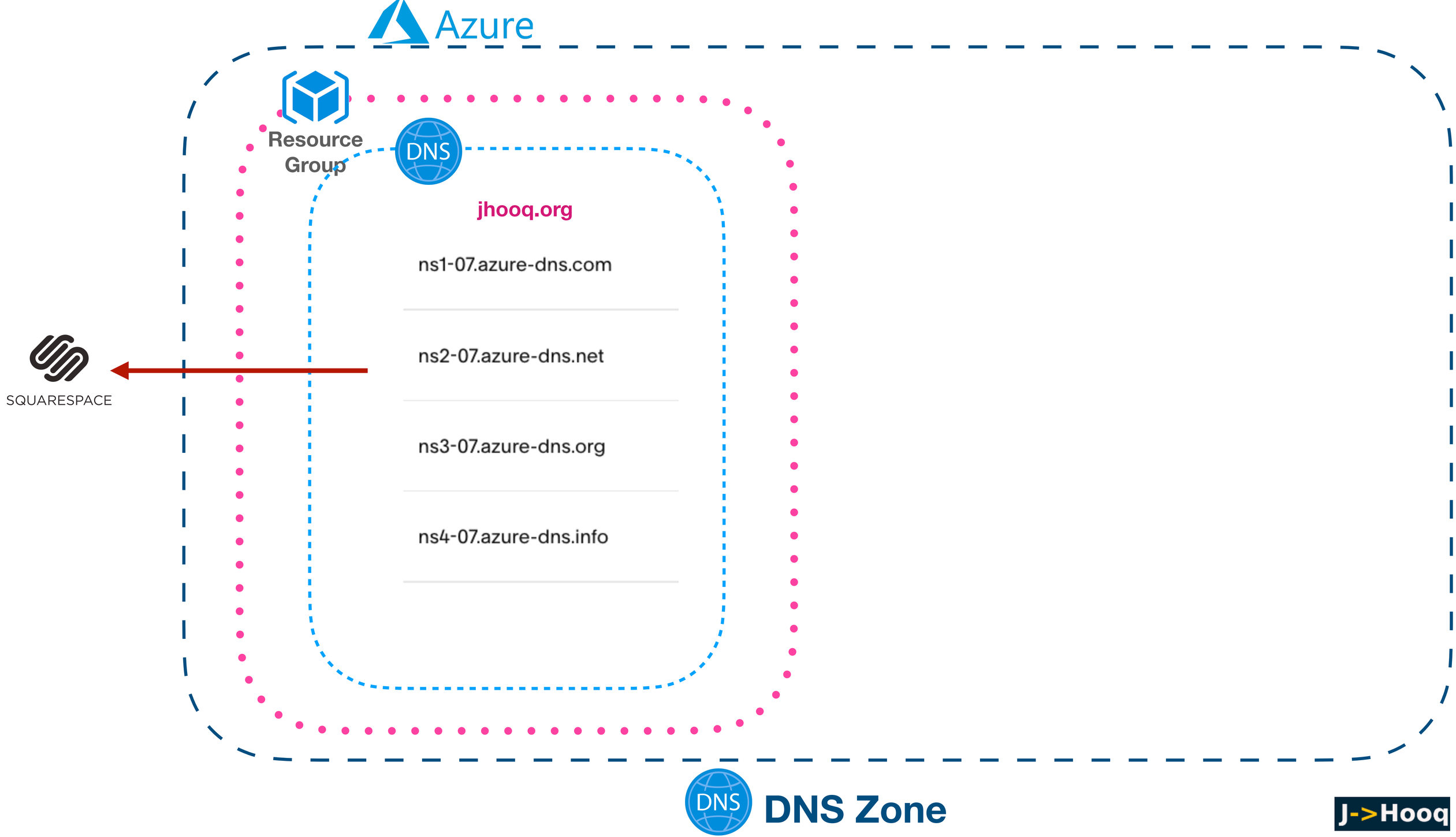
DNS Zone

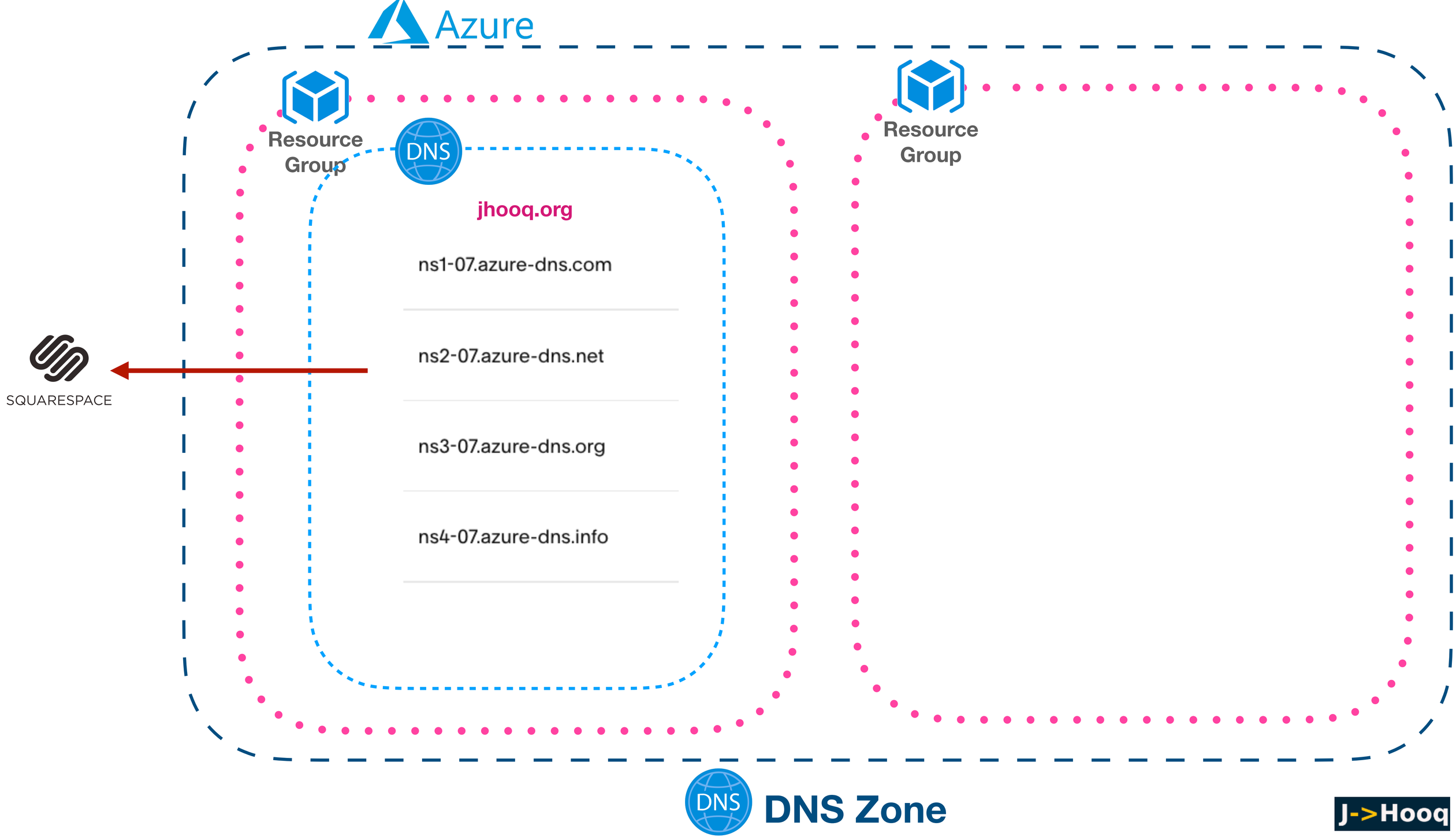
J->Hooq



DNS Zone

J->Hooq


Azure
Resource Group
DNS
jhooq.org
ns1-07.azure-dns.com
ns2-07.azure-dns.net
ns3-07.azure-dns.org
ns4-07.azure-dns.info
SQUARESPACE
DNS
DNS Zone
J->Hooq


Azure
Resource Group
DNS
jhooq.org
ns1-07.azure-dns.com
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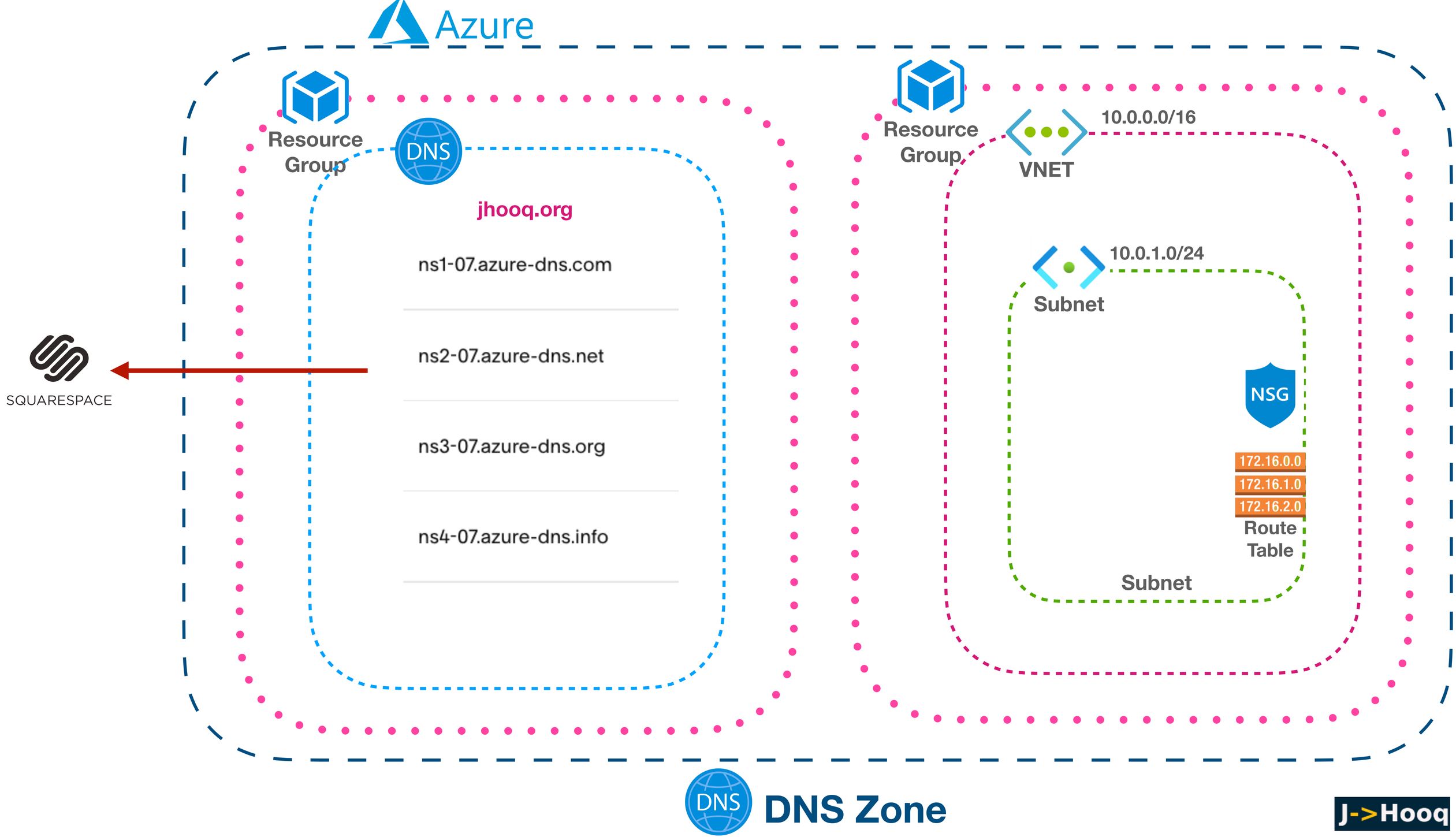
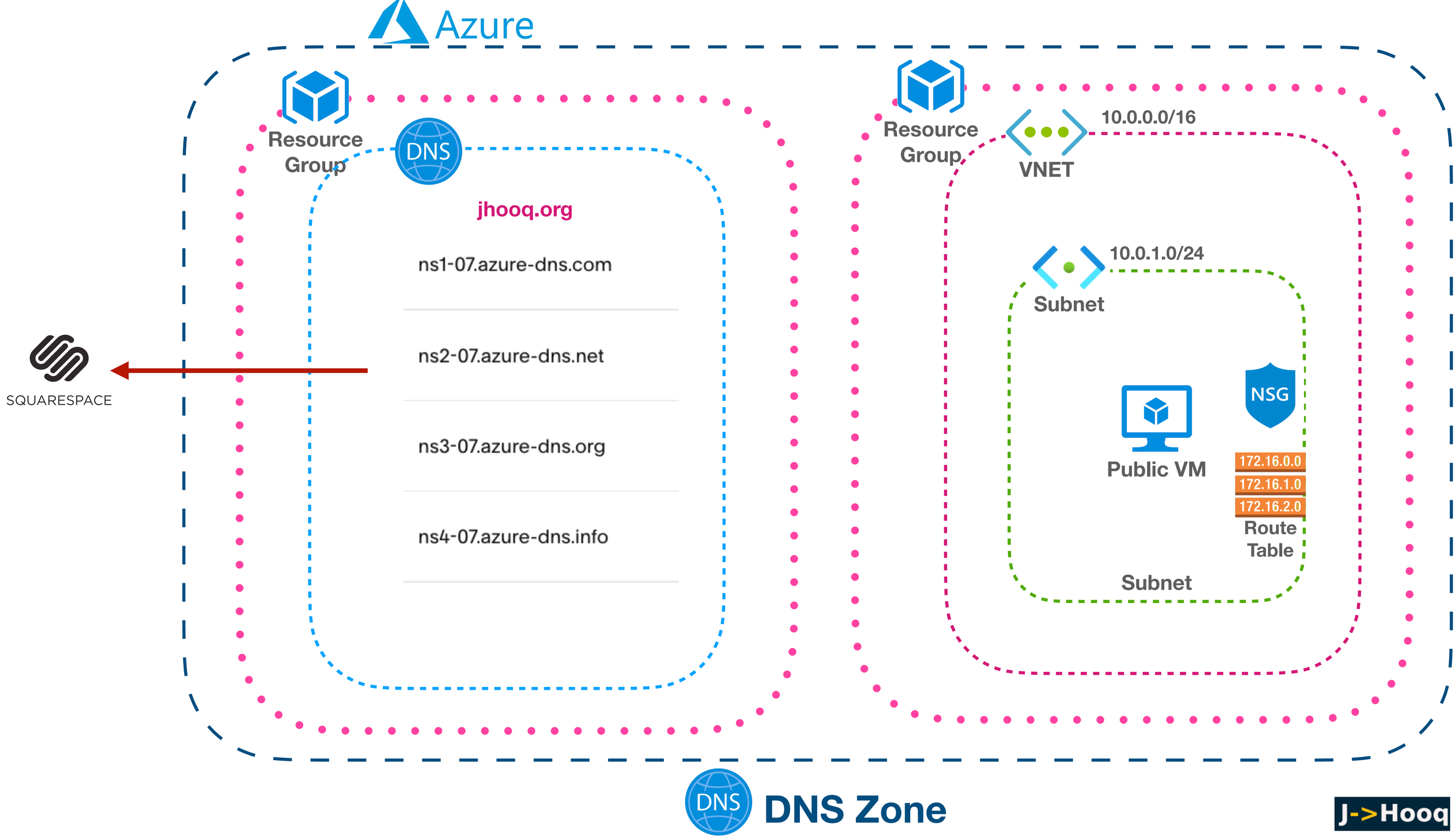
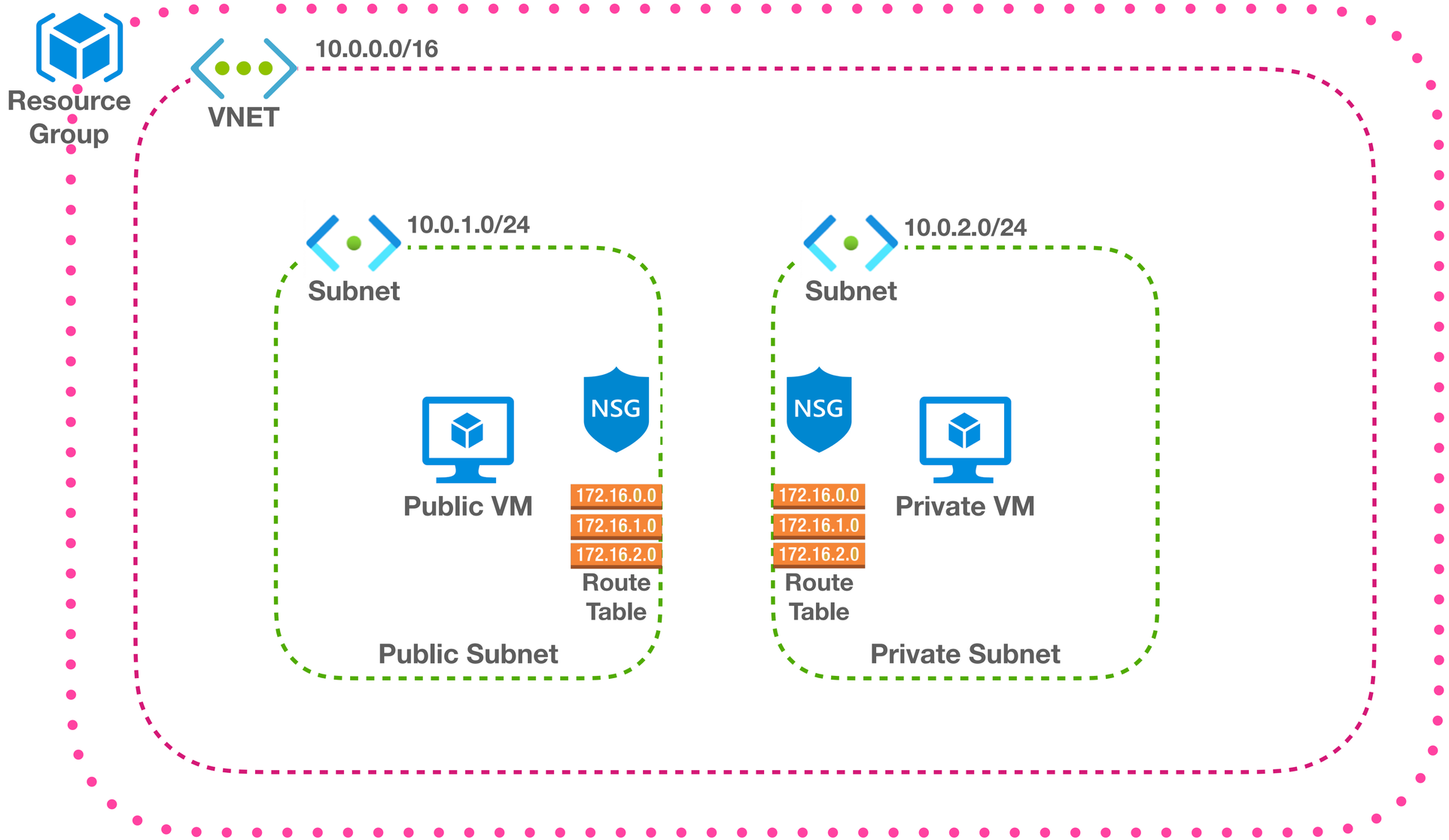
Resource Group
SQUARESPACE
DNS
DNS Zone
J->Hooq


Azure
Resource Group
DNS
jhooq.org
ns1-07.azure-dns.com
ns2-07.azure-dns.net
ns3-07.azure-dns.org
ns4-07.azure-dns.info
SQUARESPACE
Resource Group
VNET
10.0.0.0/16
10.0.1.0/24
Subnet
NSG
172.16.0.0
172.16.1.0
172.16.2.0
Route Table
Subnet
DNS Zone
J->Hooq


Azure
Resource Group
DNS
jhooq.org
ns1-07.azure-dns.com
ns2-07.azure-dns.net
ns3-07.azure-dns.org
ns4-07.azure-dns.info
SQUARESPACE
Resource Group
VNET
10.0.0.0/16
10.0.1.0/24
Subnet
Public VM
NSG
172.16.0.0
172.16.1.0
172.16.2.0
Route Table
Subnet
DNS Zone
J->Hooq

Azure Tags



J->Hooq

Azure Resource Lock

1. Types of Lock

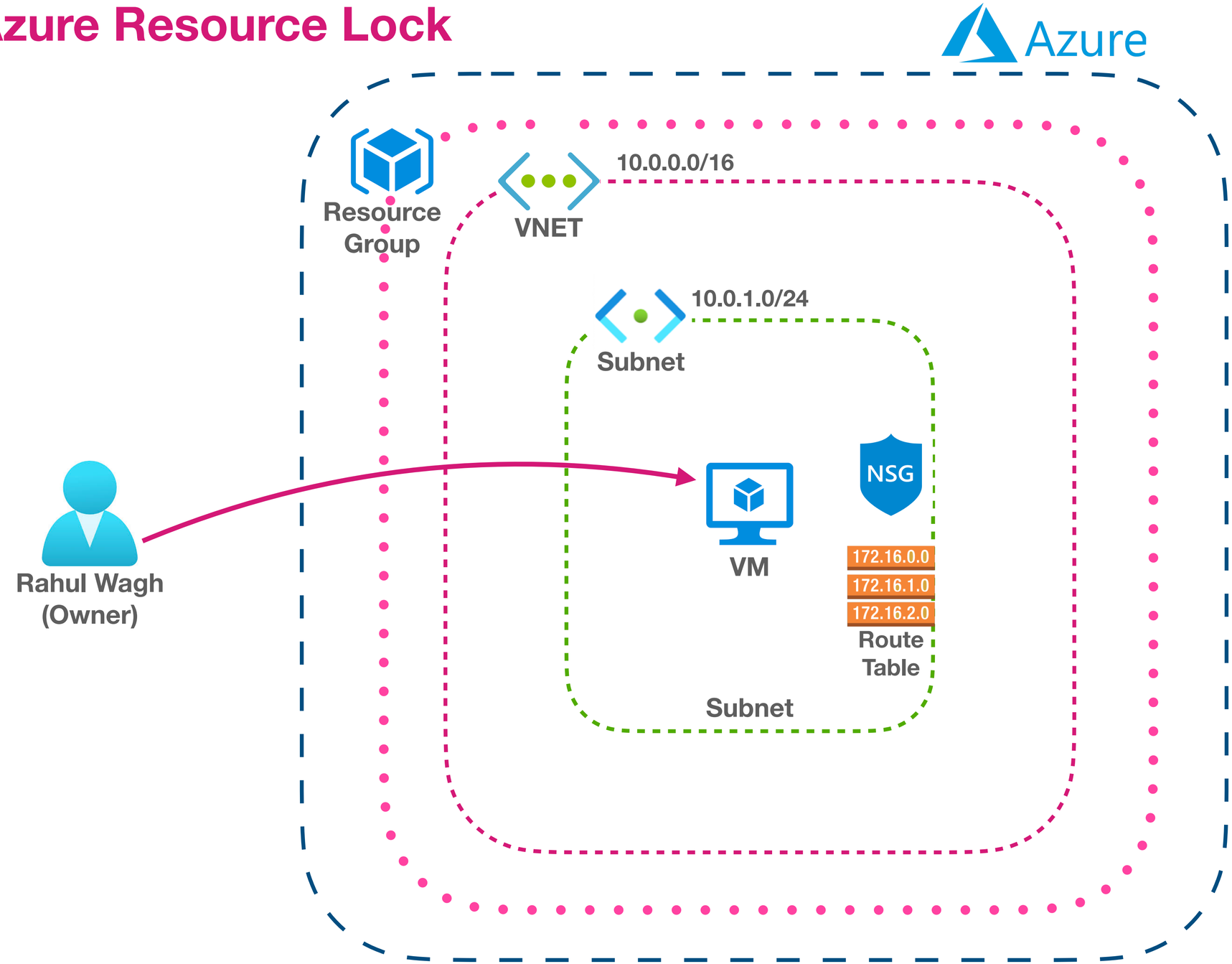
- **ReadOnly**
- **CanNotDelete**

Rahul Wagh
(Owner)

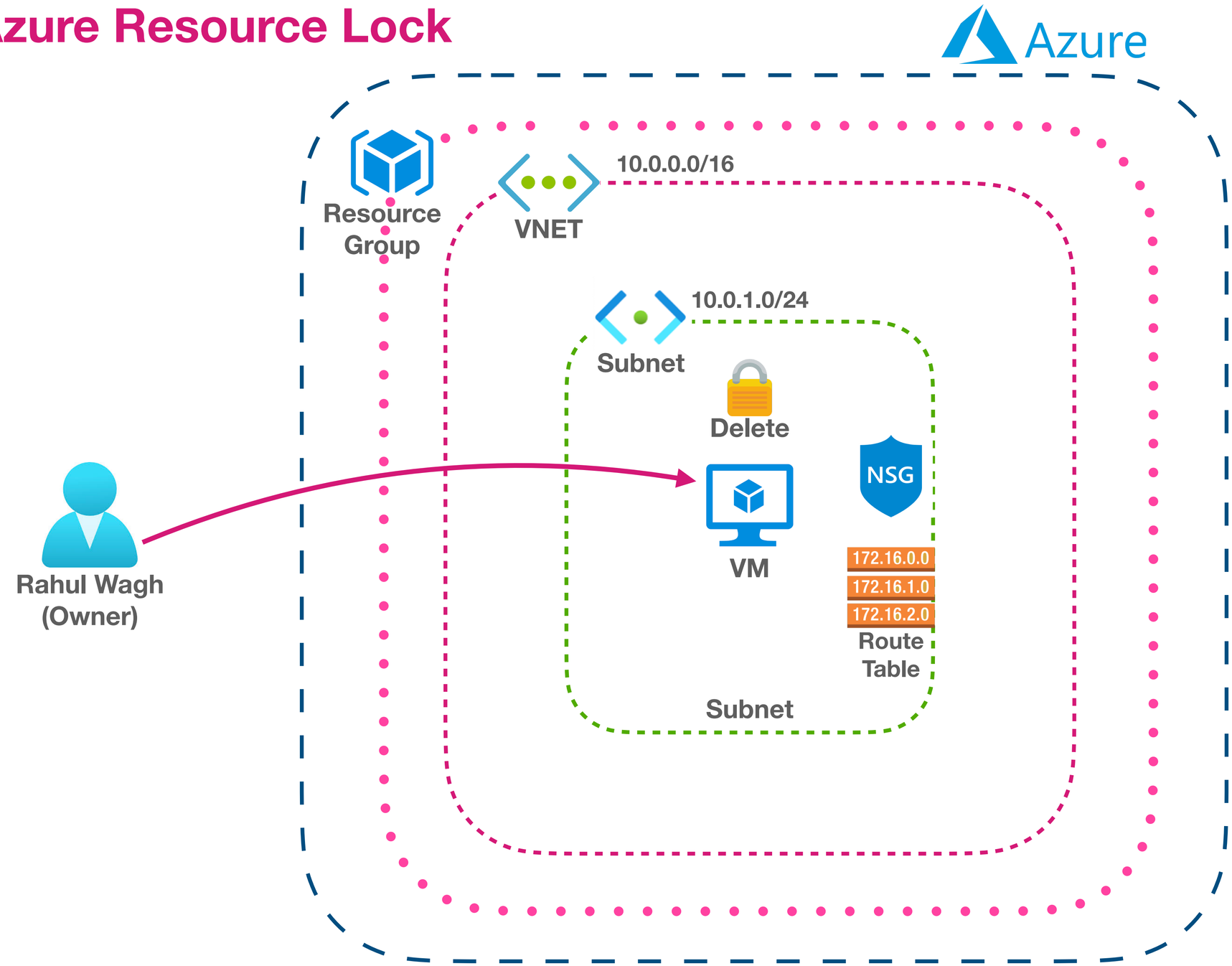
test.user
(Virtual Machine Contributor)

Name	Type	Role	Scope	Condition
Owner (1)				
RW Rahul Wagh 11f56981-0d42-41bd-90e1-130de...	User	Owner	Subscription (Inherited)	None
Virtual Machine Contributor (1)				
TE test.user 132736bc-1e34-42f8-964e-debcc...	User	Virtual Machine Contributor	This resource	None

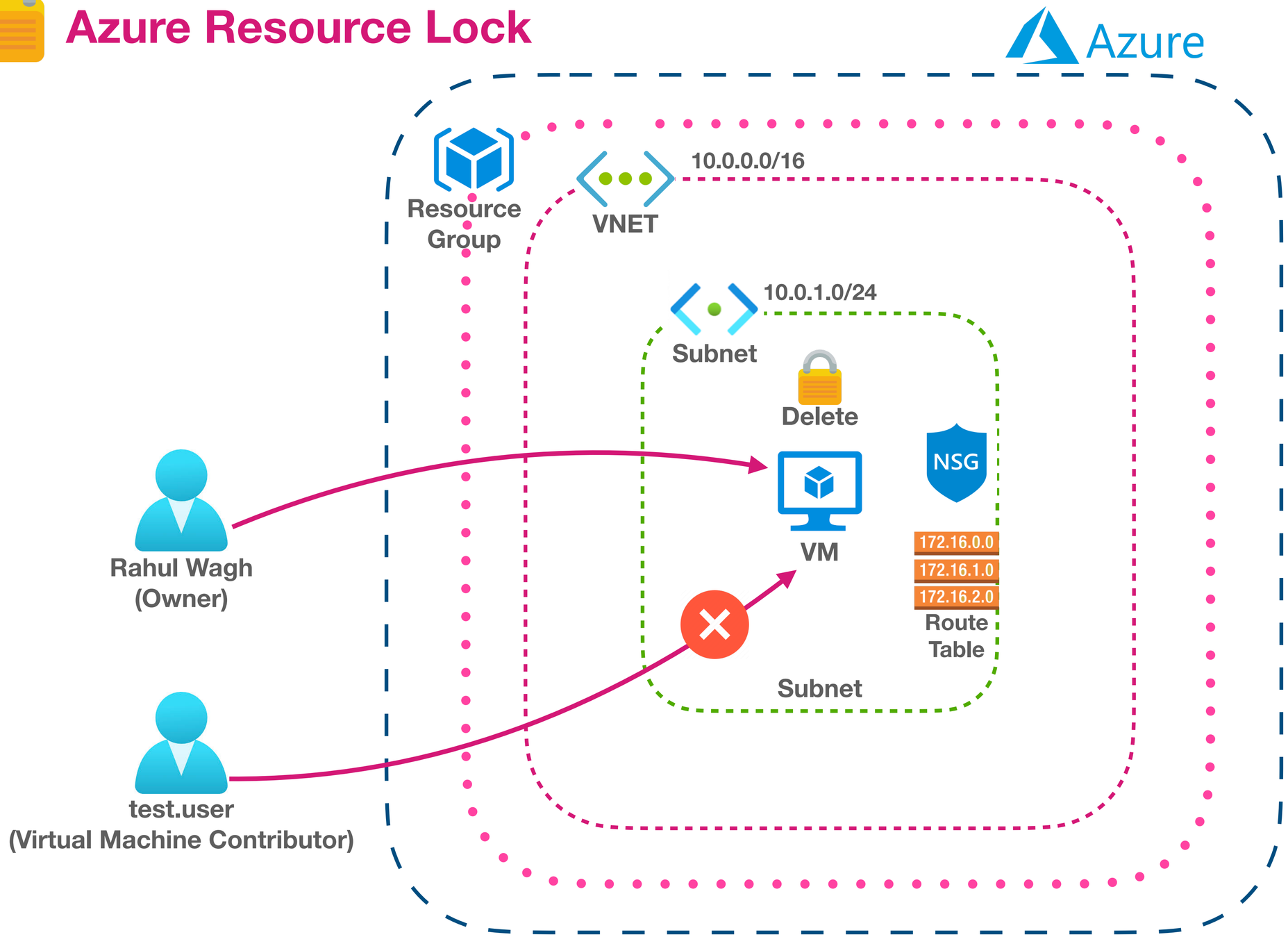
Azure Resource Lock



Azure Resource Lock



Azure Resource Lock



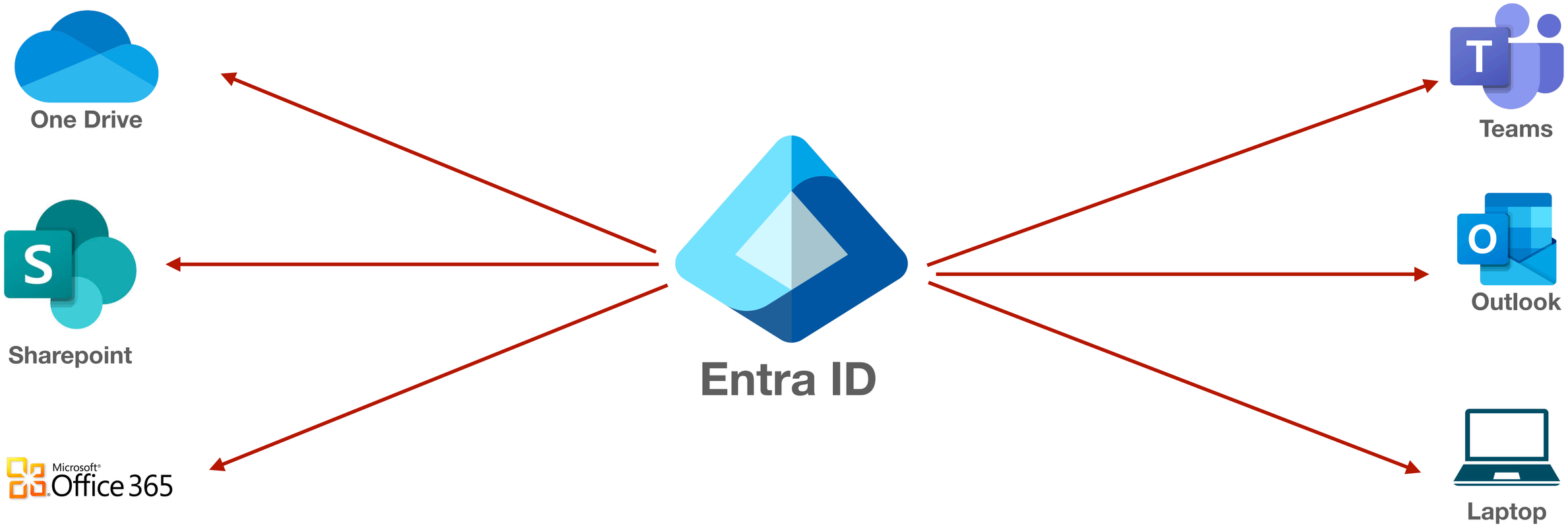
Microsoft Entra ID

1. What is Microsoft Entra ID?

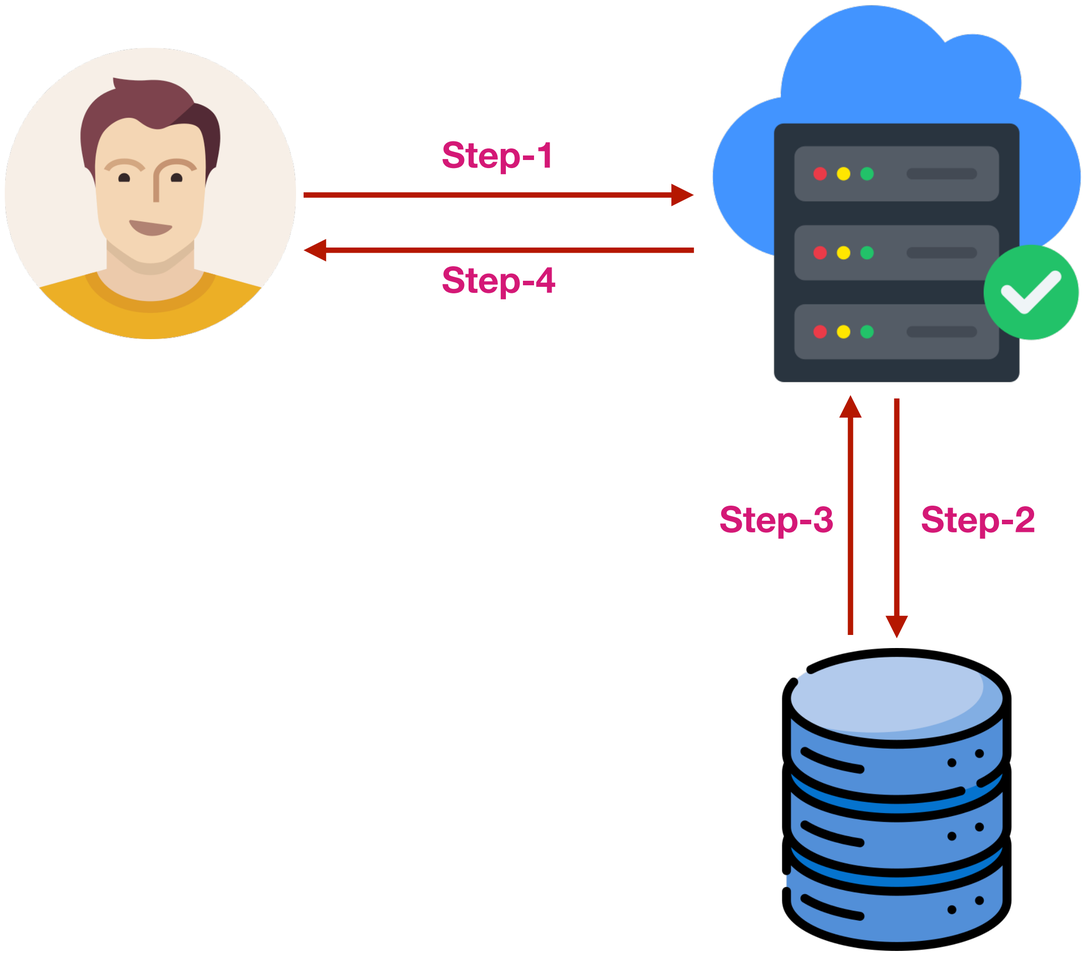
- Cloud Based Identity Gatekeeper
- Ensure Right People get access to Right Resources

Gatekeeper

Microsoft Entra ID

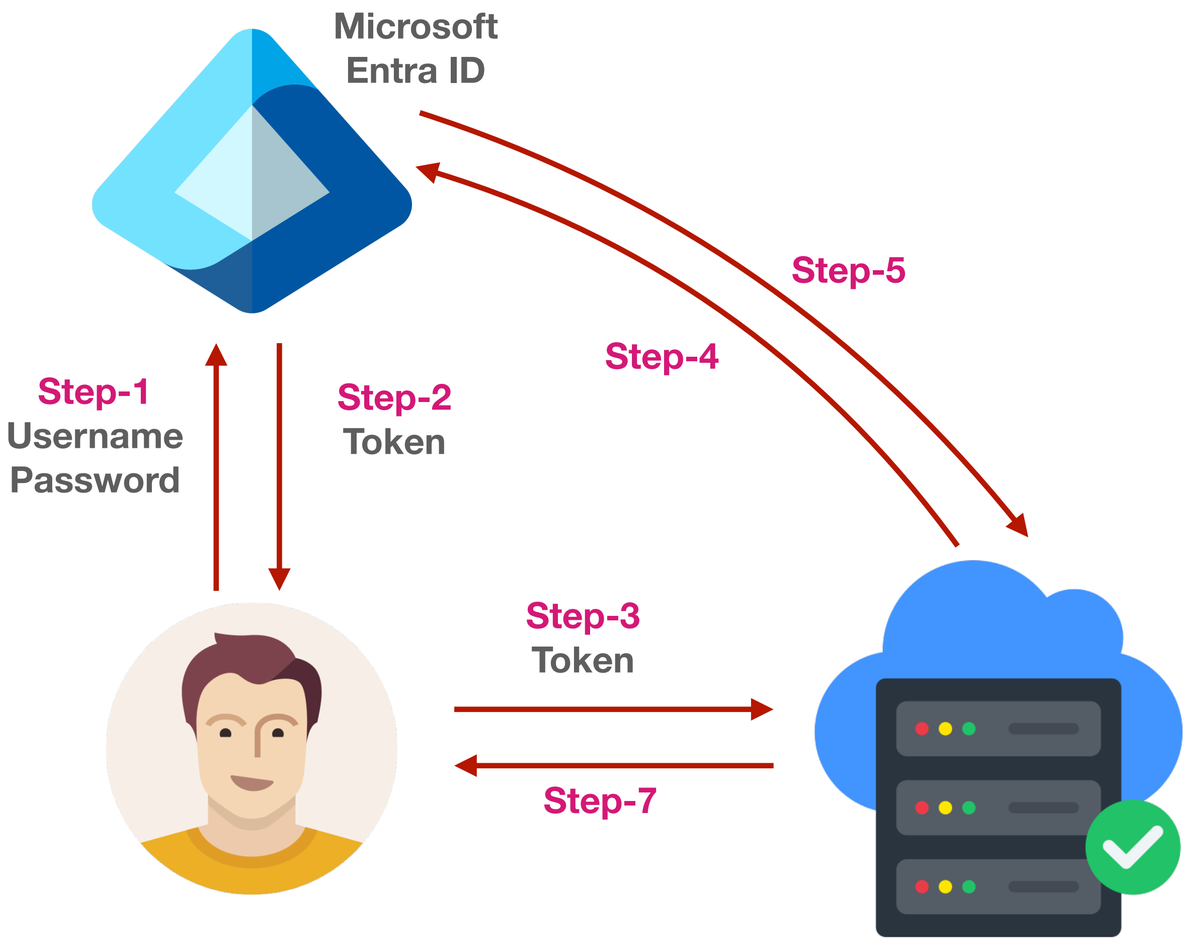


Microsoft Entra ID



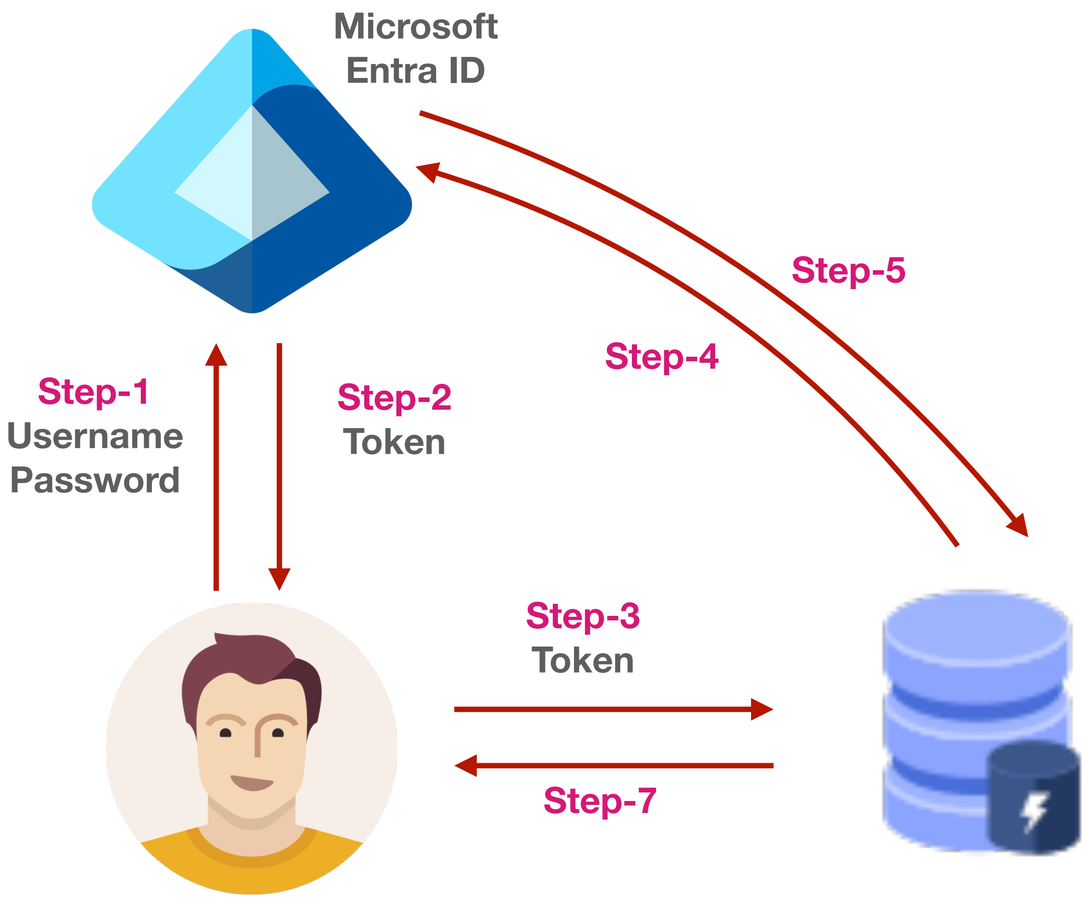
Classical Authentication

Microsoft Entra ID



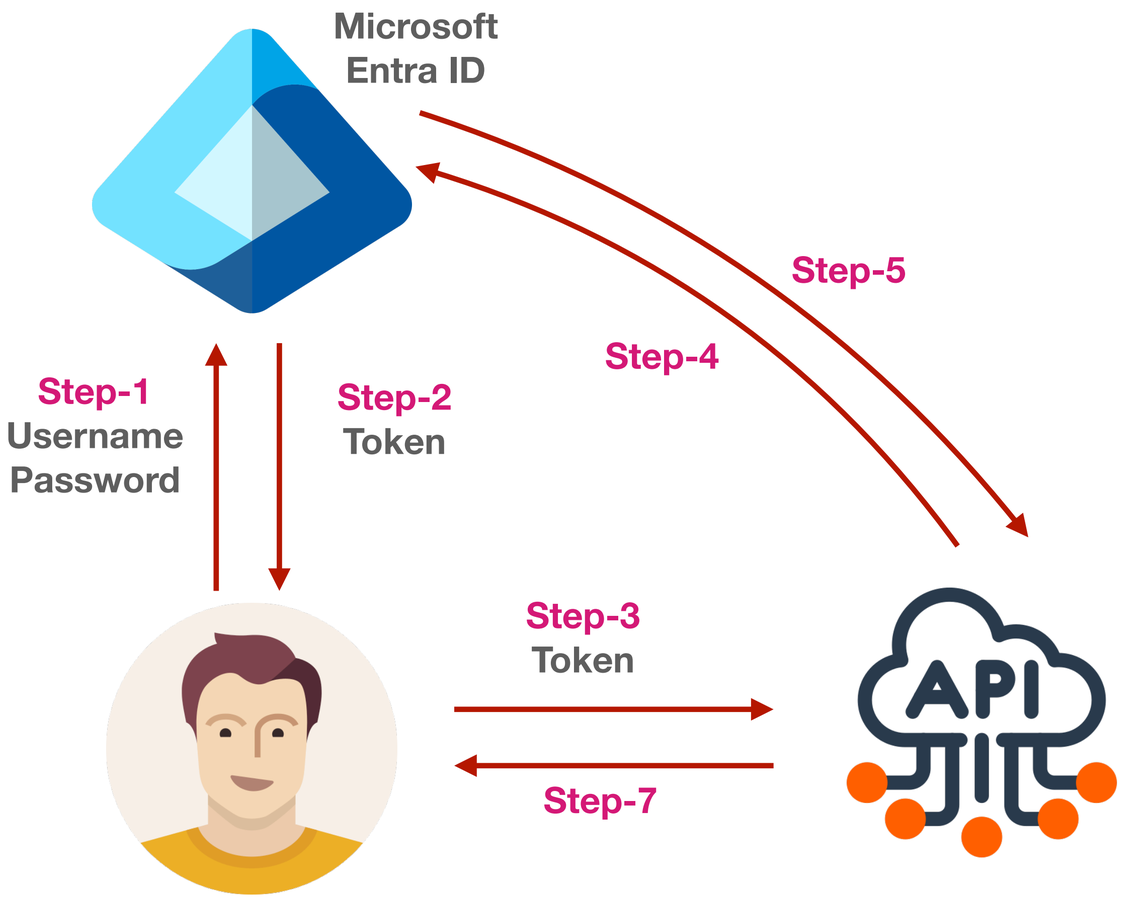
Microsoft Entra ID Authentication

Microsoft Entra ID



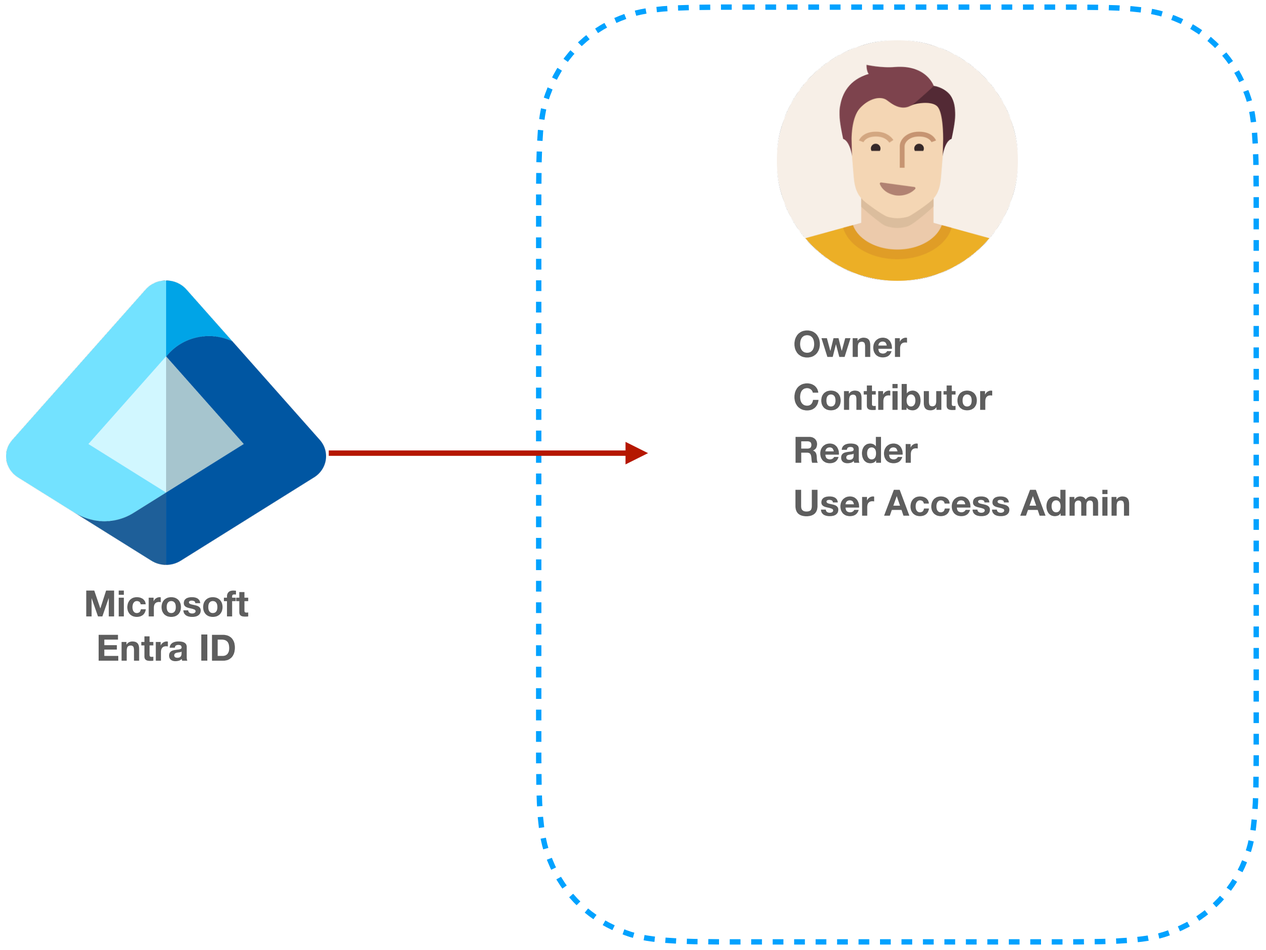
Microsoft Entra ID Authentication

Microsoft Entra ID



Microsoft Entra ID Authentication

Microsoft Entra ID



Permissions